

Electronics and Communication Systems

Electronics Systems

Master Degree in **Computer Engineering**

<https://computer.ing.unipi.it/ce-lm>

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Outline

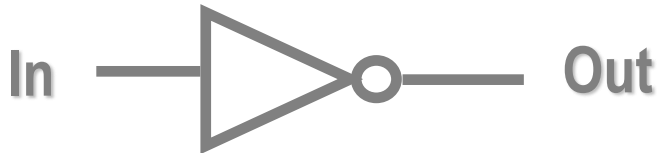
- Electronic Abstraction Level Design
- Integrated Circuit Design Styles
- Design Methodologies and Flows
- Cost-Performance Trade-off
- Field Programmable Gate Array
- Design Productivity
- Evolution of EDA Industry
- How to map a system/algorithm to a System on Chip

Inverter: Electronic Abstraction Levels

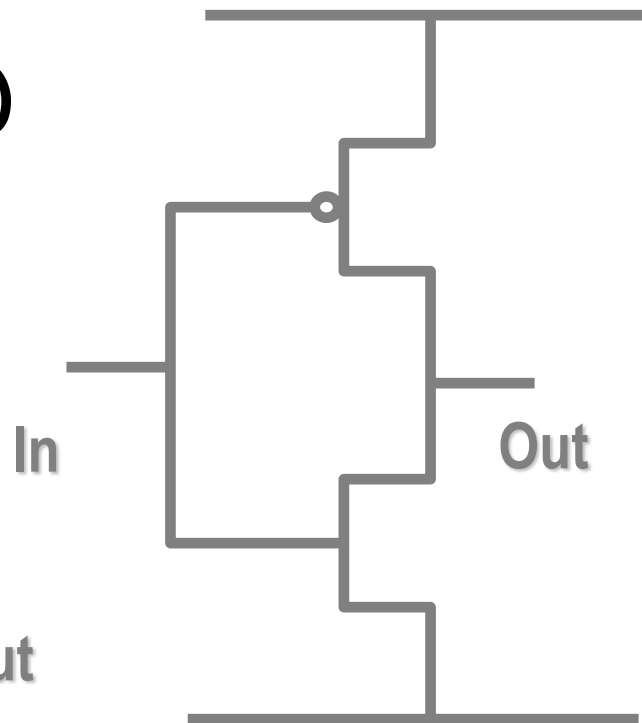
RTL

Out <= NOT (In)

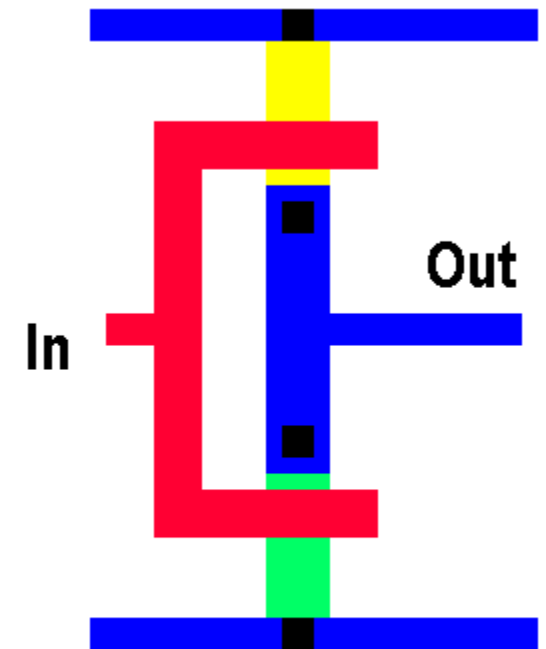
Gate



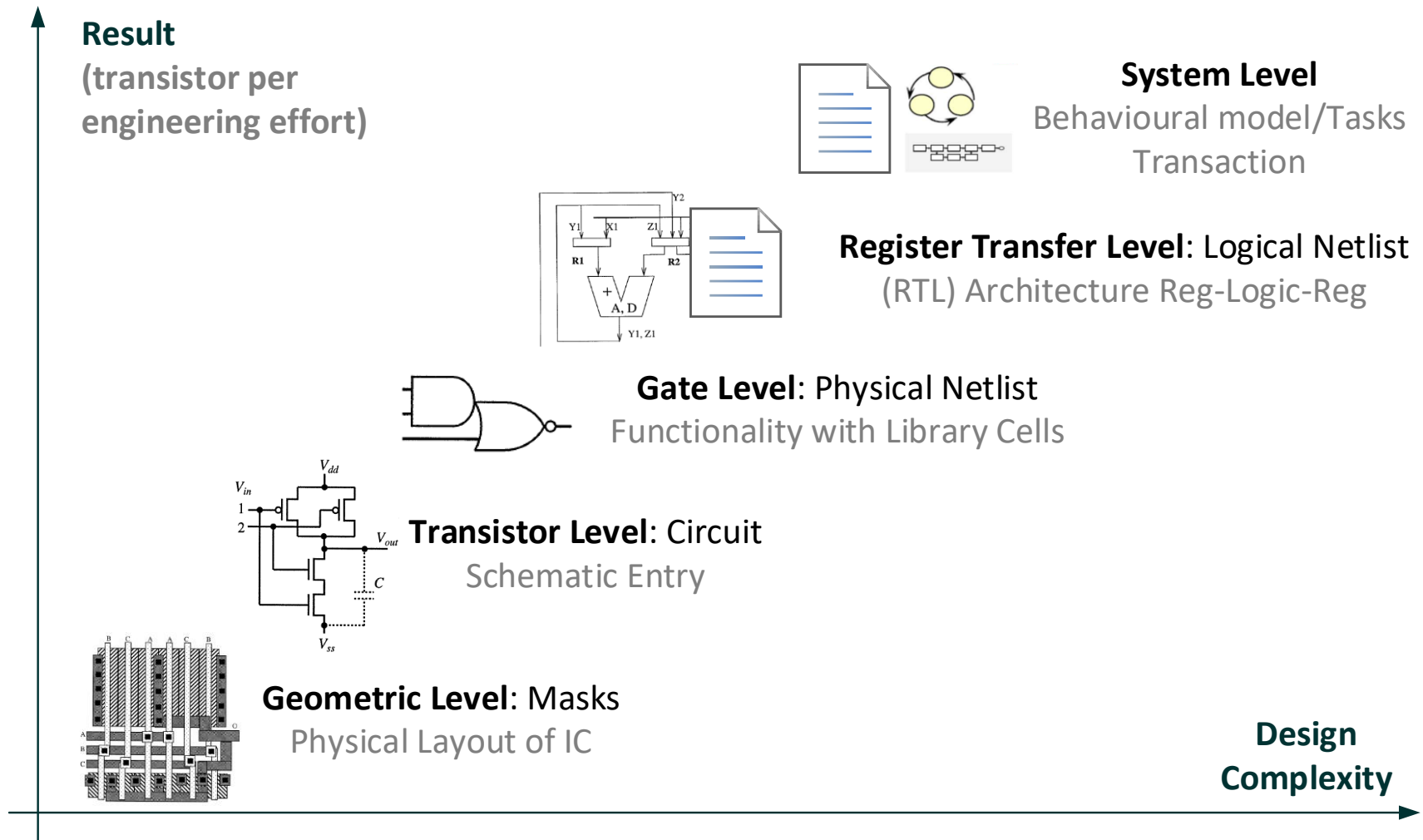
Transistor



Lay-out



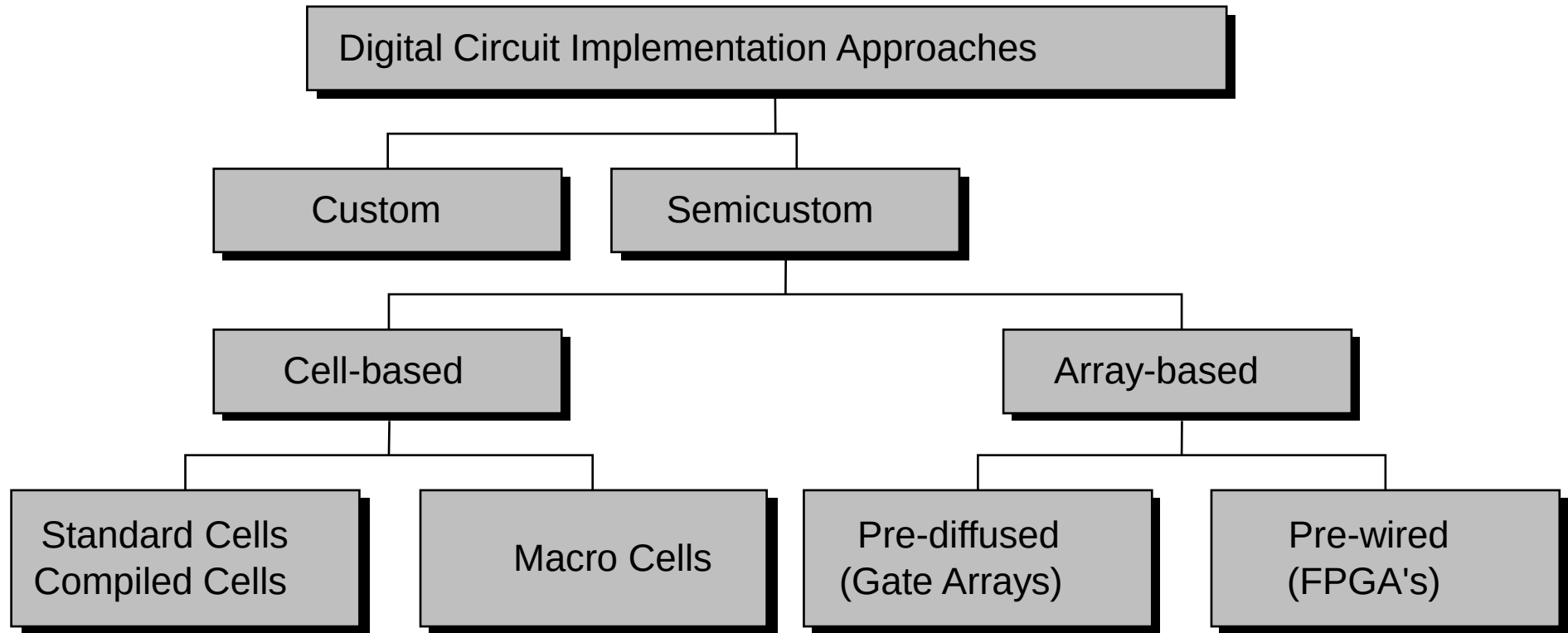
Electronic Abstraction Level Design



Outline

- Electronic Abstraction Level Design
- Integrated Circuit Design Styles
- Design Methodologies and Flows
- Cost-Performance Trade-off
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Integrated Circuit Design Styles



Let's see the main difference among these implementation approaches !!

ASIC Design Approaches

- full-custom

- Maximum Design Freedom: sizing, placement and interconnection for each single transistor

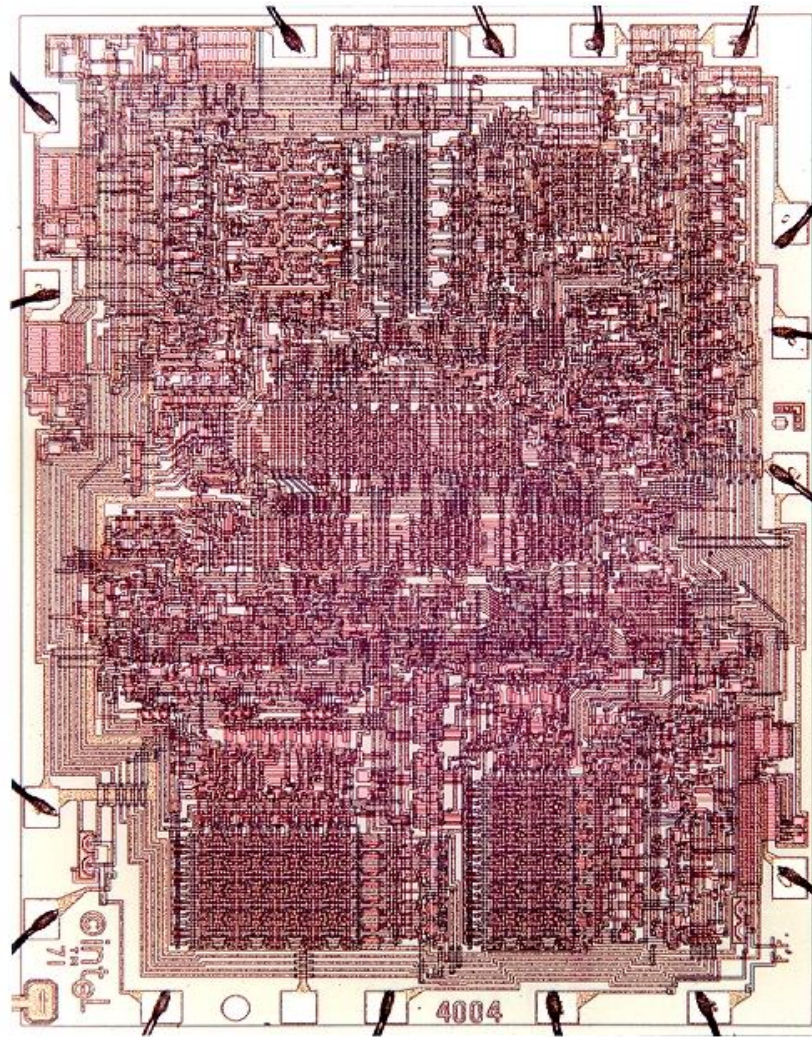
- semi-custom

- Reduction of some Design Freedom: based on pre-designed and/or pre-manufactured logic structure (transistor, gate, logic cells) and defining only their interconnection

ASIC full-custom

- Maximum Design Freedom
- Maximum Potential of Optimization
 - speed
 - area complexity
- High Cost and Development time
- High skills required on ASIC technology, Electronic Circuit and Architecture

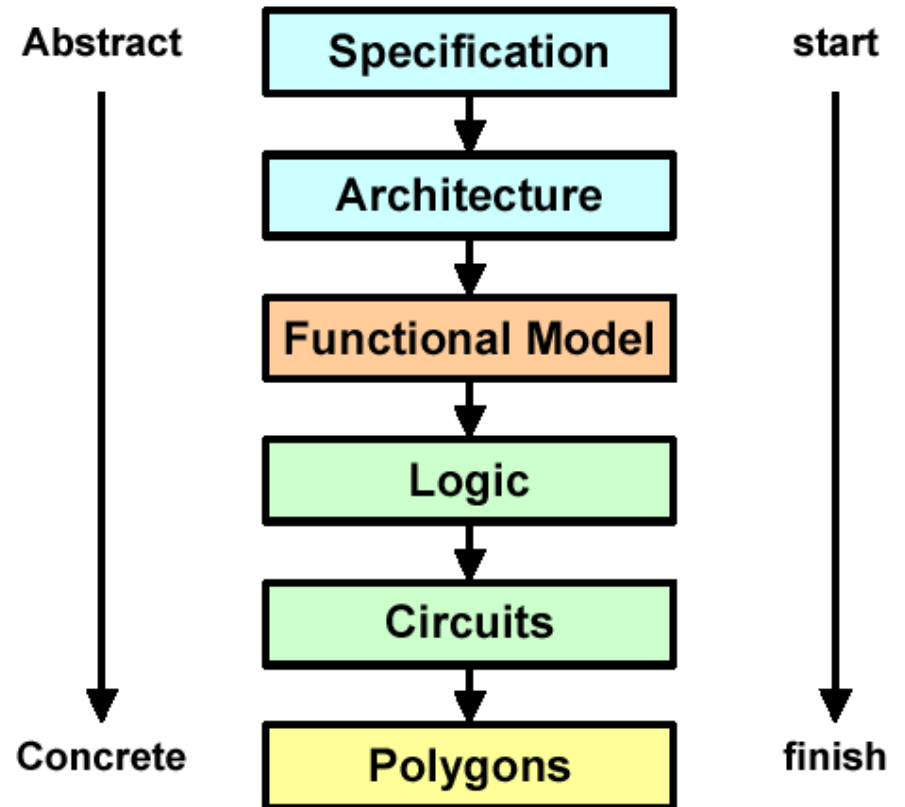
The Custom Approach



Intel 4004

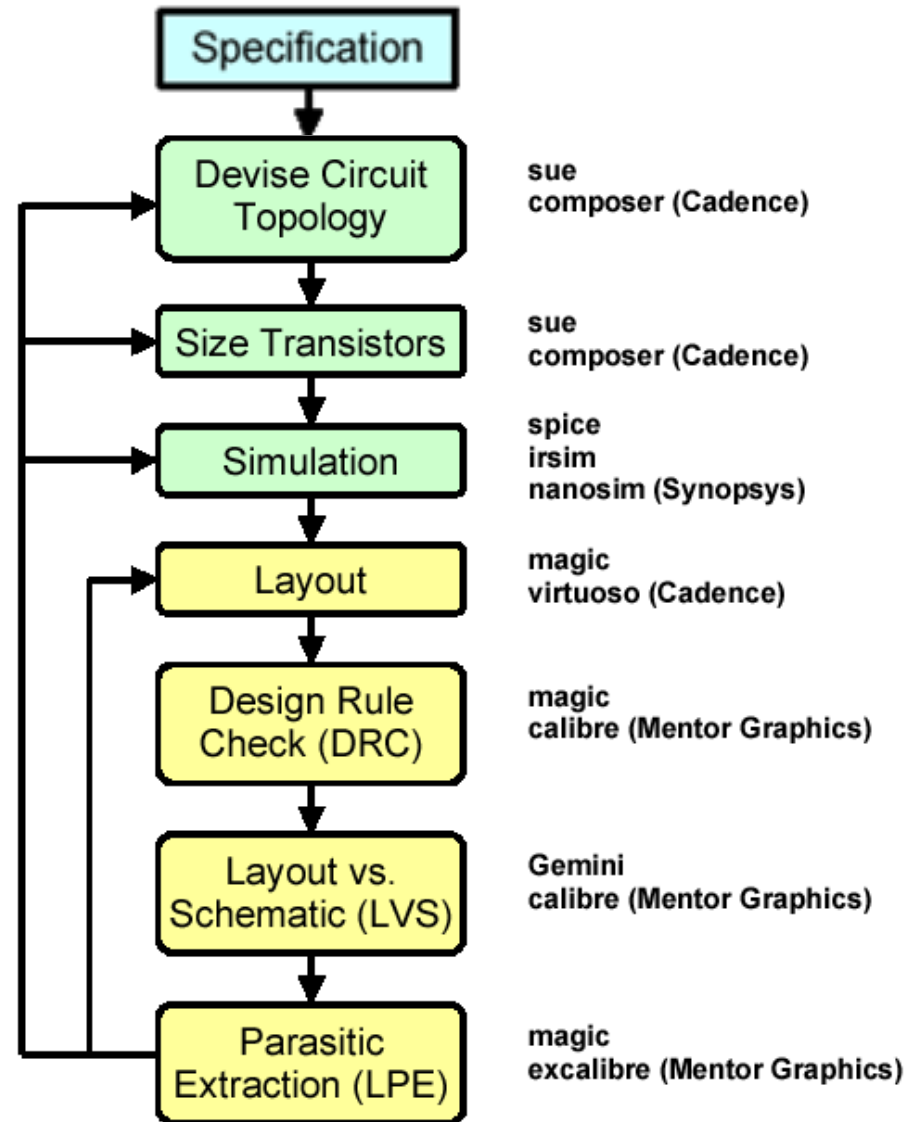
Levels of Design Abstraction

- Consider entire design at top levels only
- Increasing amounts of automation possible further down the stack

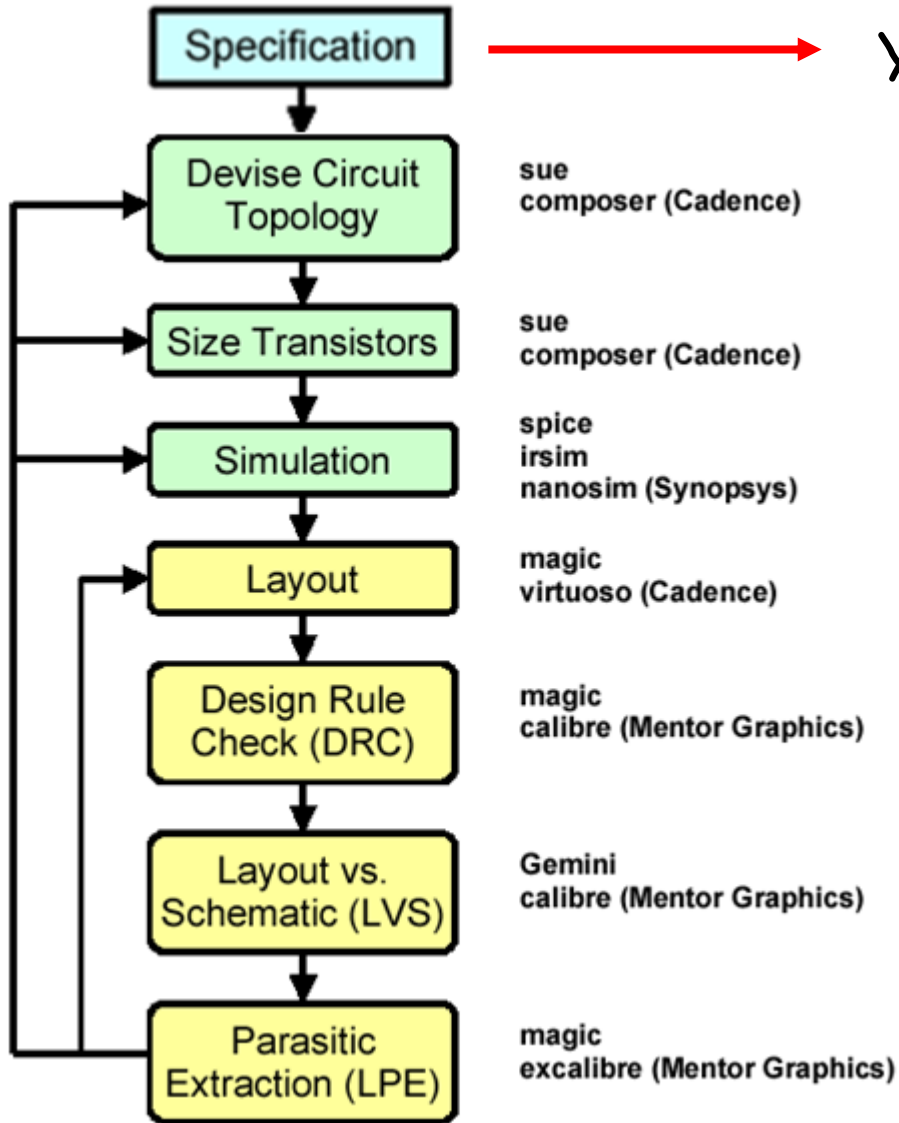


Full Custom Design Flow

- Has the best performance
- Is the most labor intensive



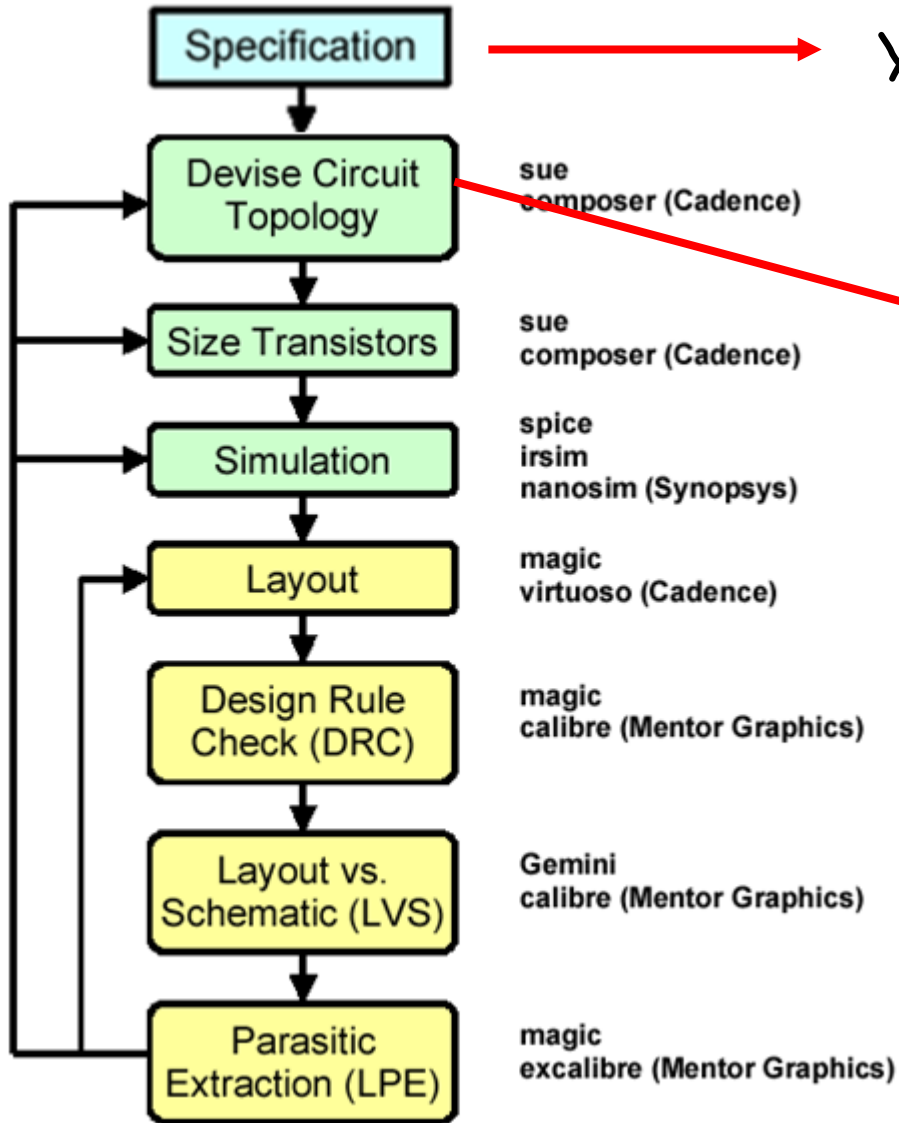
Full Custom Design Flow



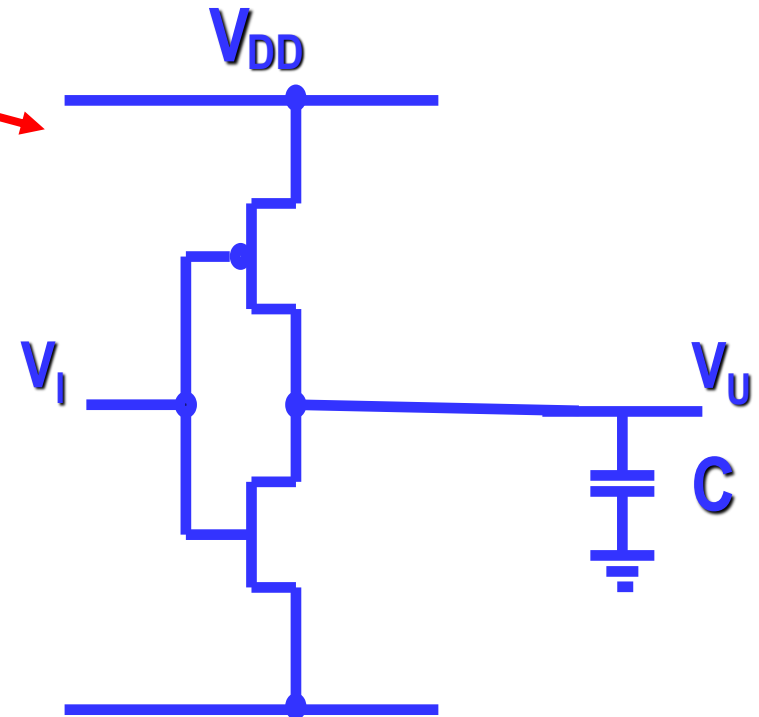
$$Y = \text{NOT}(X)$$

- Timing
- Area
- Power Consumption
- Etc.

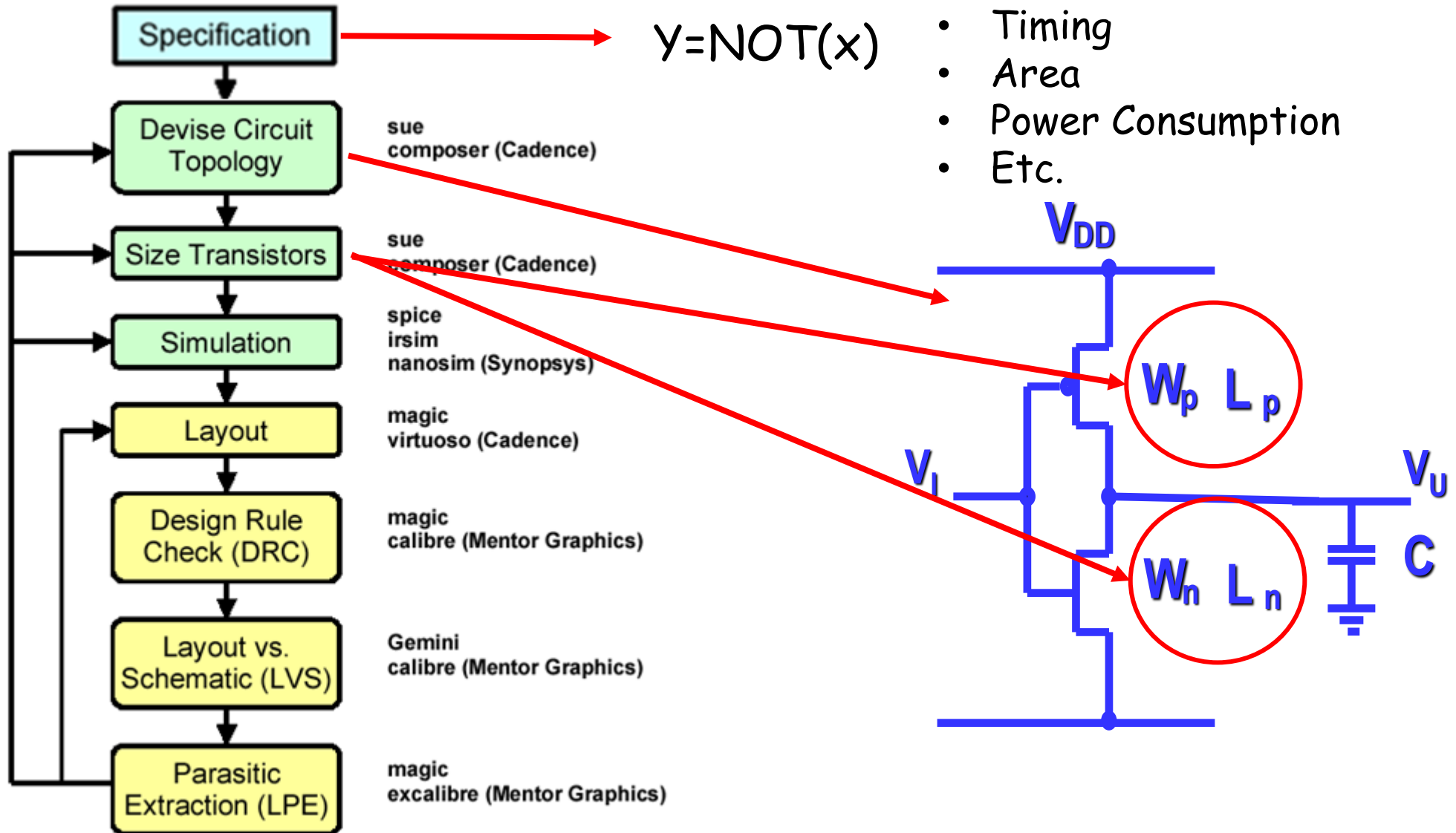
Full Custom Design Flow



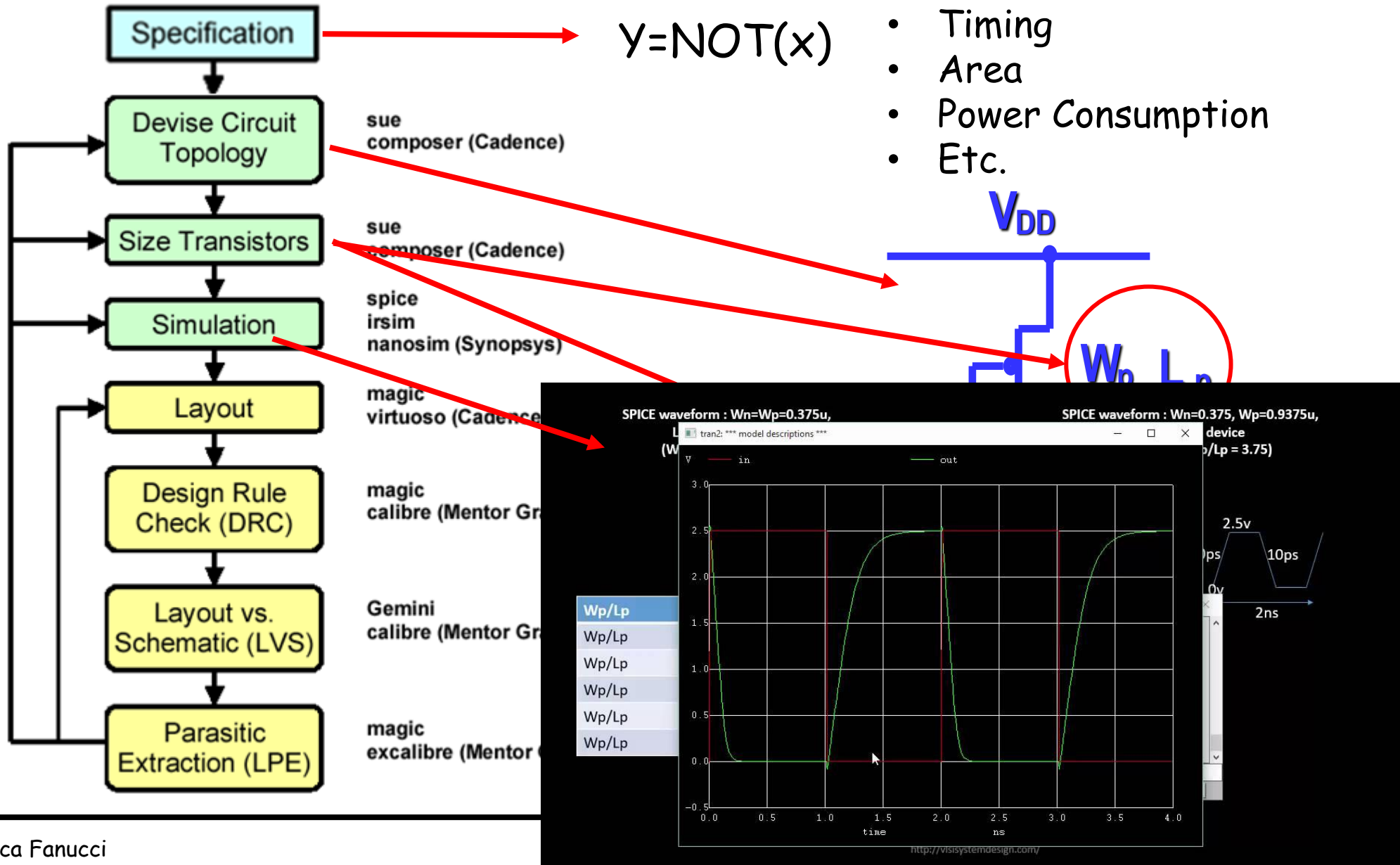
- Timing
- Area
- Power Consumption
- Etc.



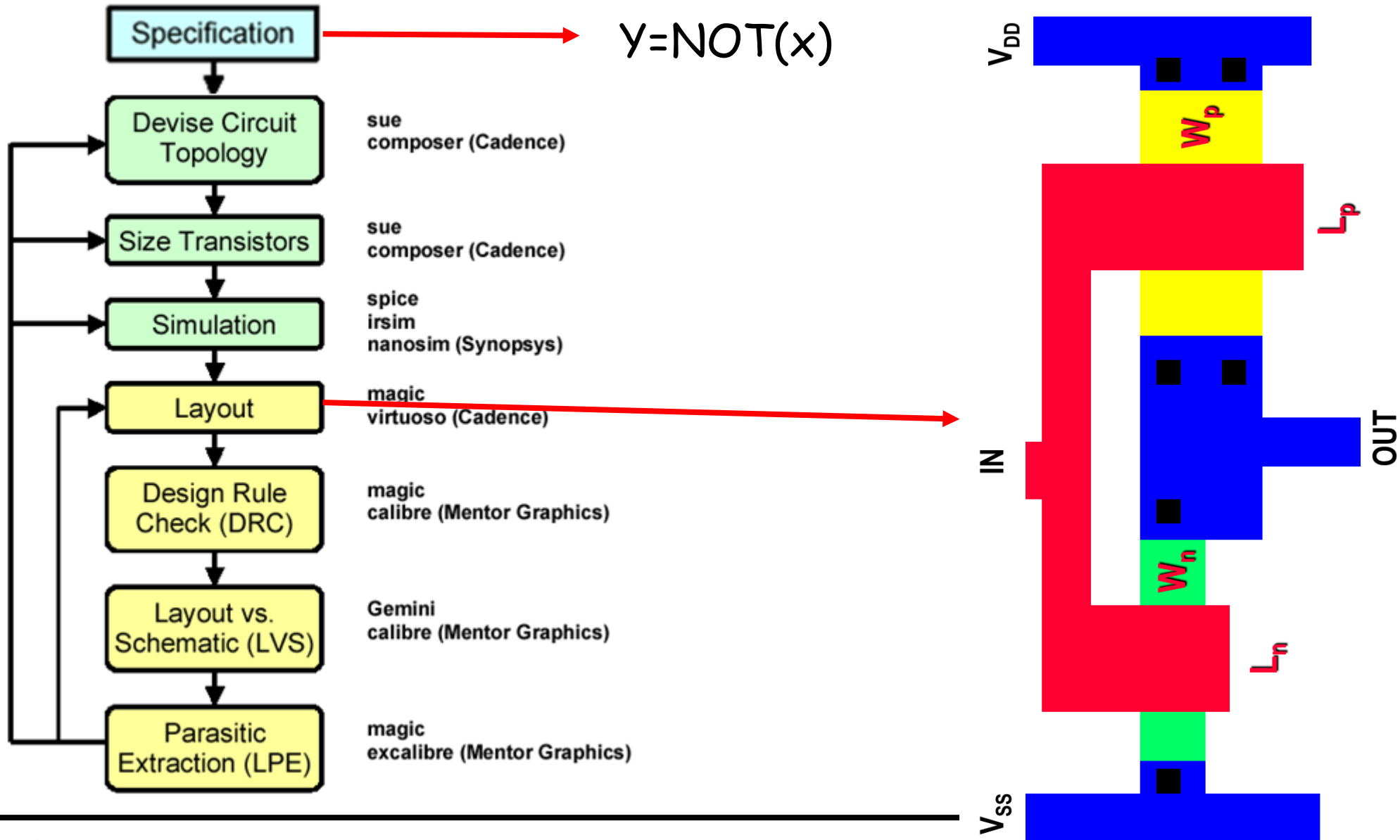
Full Custom Design Flow



Full Custom Design Flow

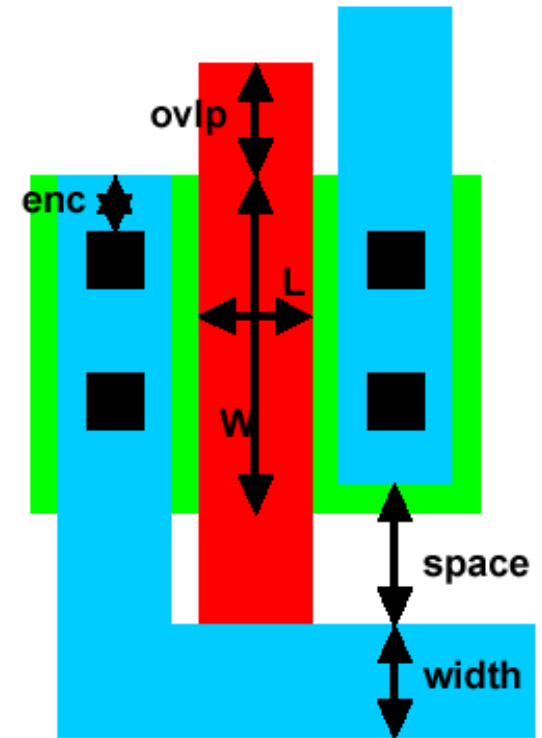


Full Custom Design Flow



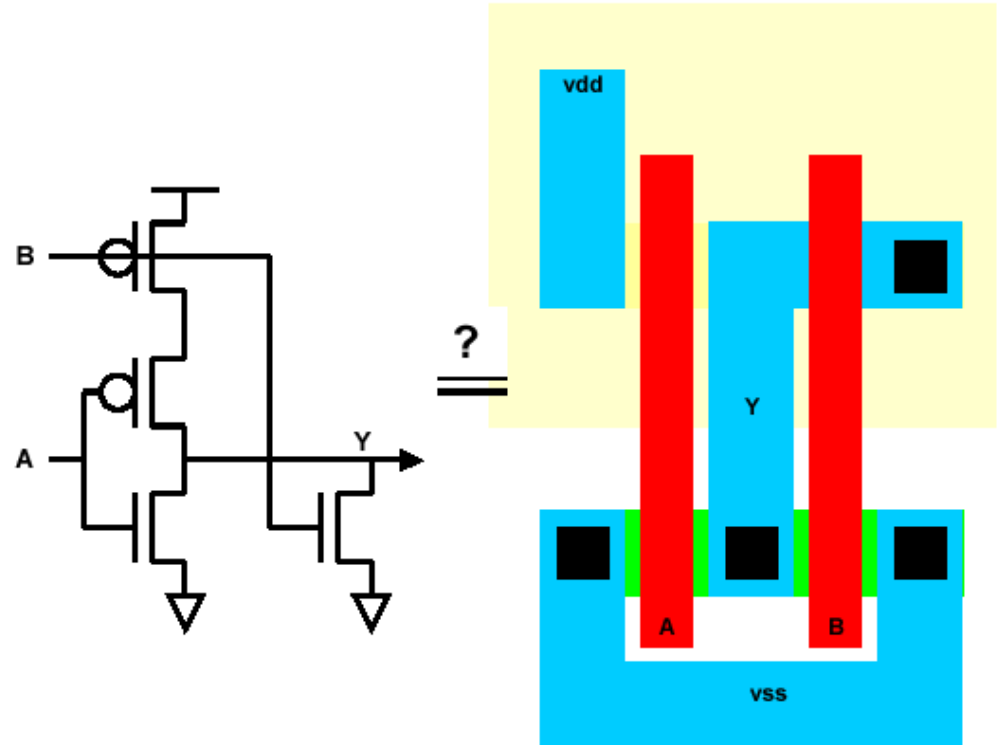
Design Rule Checking (DRC)

- Fab has rules for relationships between polygons in layout
 - Required for manufacturability
- DRC checker looks for errors
 - width
 - space
 - enclosure
 - overlap
- Violations flagged for later fixup

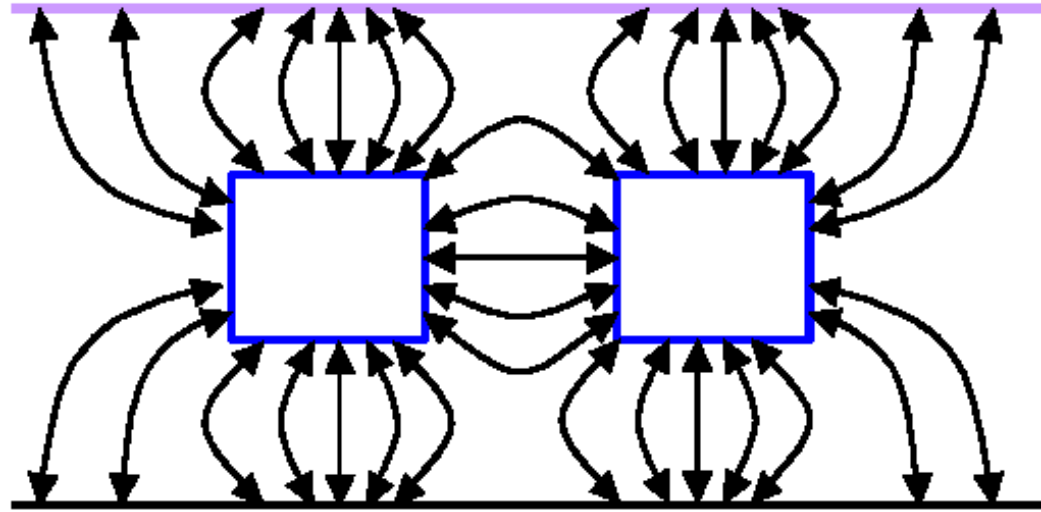


Layout Versus Schematic (LVS)

- Extracts netlist from layout by analyzing polygon overlaps
- Compares extracted netlist with original schematic netlist
- When discrepancies occur, tries to narrow down location

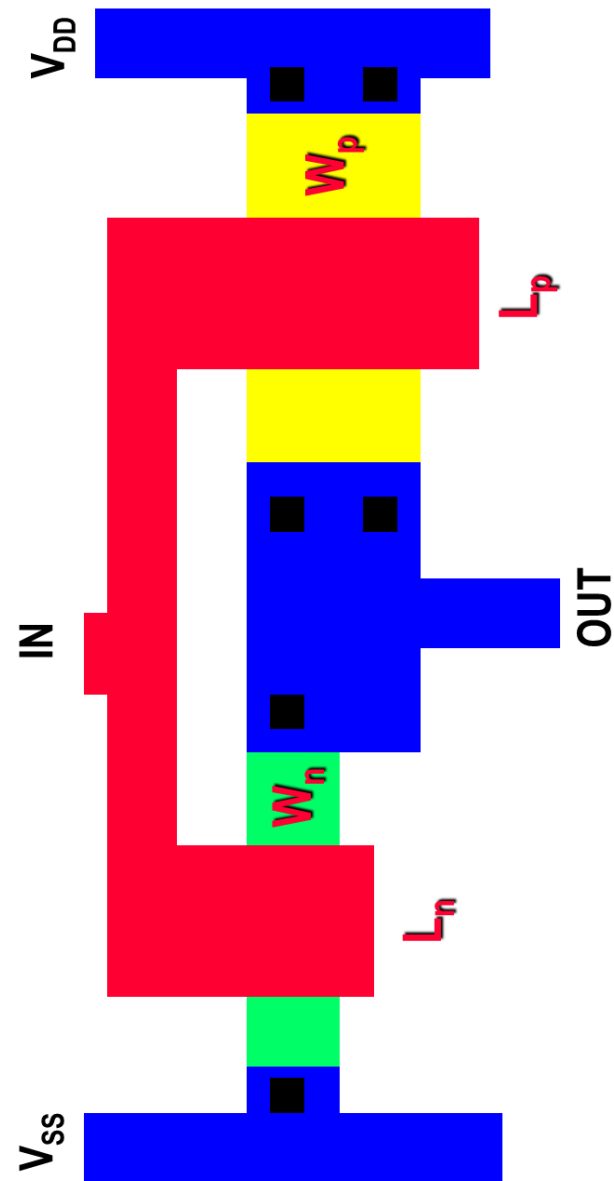
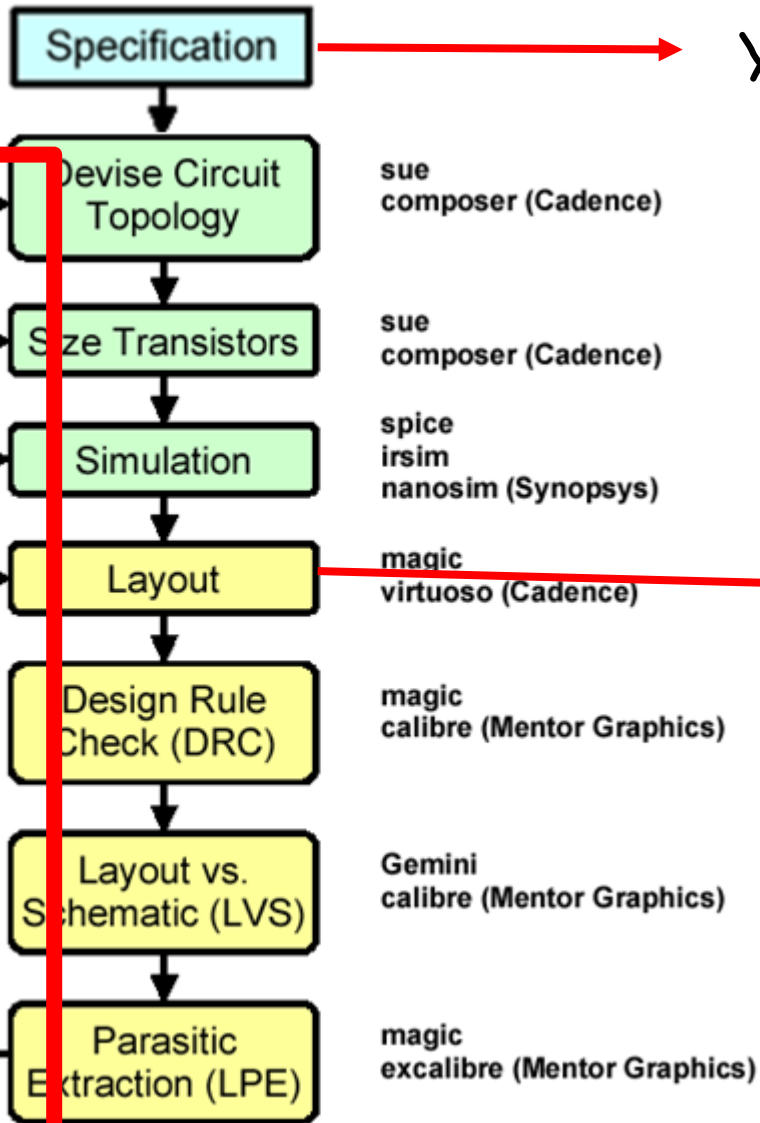


Layout Parasitic Extraction (LPE)



- Estimates capacitance between structures in the layout
- Calculates resistance of wires
- Output is either a simulation netlist or a file of interblock delays

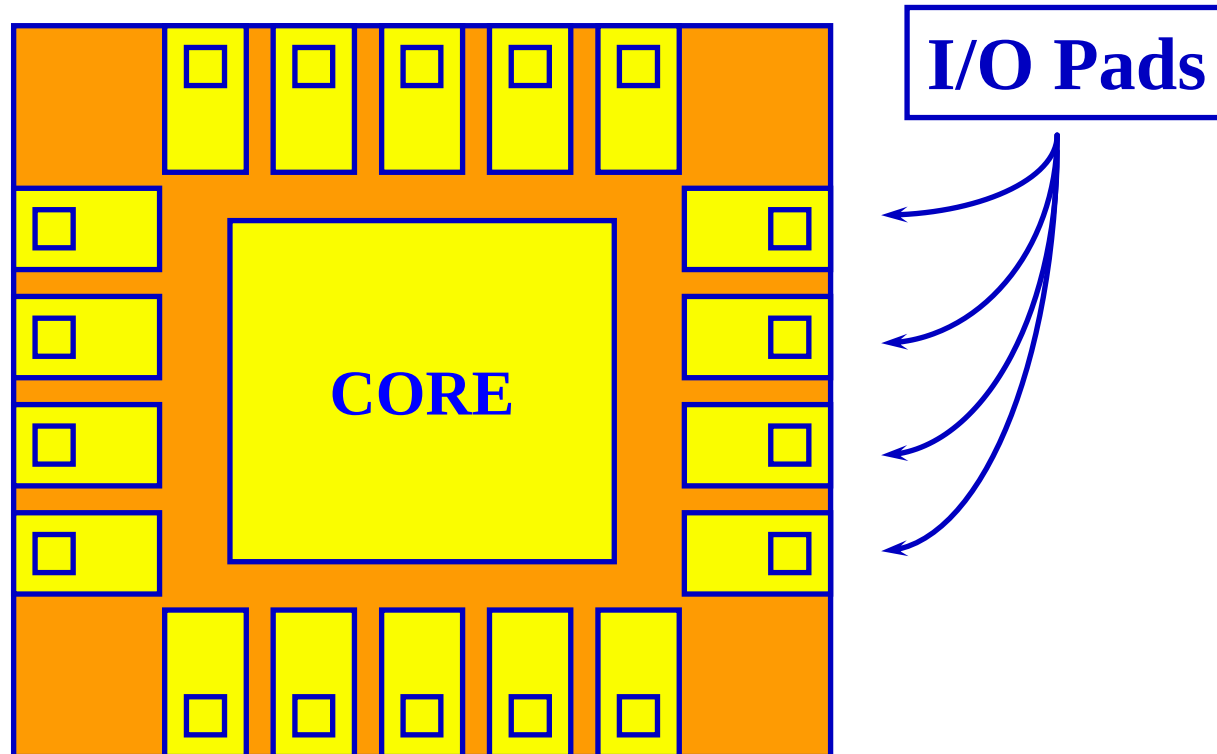
Full Custom Design Flow



ASIC semi-custom

- Designer defines interconnection among pre-designed and/or pre-manufactured logic blocks
- No global optimization
- Reduced cost and development time
- Not high skill required on technology and electronics circuits

ASIC Structure



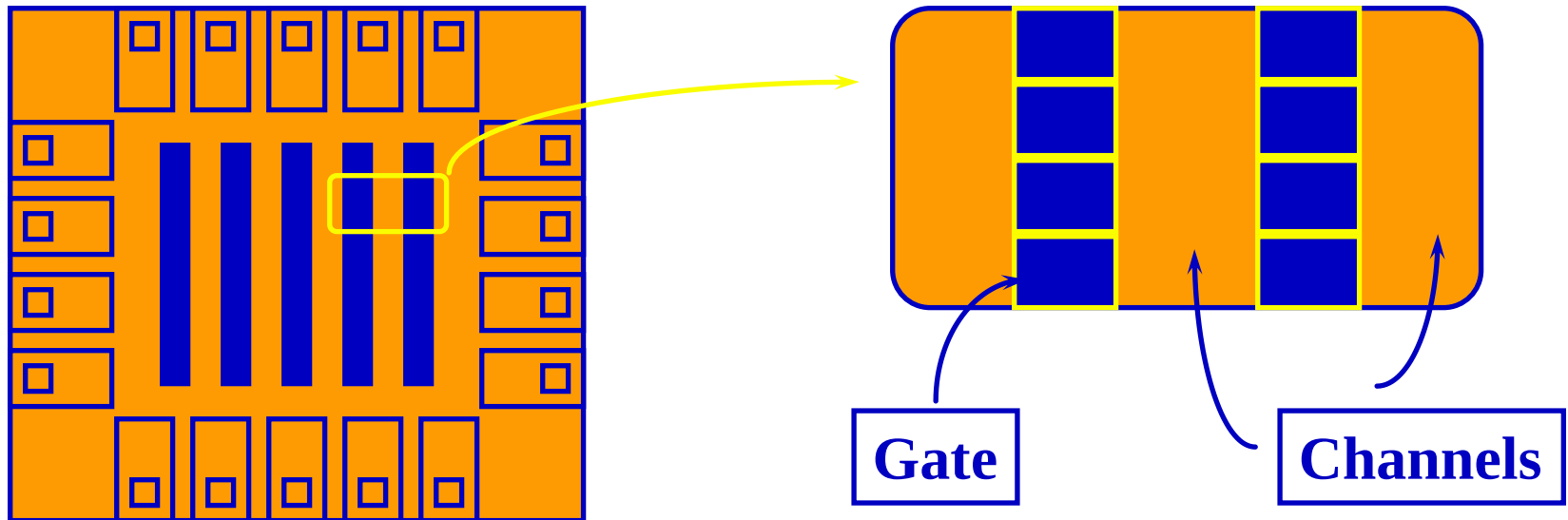
semi-custom technologies

- Gate array or channelled gate-array
- Sea of gate or channelless gate-array
- Standard cell

Gate arrays

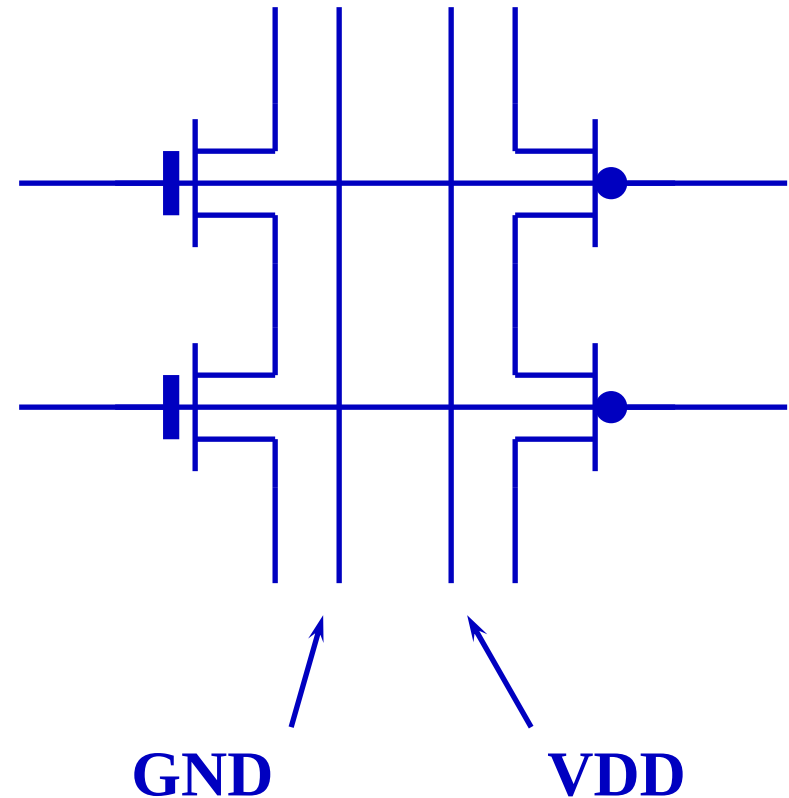
- Integrated circuits based on pre-manufactured transistors organized as a matrix
- Customization for the Specific Function is performed via metal layers (last steps of the semiconductor process)
- Size of the circuits are fixed
- I/O pads number and position is fixed
- Incomplete use of the overall transistors (<80%)

Gate array Structure

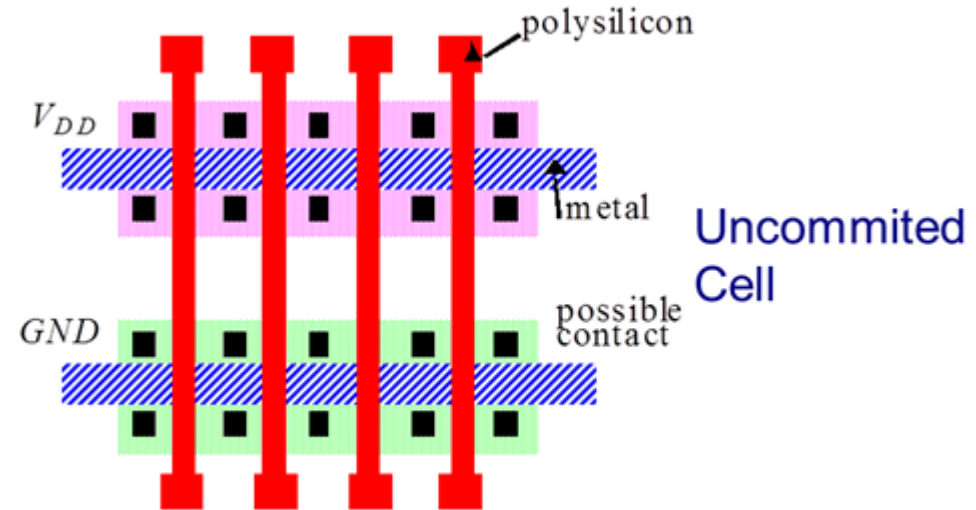
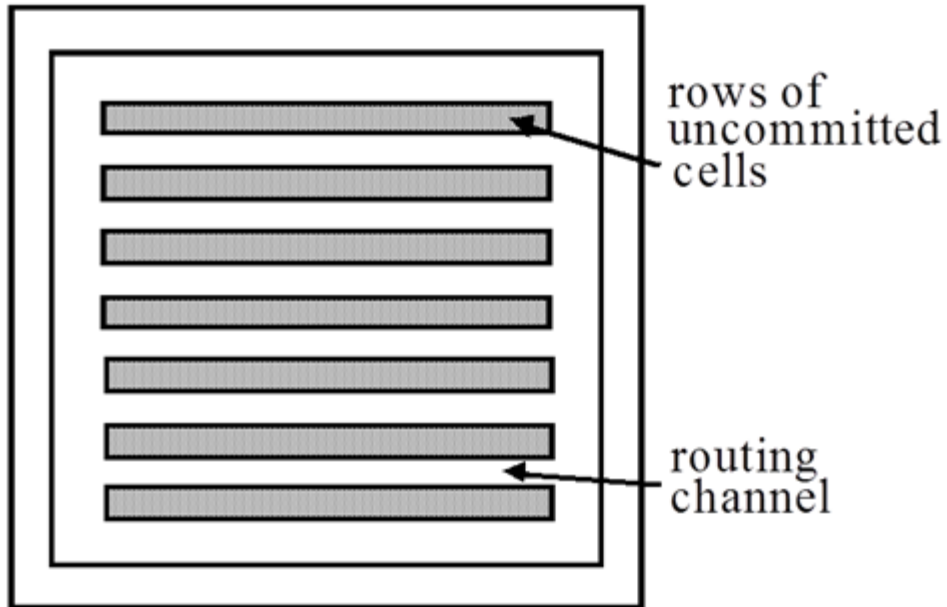


Single Gate Structure

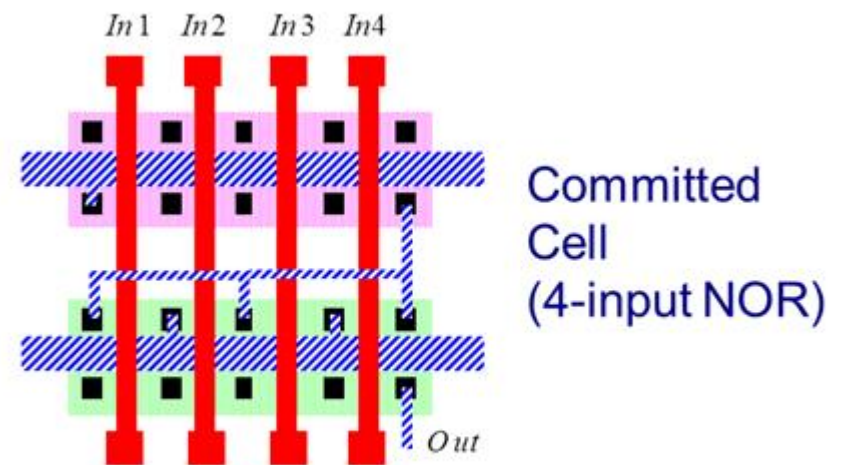
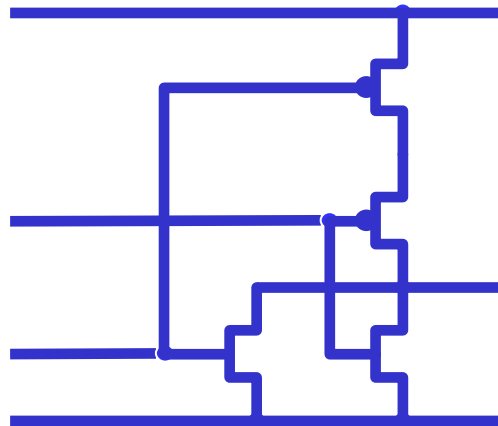
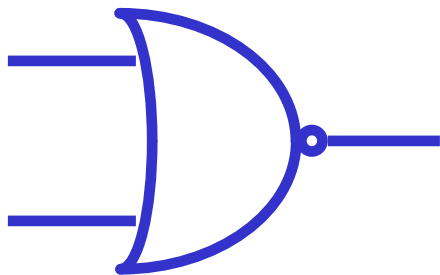
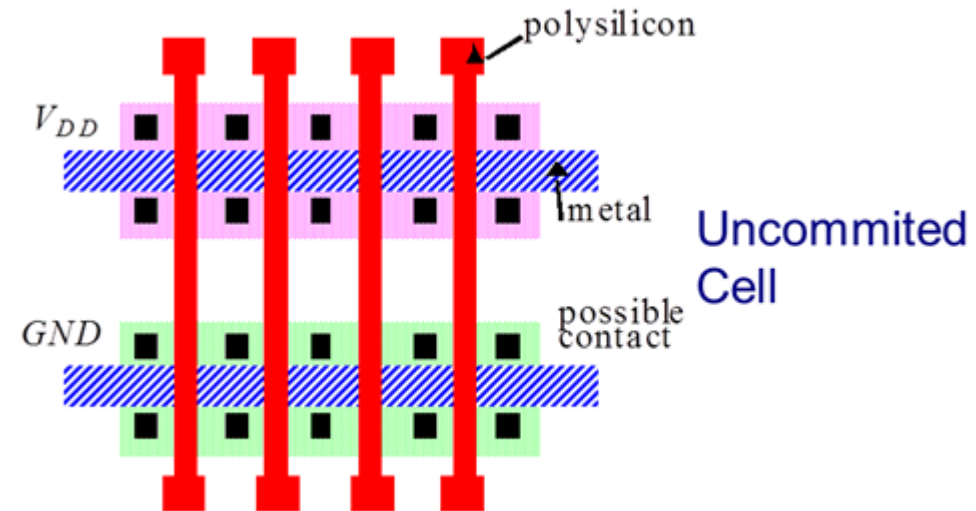
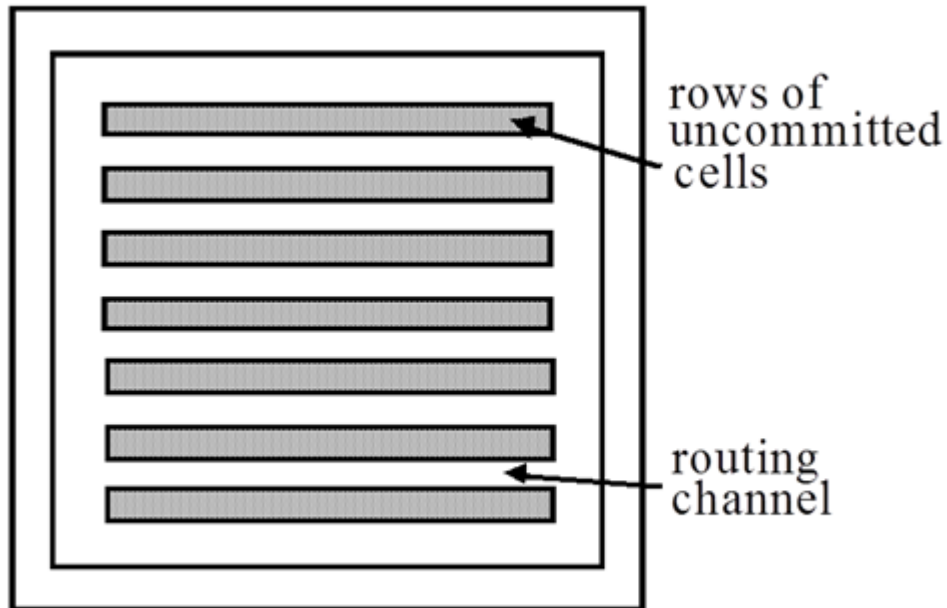
- 2 n-transistors and 2 p-transistors per gate
- Possible interconnection from the channel to drain, source and gate terminal of the transistor



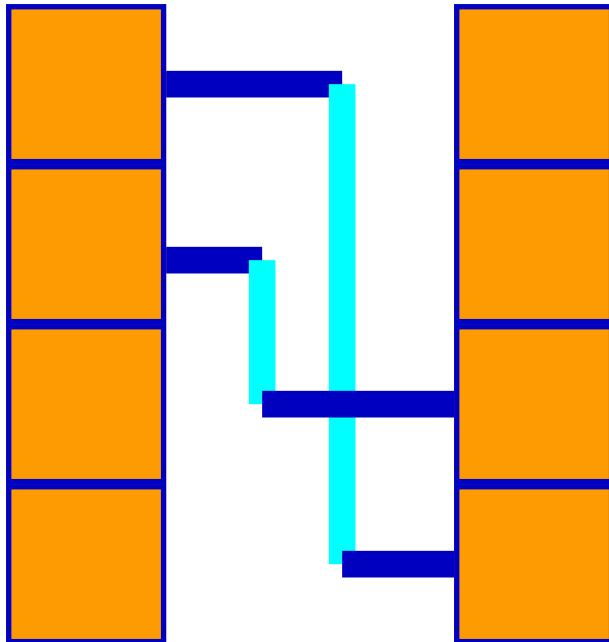
Single Gate Structure



Single Gate Structure

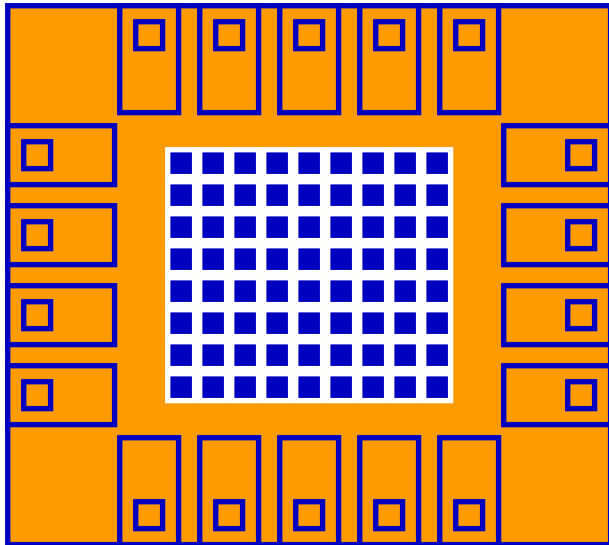


Gate-array Interconnection



- Example of a two metal layers interconnection
- Interconnection Channel (routing) of fixed size
- Sometime gate-array is not routable for channel congestion

Sea-of-gate structure

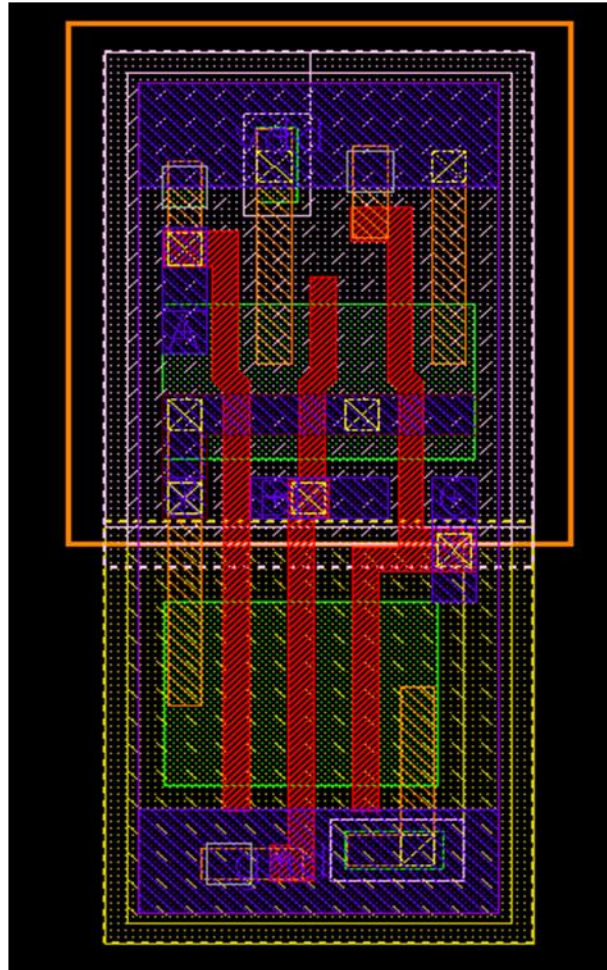


- No dedicated channels for routing
- Interconnections on top of gates, which are then not usable
- Greater flexibility

Standard cell

- Integrated Circuits **manufactured ex-novo**
- Based on standard cell, i.e. Circuit pre-designed and layout optimized with fixed height and variable width
- Organized as standard-cell rows separated by interconnection channels of variable sizes
- Possible to include macrocell such as DAC, ADC, ROM, RAM, etc.

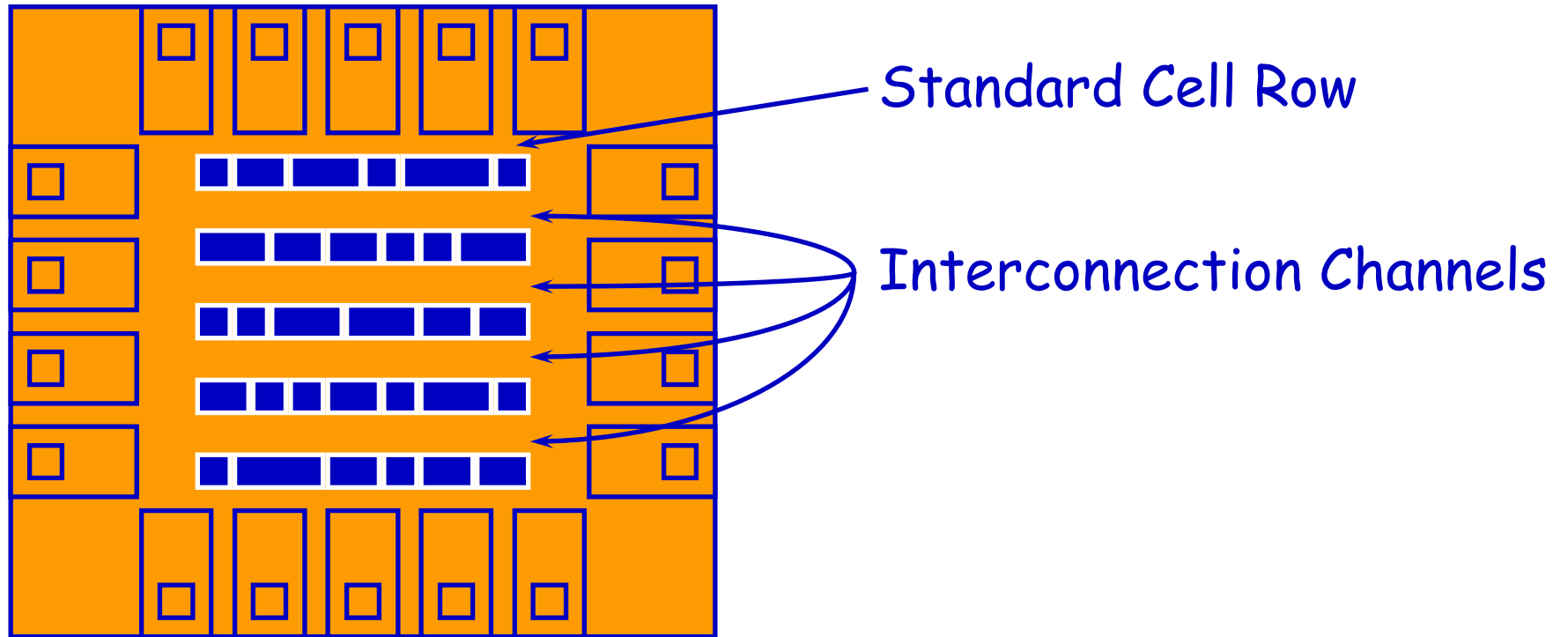
Standard cell



Path	1.2V - 125°C	1.6V - 40°C
$In1-t_{pLH}$	$0.073+7.98C+0.317T$	$0.020+2.73C+0.253T$
$In1-t_{pHL}$	$0.069+8.43C+0.364T$	$0.018+2.14C+0.292T$
$In2-t_{pLH}$	$0.101+7.97C+0.318T$	$0.026+2.38C+0.255T$
$In2-t_{pHL}$	$0.097+8.42C+0.325T$	$0.023+2.14C+0.269T$
$In3-t_{pLH}$	$0.120+8.00C+0.318T$	$0.031+2.37C+0.258T$
$In3-t_{pHL}$	$0.110+8.41C+0.280T$	$0.027+2.15C+0.223T$

3-input NAND cell
 (from ST Microelectronics):
 C = Load capacitance
 T = input rise/fall time

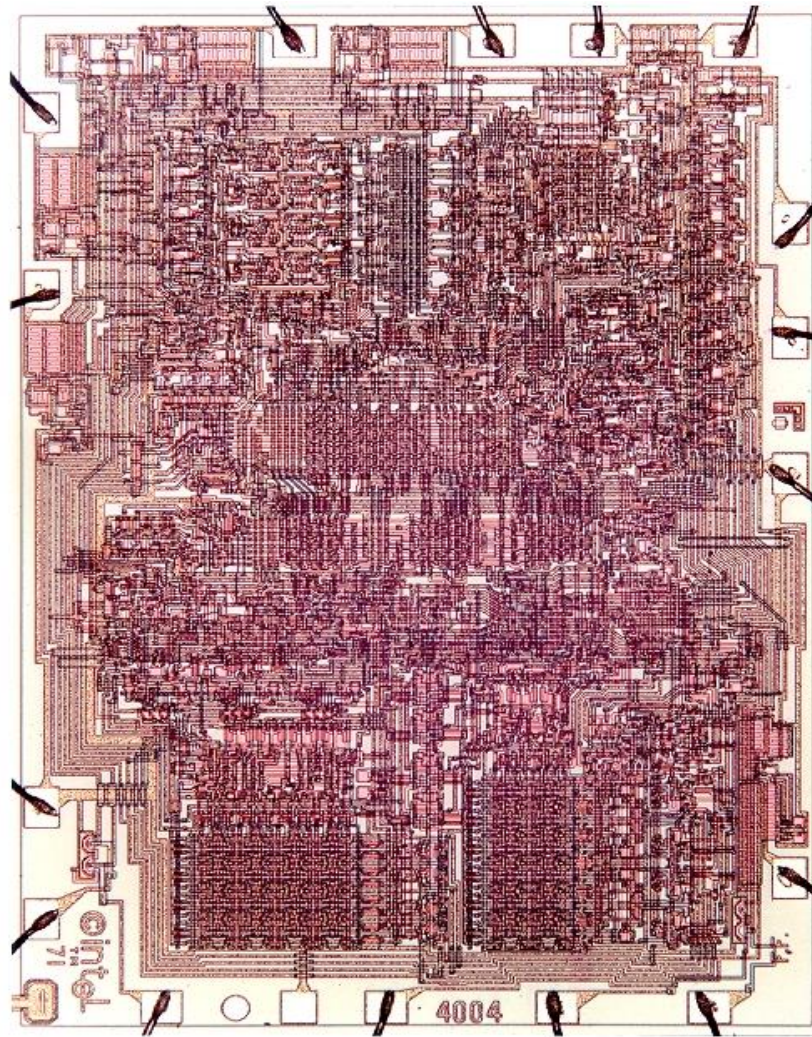
Standard cell Structure



Standard cell Characteristics

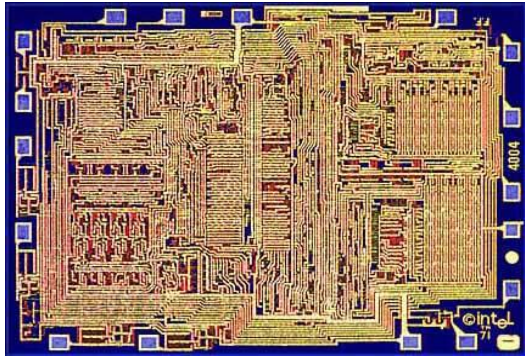
- Based on optimized cell in area and speed
- Availability of macrocells (RAM, ROM, ADC, DAC, OpAmp, etc.)
- I/O number and position decided by the designer
- Greater cost and development time w.r.t. gate array

The Custom Approach

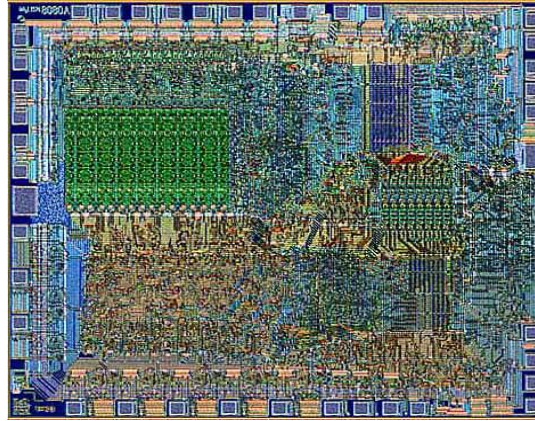


Intel 4004

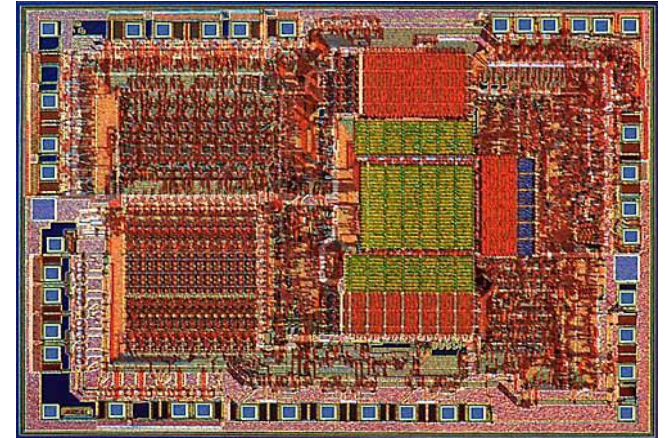
Transition to Automation and Regular Structures



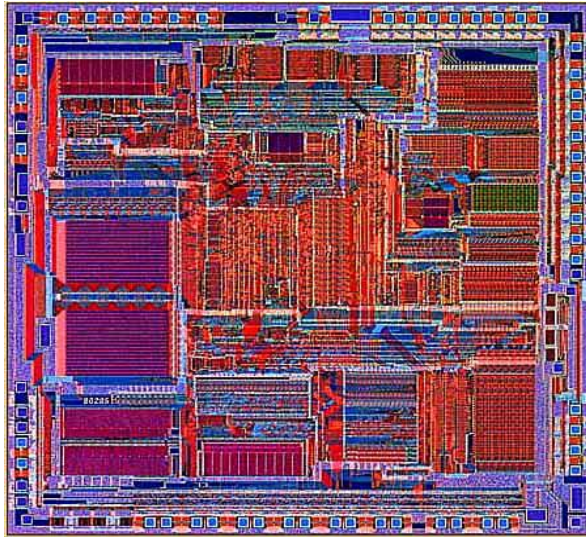
Intel 4004 ('71)



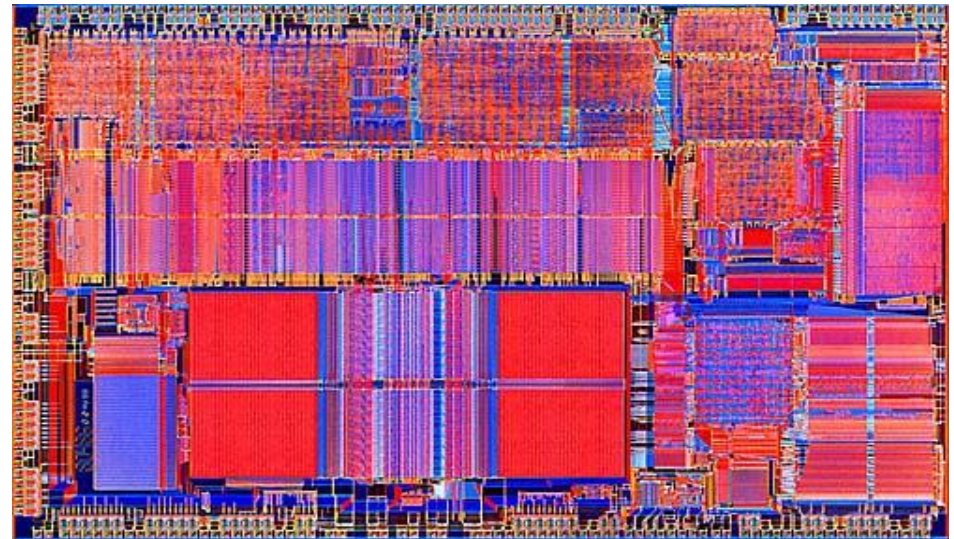
Intel 8080



Intel 8085



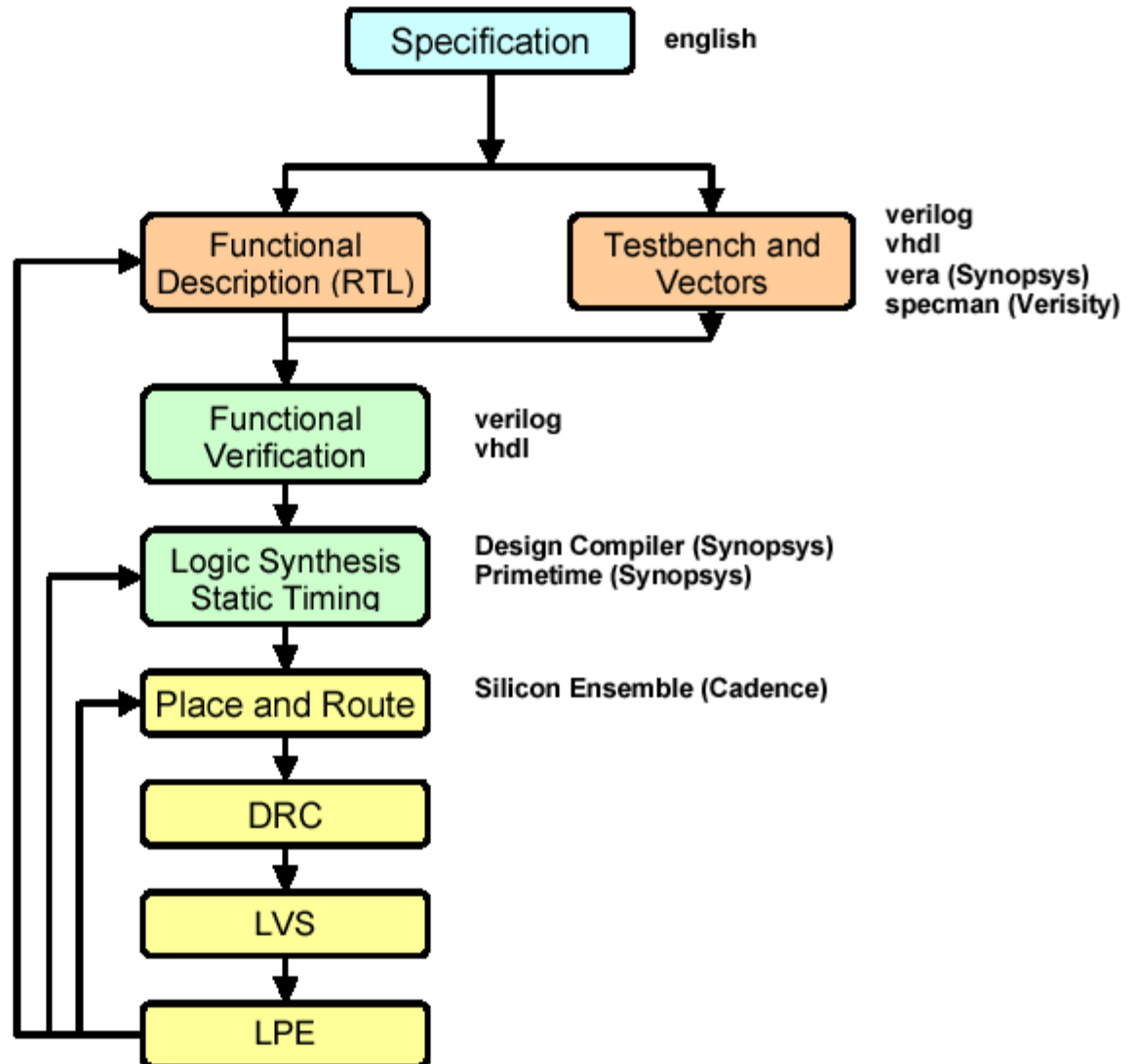
Intel 8286



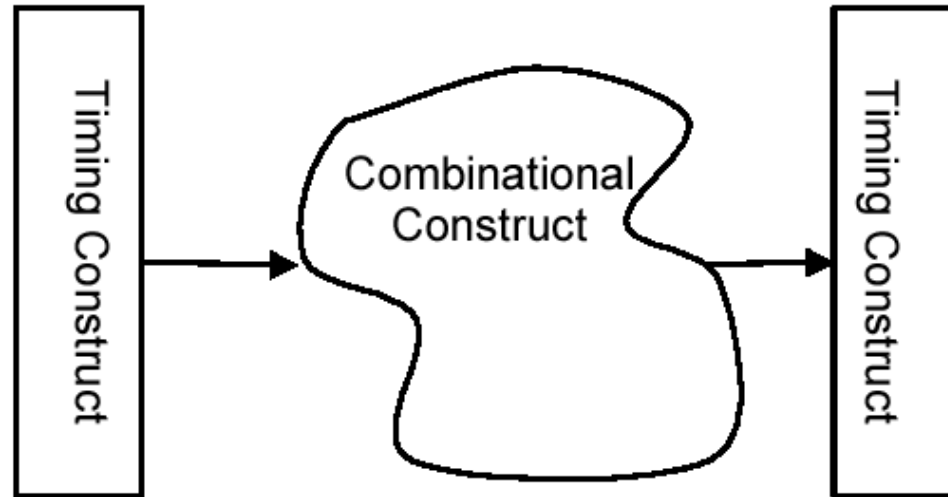
Intel 8486

"Semi-Custom" Design Flow

- Separate teams to design and verify
- Physical design is (semi-) automated
- Loops to get device operating frequency correct can be troubling



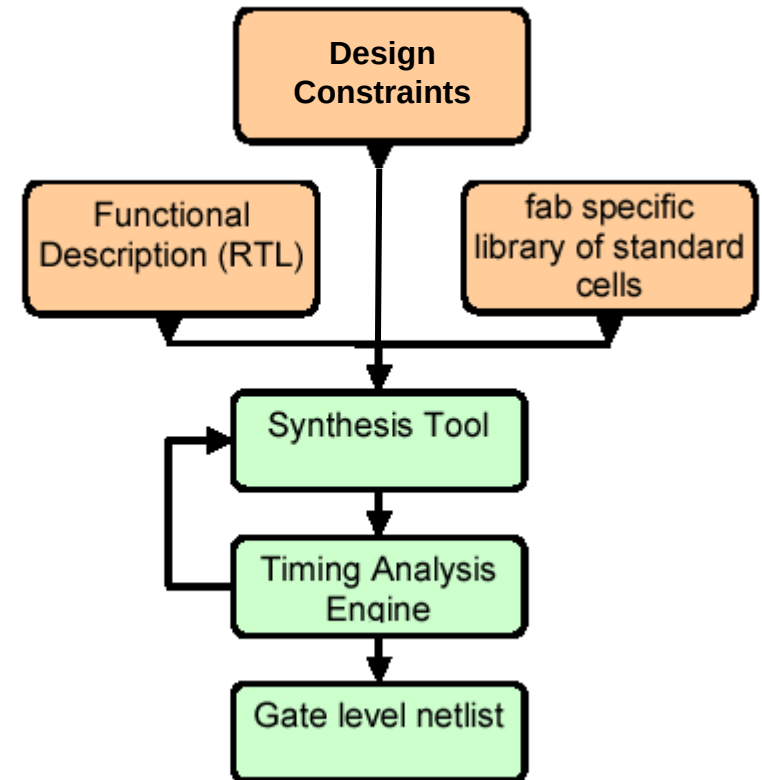
Register Transfer Level (RTL)



- Sections of combinational Constructs separated by timing statements
 - Defines behavior of part on every clock cycle boundary

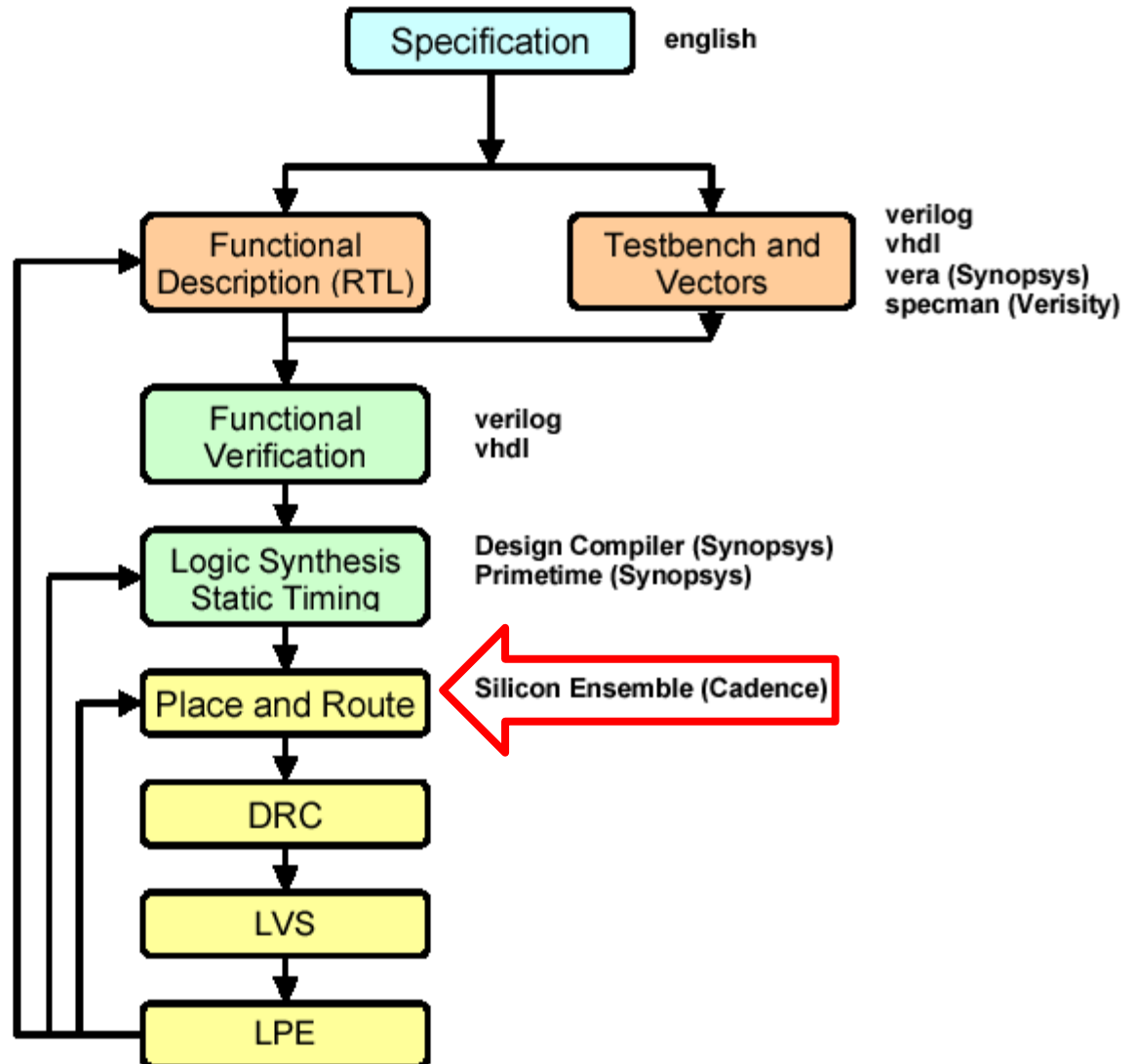
Logic Synthesis

- Changes cloud of combinational functionality into standard cells (gates) from fab-specific library
- Chooses standard cell flip-flop/latches for timing statements
- Attempts to minimize delay and area of resulting logic



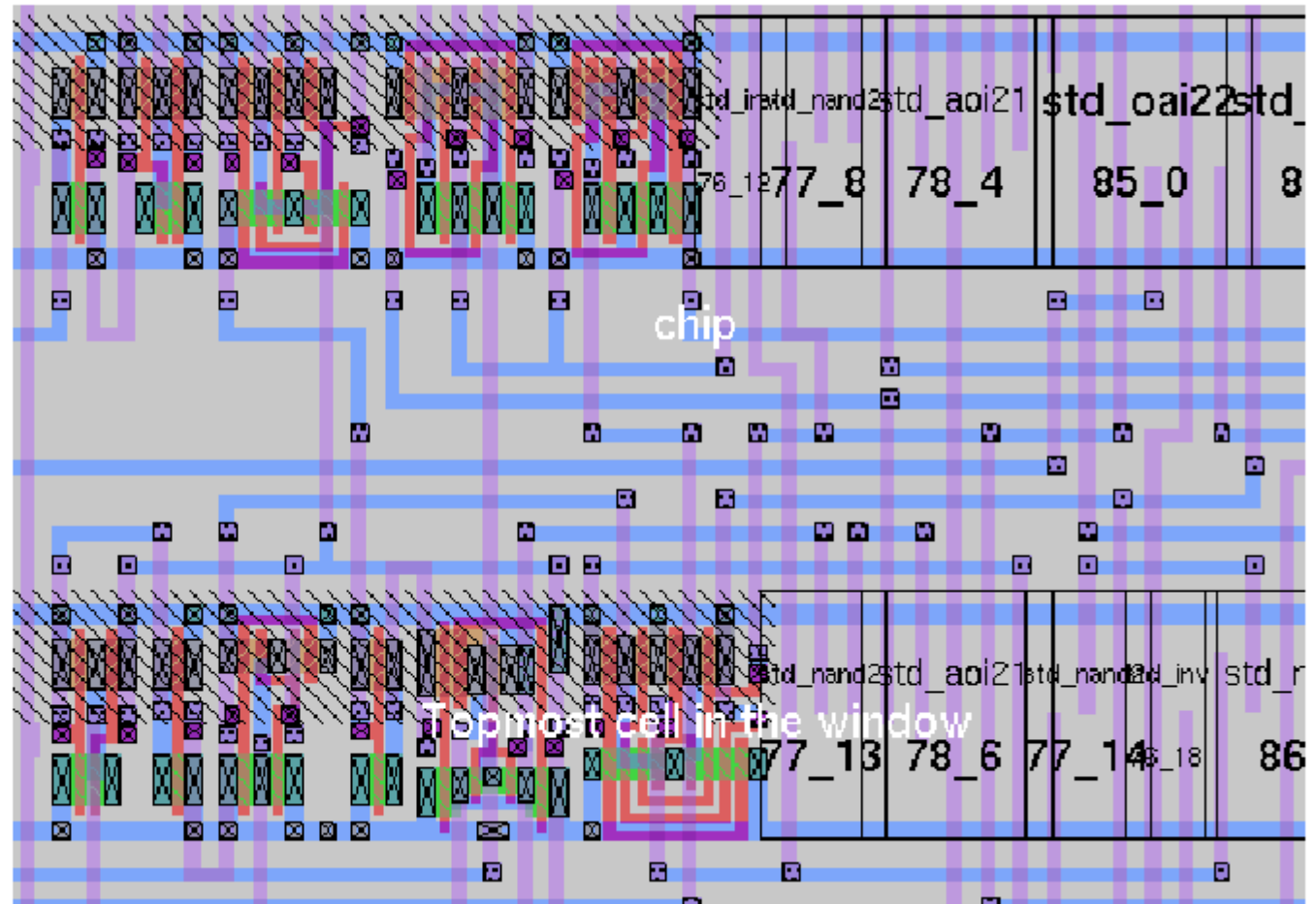
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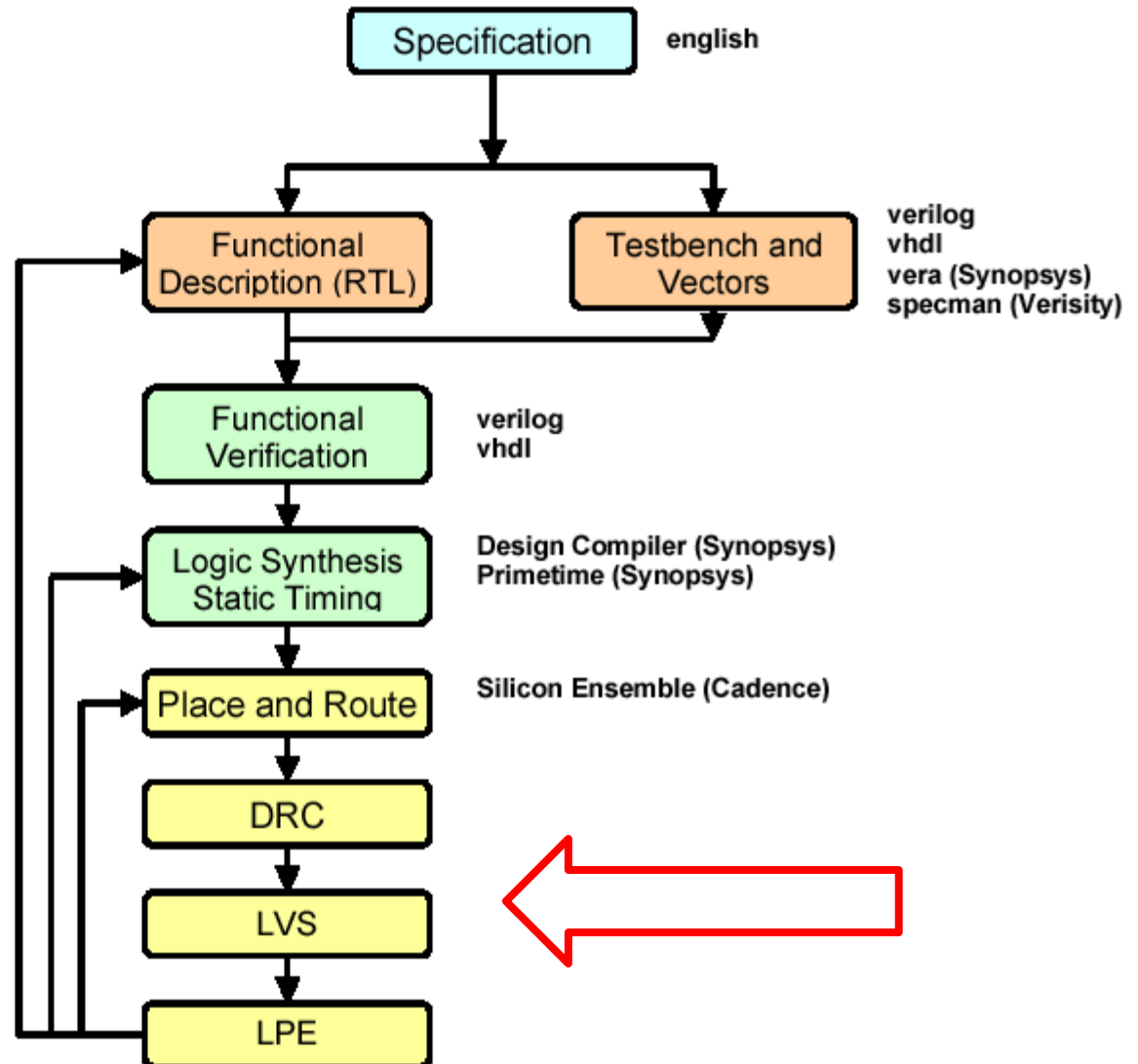
Standard Cell Placement and Routing

- Place layout for each gate (“cell”) in design into block
- Rearrange cell layouts to minimize routing
- Connect up cells



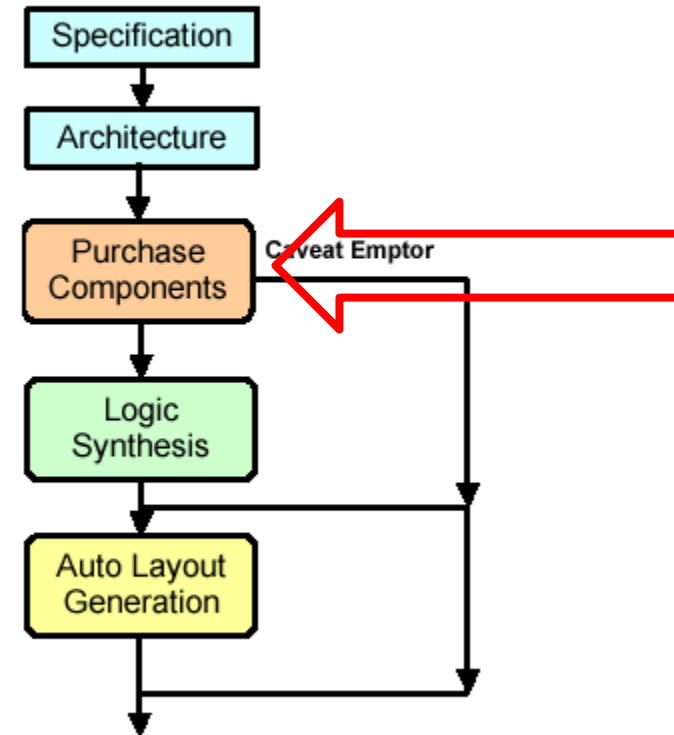
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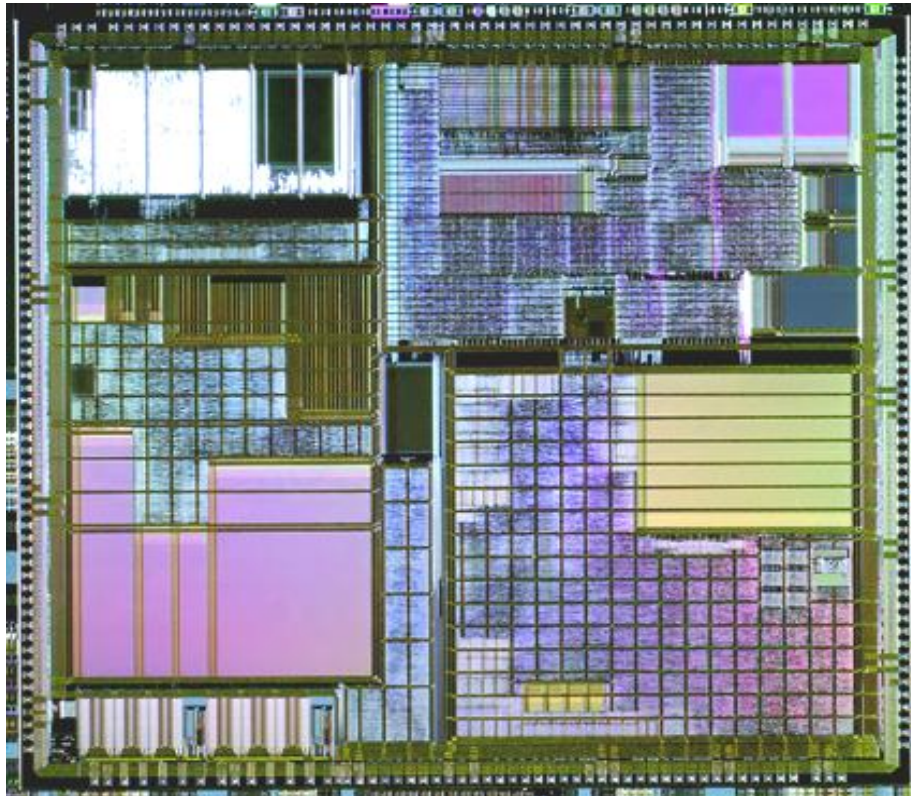
System On Chip Design Flow

- Can Buy “Intellectual Property” (IP) from various vendors
- “Soft IP”: RTL or gate level description
 - Synthesize and Place and Route for your process.
 - Examples: Ethernet MAC, USB
- “Hard IP”: Polygon level description
 - Just hook it up
 - Examples: micro/DSP cores embedded DRAM
- Also: Standard cell libraries for ASIC flow



Chip Assembly System

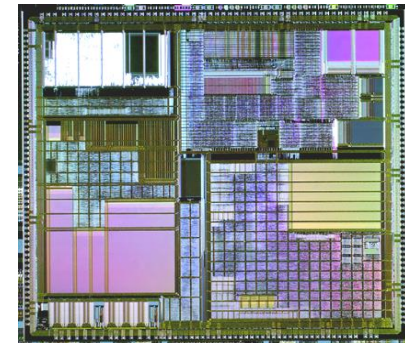
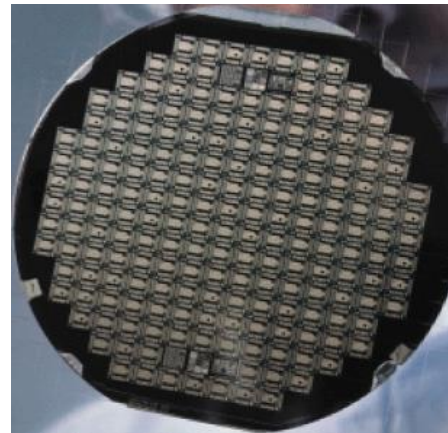
- Integrate blocks from previous steps into final layout
- Early Floorplanning is key



Tapeout

- Used to write 9-track computer tapes for mask making
- Now, Transfer polygons to fabrication company via ftp
- You're done! (Except for documentation, test vector generation, device bringup, skew lots, reliability tests, burnin...)

Foundry

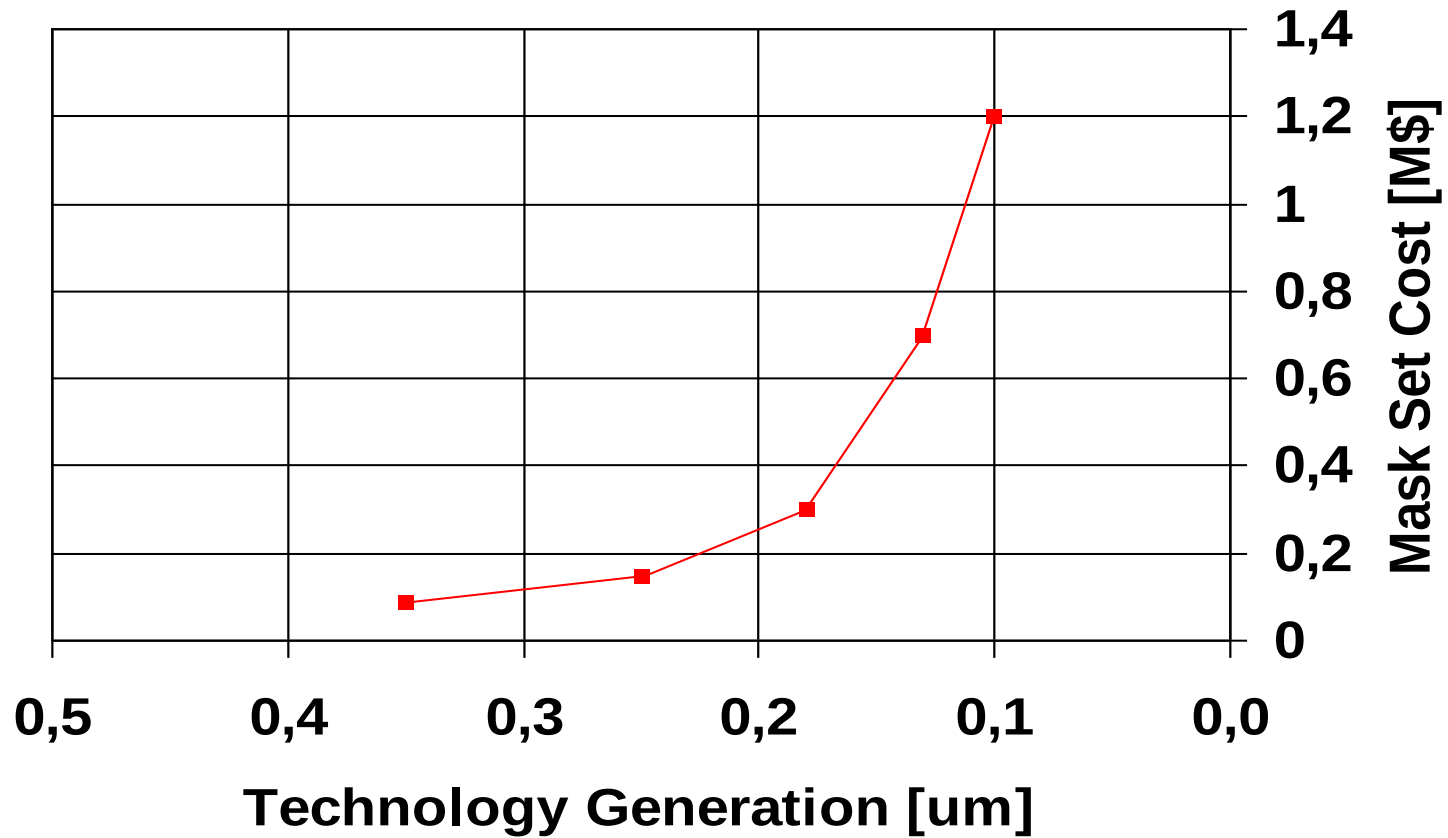


AMS 200 mm Production Line - (0,35 up to 0,18 μm)

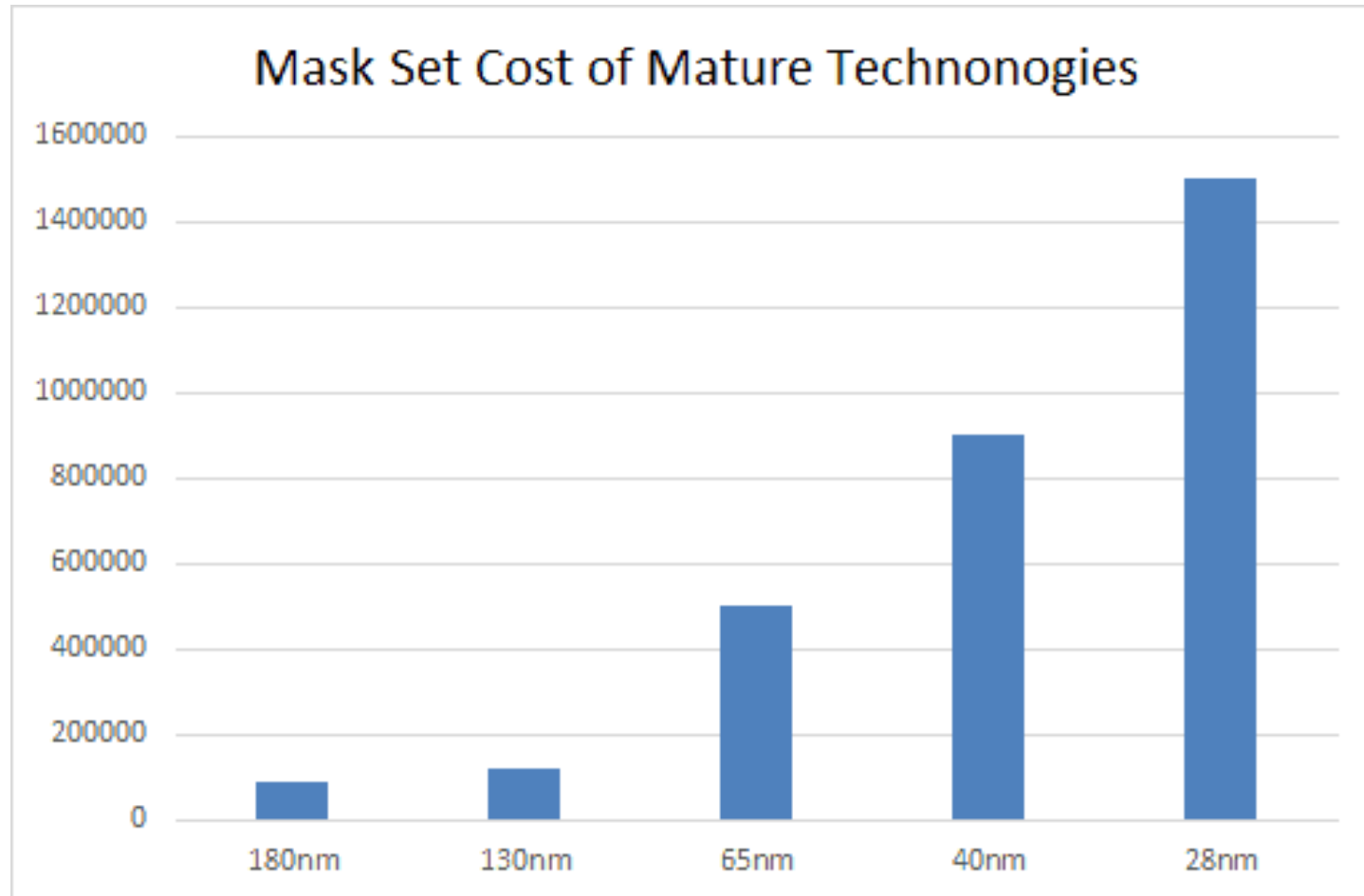
Production Capacity: 3.000 Wafer Starts Per Week

Clean room Area: 2.600 m²

Mask Costs



Mask Costs



source: friends of AnySilicon



Testing the manufactured ASIC



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- Electronic Abstraction Level Design
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- **Cost-Performance Trade-off**
- Field Programmable Gate Array
- Design Productivity
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ASIC solution costs

- Major drawback is the **overall cost** for development and manufacturing
- Requires significant investment and risks
- High barrier especially for small-medium enterprise in the use of ASIC technology
- Requires a detailed cost estimation

Costs

- Fixed cost for investment, design and manufacturing
- Recurrent cost for each product unit

Number of product units

$$\text{Total Cost} = \text{Fixed Cost} + n \times \text{Recurrent Cost}$$


NRE Costs

- Fixed cost or NRE (non-recurrent engineering)
 - Design Costs
 - » Designer man months
 - » CAD tool machine time
 - » hardware and software amortization
 - » ...
 - Manufacturing and Testing Fixed Costs
 - » Masks for the semiconductor process
 - » Board and tool for ASIC testing

RE Costs

- Cost repeated for each unit or RE (recurrent engineering)
 - Manufacturing Process Cost
 - » Manufacturing cost for the single component
 - » Proportional to the circuit area (wafer cost divided by numebr of dies per wafers)
 - » package
 - Testing costs
 - » Tester Machine usage time for the single circuit

Unit Cost: C_1

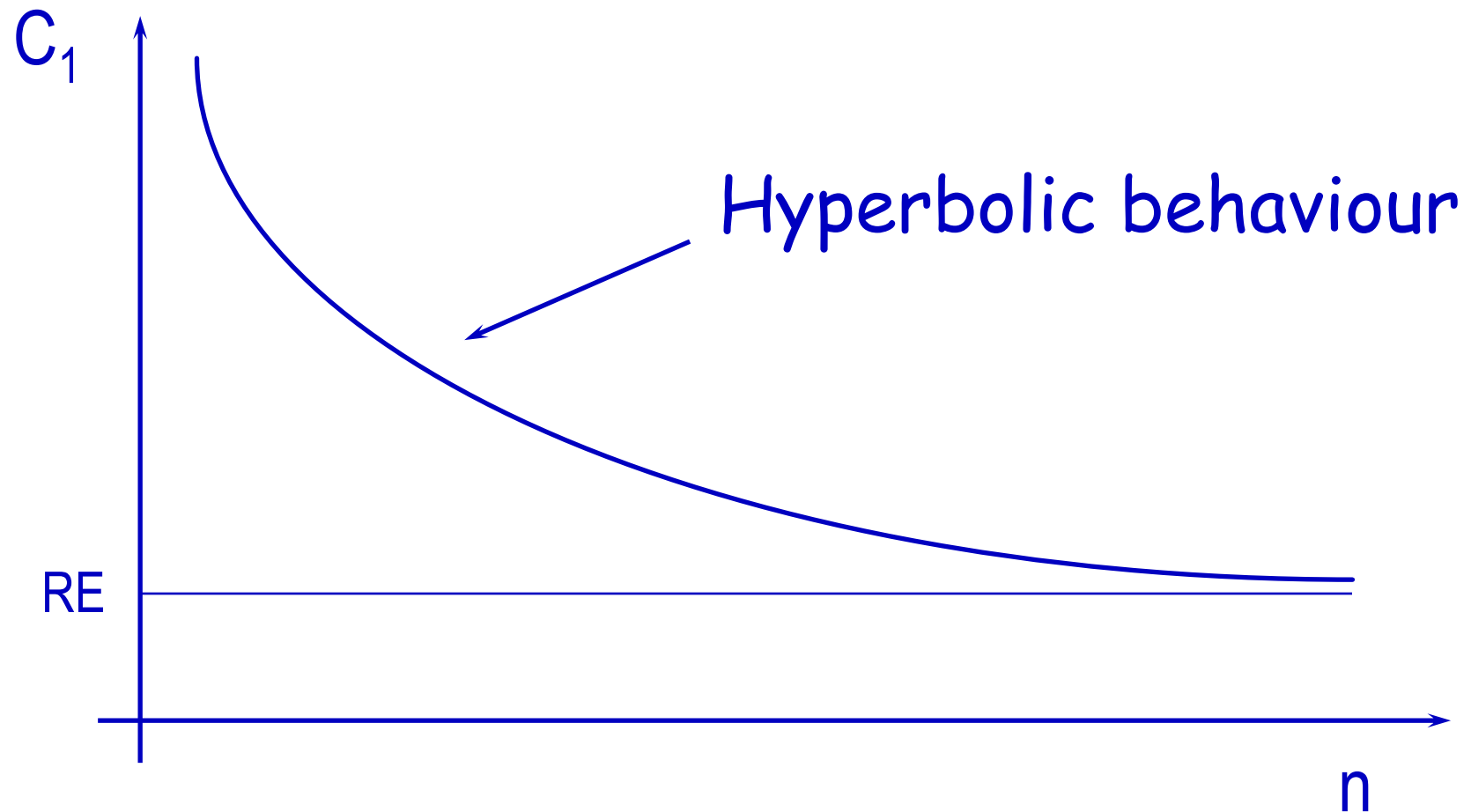
Total Cost = Fixed Cost + n x Recurrent Cost

$$C_T = NRE + n \times RE$$

Dividing by n we have the final cost for single unit

$$C_1 = NRE / n + RE$$

Cost Curve



Example

- Value for n in order to have an unit cost of 2,00 € in case $NRE = 200 \text{ K€}$ and $RE = 0,75 \text{ €}$?
- $C_1 = NRE / n + RE \rightarrow C_1 - RE = NRE / n$
- $n = NRE / (C_1 - RE) = 200 \cdot 10^3 / (2,00 - 0,75) = 160 \text{ K}$

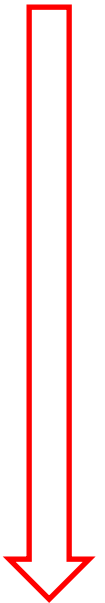
Comparison of NRE Costs

- Design Time
 - full-custom involves the greatest design time
 - Different semi-custom have similar design time
- Fixed Manufacturing costs
 - full-custom and standard-cell are similar, since both are manufactured from scratch (**ex-novo**)
 - gate-array is smaller (most masks are shared among many users, only metal layers are application specific)

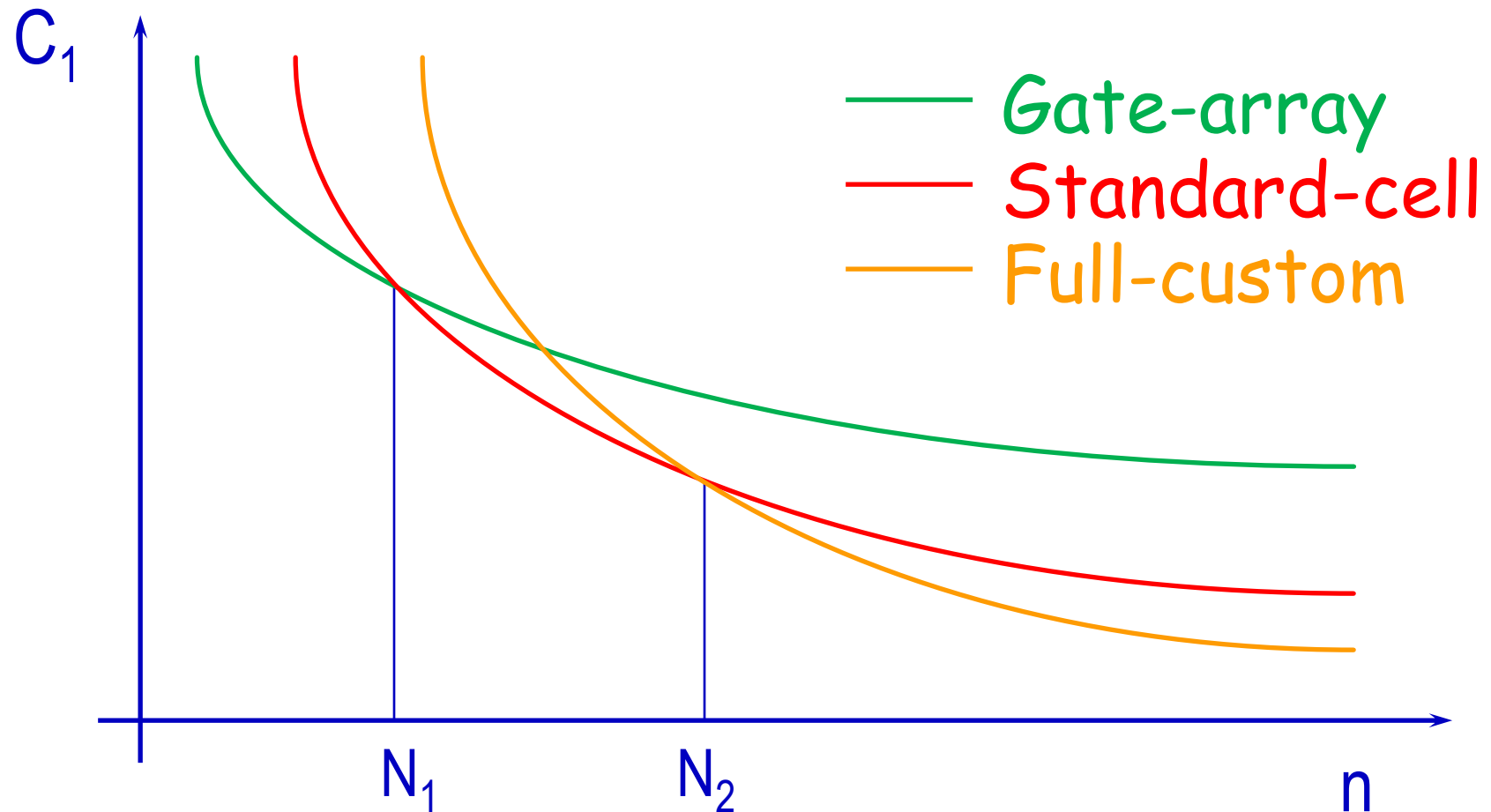
Comparison of NRE Costs

- Manufacturing costs is proportional to area
 - **Full-custom**: optimized area
 - **Standard-cell**: greater than full-custom but still with maximum freedom in the interconnection and complete used of all implemented resources
 - **Gate array**: fixed sizes with partial use of implemented resources, greater are than the other approaches

AREA



Cost curves



Notes (1/2)

- Three zones for three most convenient solution
 - $n < N_1$ gate-array
 - $N_1 < n < N_2$ standard-cell
 - $n > N_2$ full-custom

Notes (2/2)

- Unit cost is not all. **Other metrics** could be important such as speed performance, size, power consumption, etc.
- For application with low or medium volume it is important to keep **NRE costs** to a minimum
- **Time-to-market**: full-custom and standard-cell technologies have a great impact on time-to-market

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FPGA technologies

- Characteristics of Field Programmable Gate-Array (FPGA)
 - Comparison with full-custom and semi-customs ASIC
- FPGA Architectures available on the market
 - Logic Cell Array
 - Gate Array

FPGA

- Field Programmable Gate Array (FPGA)
 - Internal logic is programmable
 - Interconnections are programmable
- FPGA Architecture
 - Gate-array type
 - Logic Cell Array type

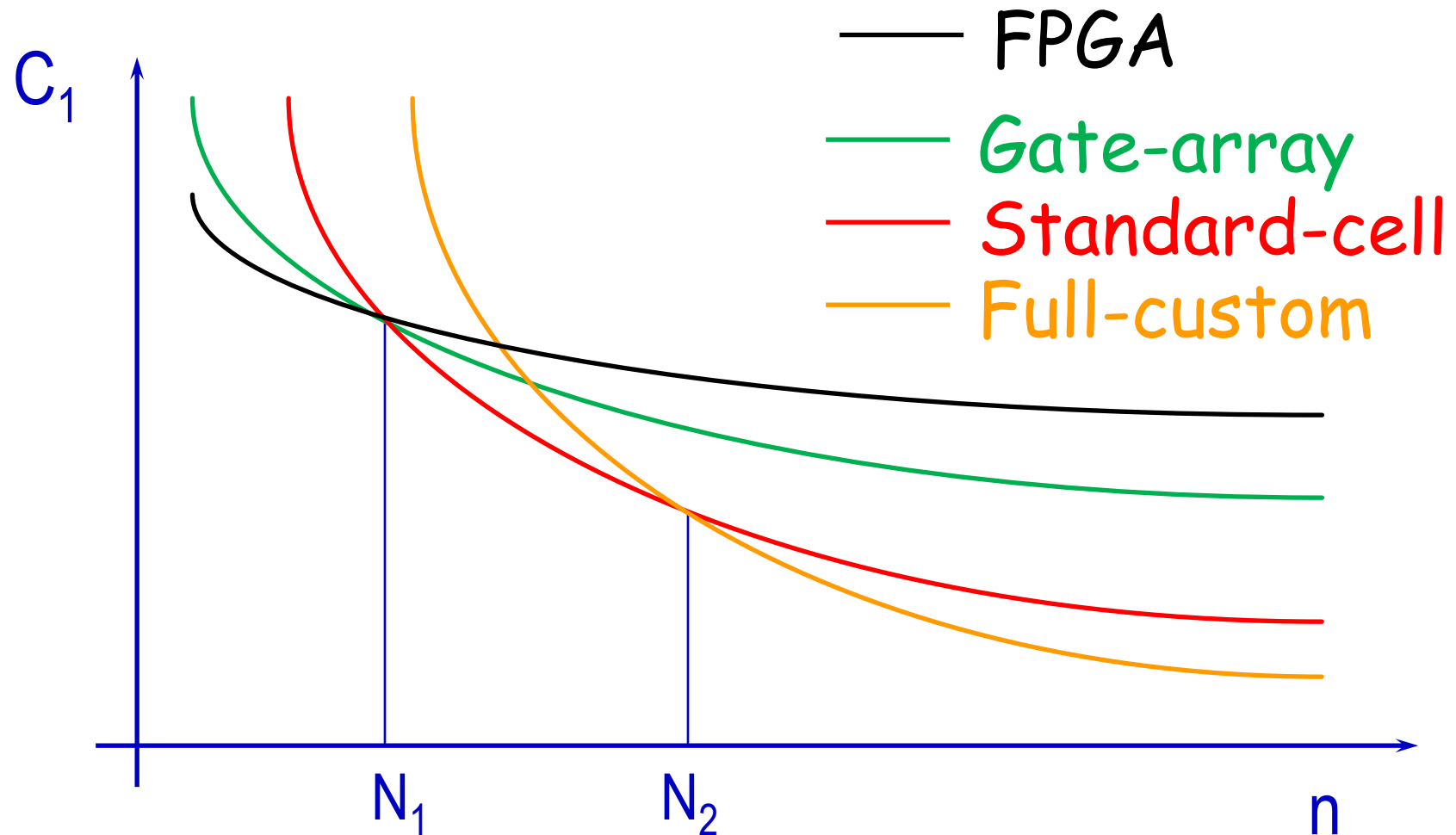
FPGA pro

- Programmable
 - Design Flexibility
 - Design updatable
- Zero turn-around time
- NRE costs smaller than ASIC

FPGA contra

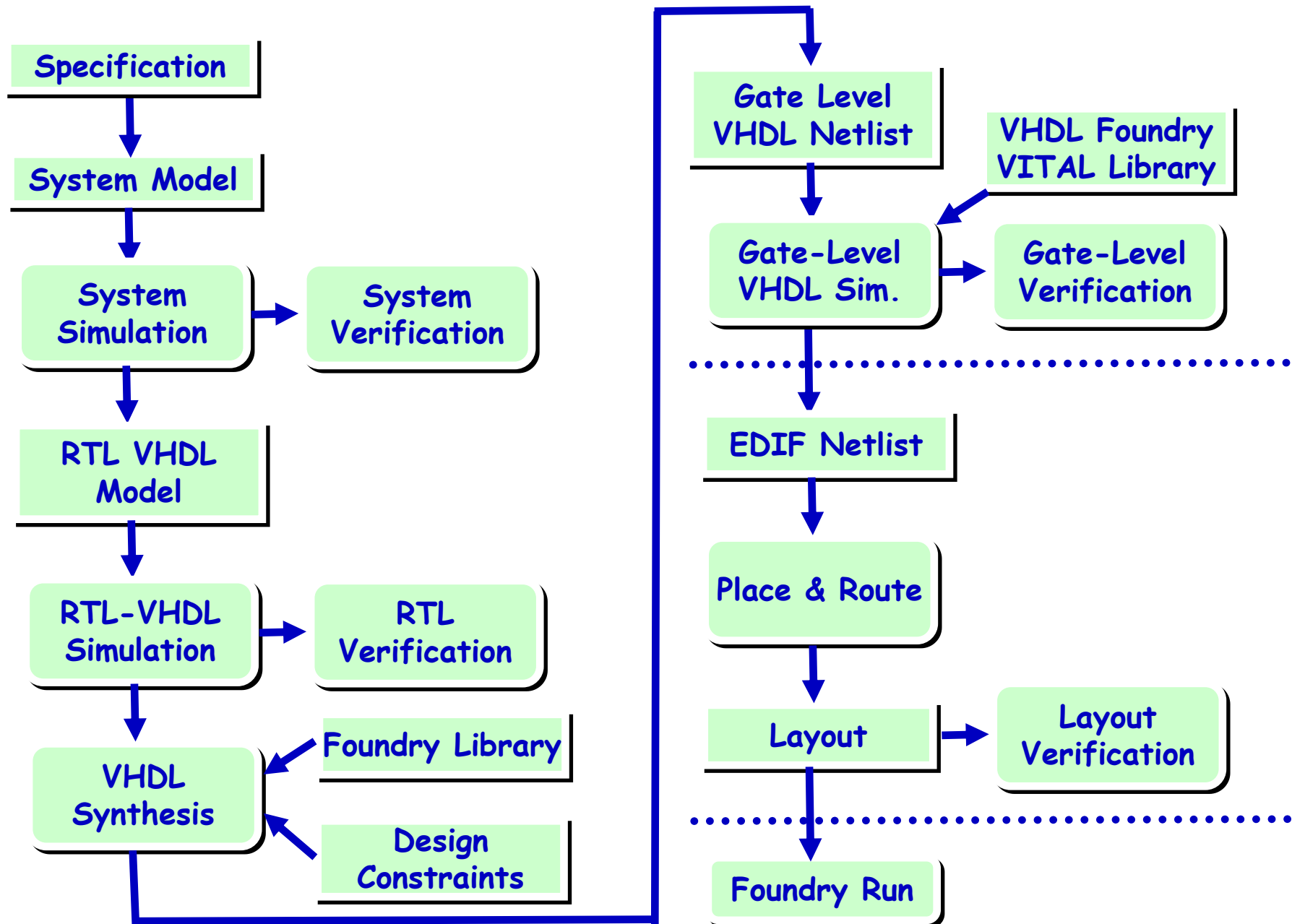
- No optimized Device
 - Manufactured in fixed size (area are not fully used)
 - No optimized performance (collegamenti e logica interna ridondanti)
- RE costs greater than ASIC
 - No convenient for huge volume

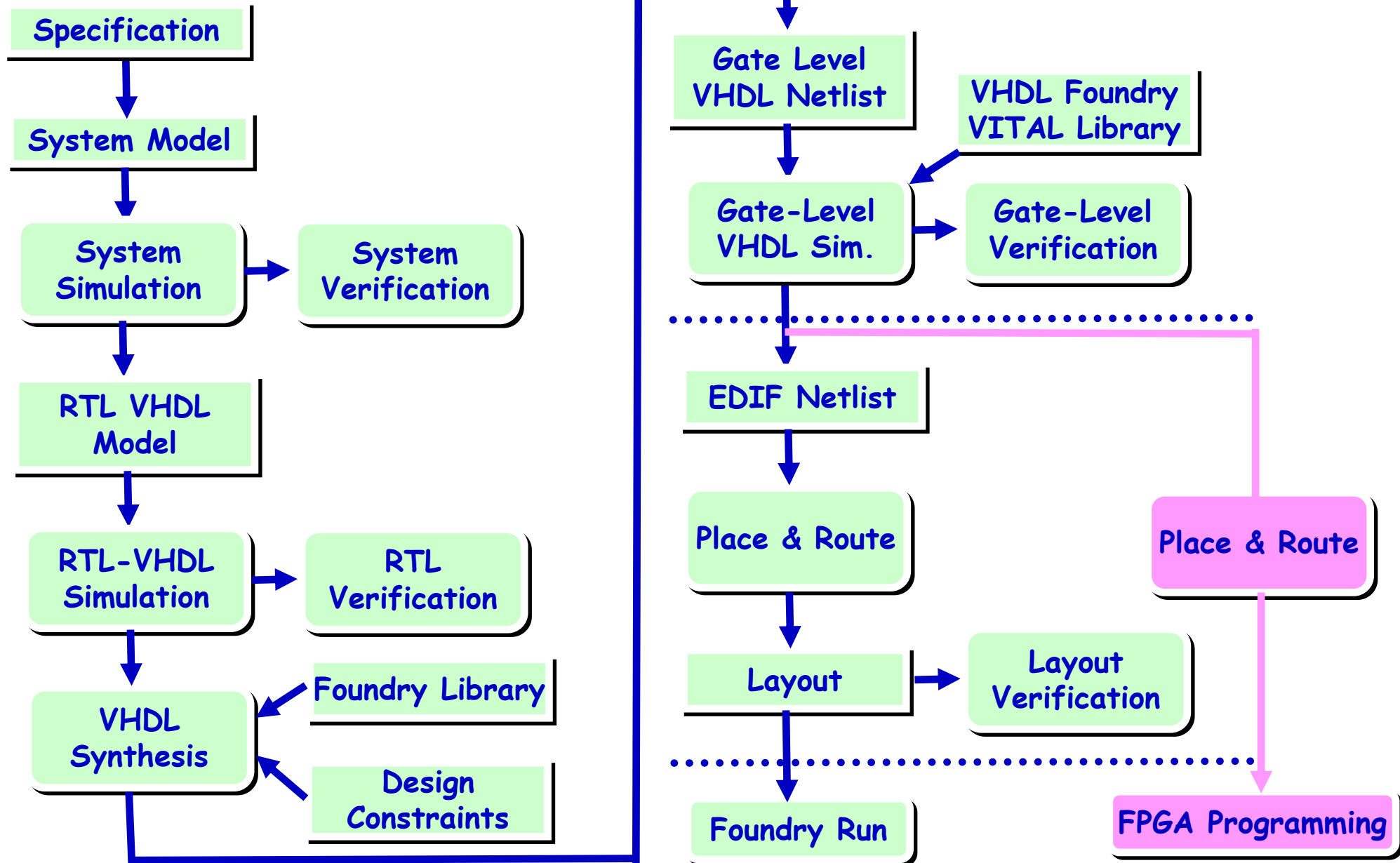
Comparison with ASIC



When FPGA are advisable

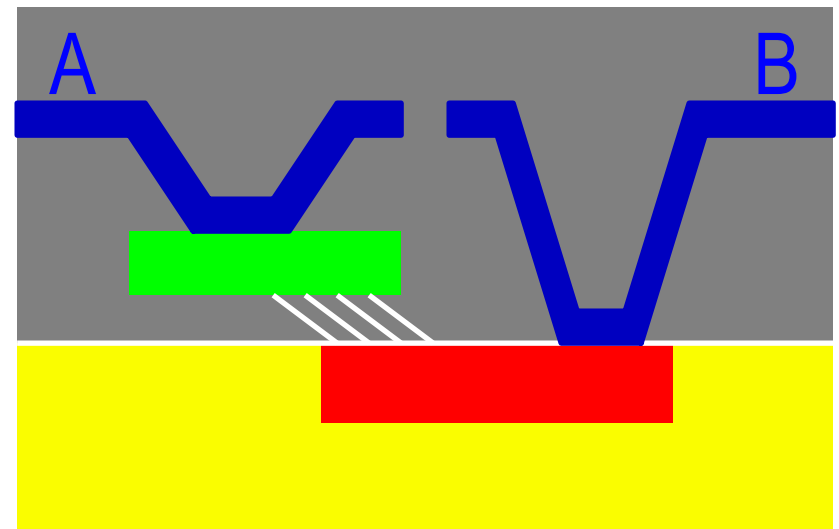
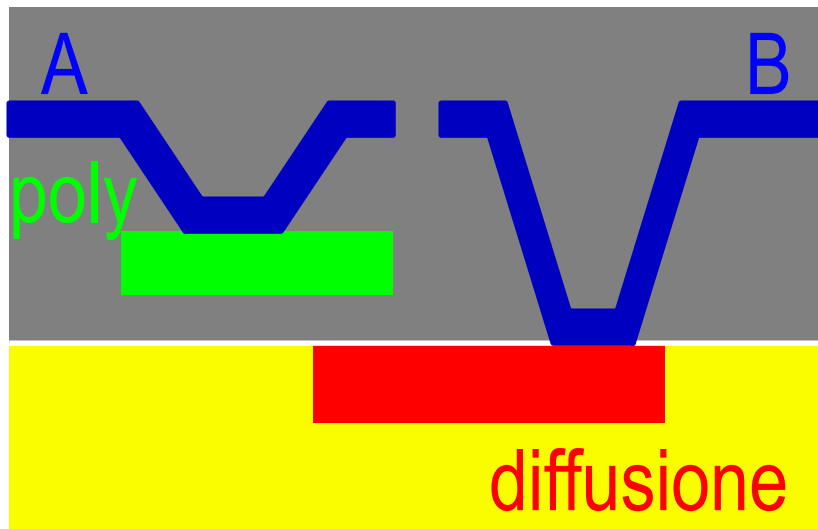
- Insufficient budget to cover ASIC NRE costs
- Small and Medium production volume
- System spec not completely frozen
- Prototyping
 - FPGA prototype for proof-of-concept
 - Easy and fast modification
 - Technology re-targeting to ASIC technology for volume production





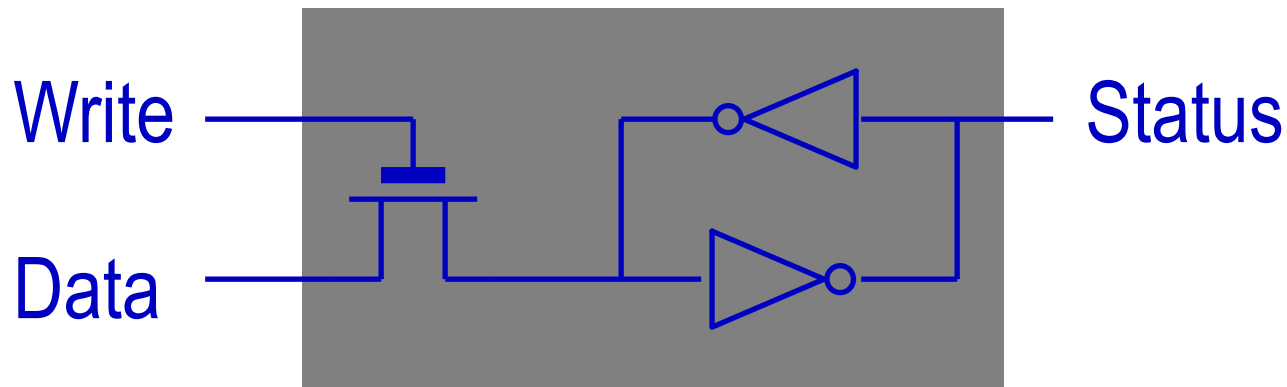
Programming Method: Anti-Fuse

- **Anti-fuse programming**
 - Two nodes are electrically isolated
 - When programmed two nodes are closed



Programming Method: SRAM based

- Programming with SRAM memory cell
 - Status of programmable unit depend on the data stored in the memory cell



FPGA SRAM-based

- Volatile Static Memory
- ☹ Configuration Load at power-on
- ☹ Sensibility on external noise (soft errors)
- ☺ Re-programmable
 - Easy and fast modification to the design
- ☺ Run time Re-programmability
 - Part of the FPGA can be re-programmed on the fly

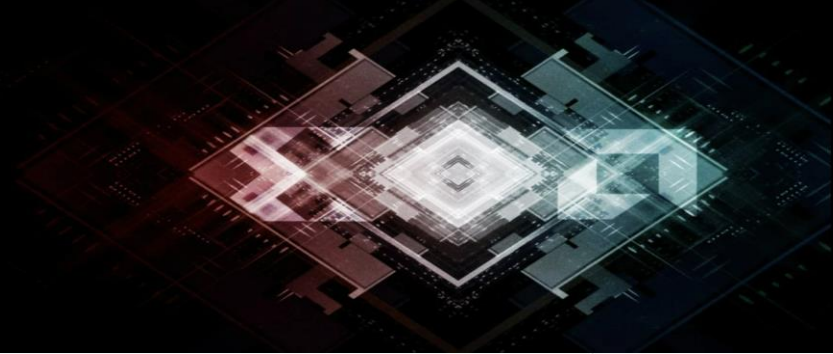
FPGA Architectures

- Matrix of Logic Cell with different complexity/granularity
 - Row of logic cells
 - » **ACTEL** (www.actel.com) → www.microsemi.com (a microchip company)
 - Matrix of logic cells
 - » **XILINX** (www.xilinx.com)

AMD Acquires Xilinx

Creating the Industry's High Performance and Adaptive Computing Leader

READ THE PRESS RELEASE



- Transaction creates high-performance and adaptive computing leader
- AMD to offer industry's strongest portfolio of leadership CPUs, GPUs, FPGAs and Adaptive SoCs to address approximately \$135 billion market opportunity

SANTA CLARA, Calif. 02/14/2022

AMD (NASDAQ: AMD) today announced the completion of its acquisition of Xilinx in an all-stock transaction. The acquisition, originally announced on October 27, 2020, creates the industry's high-performance and adaptive computing leader with significantly expanded scale and the strongest portfolio of leadership computing, graphics and adaptive SoC products. AMD expects the acquisition to be accretive to non-GAAP margins, non-GAAP EPS and free cash flow generation in the first year.

"The acquisition of Xilinx brings together a highly complementary set of products, customers and markets combined with differentiated IP and world-class talent to create the industry's high-performance and adaptive computing leader," said AMD President and CEO Dr. Lisa Su. "Xilinx offers industry-leading FPGAs, adaptive SoCs, AI engines and software expertise that enable AMD to offer the strongest portfolio of high-performance and adaptive computing solutions in the industry and capture a larger share of the approximately \$135 billion market opportunity we see across cloud, edge and intelligent devices."

Former Xilinx CEO Victor Peng will join AMD as president of the newly formed Adaptive and Embedded Computing Group (AECG). AECG remains focused on driving leadership FPGA, Adaptive SoC and software roadmaps, now with the additional scale of the combined company and the ability to offer an expanded set of solutions including AMD CPUs and GPUs.

"The rapid expansion of connected devices and data-intensive applications with embedded AI are driving the growing demand for highly efficient and adaptive high-performance computing solutions," said Victor Peng. "Bringing AMD and Xilinx together will accelerate our ability to define this new era of computing by providing the most comprehensive portfolio of adaptive computing platforms capable of powering a wide range of intelligent applications."

Upon close, Xilinx stockholders received 1.7234 shares of AMD common stock and cash in lieu of any fractional shares of AMD common stock for each share of Xilinx common stock. Xilinx common stock will no longer be listed for trading on the NASDAQ stock market.

Transaction Website

For more information about the transaction, investors are encouraged to visit the [AMD/Xilinx Acquisition web page](#).

About AMD

For more than 50 years AMD has driven innovation in high-performance computing, graphics and visualization technologies. Billions of people, leading Fortune 500 businesses and cutting-edge scientific research institutions around the world rely on AMD technology daily to improve how they live, work and play. AMD employees are focused on building leadership high-performance and adaptive products that push the boundaries of what is possible. For more information about how AMD is enabling today and inspiring tomorrow, visit the AMD (NASDAQ: AMD) [website](#), [blog](#), [Facebook](#) and [Twitter](#) pages.

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AMD Investor Relations
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AMD Acquires Xilinx

Creating the Industry's High Performance and Adaptive Computing Leader

[READ THE PRESS RELEASE](#)

Long-Term TAMs Present Enormous Opportunity



Data Center

\$50B TAM



PCs

\$40B TAM



Embedded

\$29B TAM



Gaming

\$16B TAM

~\$135B TAM

A Track Record of Innovation

Hardware Innovations



World's First
FPGA

1984



First High-End
High-Capacity
FPGA

1999



First FPGA with
Integrated SerDes
and Processor

2001



First 3D FPGA
and Zynq Dual HW
Programmable SoC

2012



First Zynq®
MPSoC &
RFSoc

2017



ALVEO™
Data Center
Accelerator Card

2018



VERSAL®
First Adaptive
Compute Acceleration
Platform

2019



ALVEO SN100
First
Composable,
Adaptable
SmartNIC

2021



KRIA SOM
Adaptive System
on Module with
First Embedded
App Store

Software Innovations



ISE
DESIGN SUITE



VIVADO
Next-Gen Development
Environment



SDx
Editions
Programmability
for SW Developers



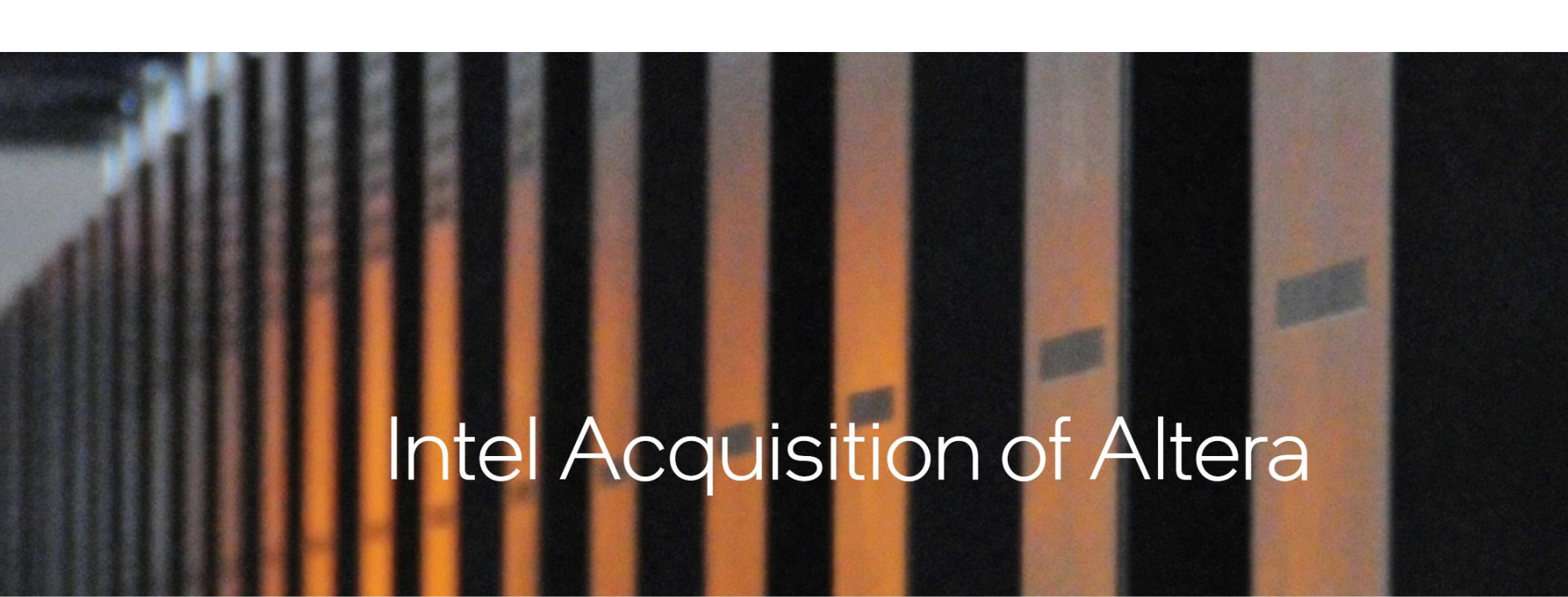
HLx
Editions
C, C++ & System
C Enabled



XILINX
VITIS™
Unified Software Platform

The Early Years (1984-1994): When Xilinx designed and marketed the first field-programmable gate array, they spawned a new industry that would enable custom solutions for various markets. A decade later, in 1994, Xilinx released the Virtex® FPGA, a breakthrough in the architecture and performance of their original FPGA.

The Evolution of a FPGA: Fast forward to 2012, Xilinx introduced the first 28nm Zynq® device, a full SoC with support for application processors and a complete subsystem of caches, memory controller, peripherals, FPGA logic, DSP blocks, and SerDes blocks. The Zynq SoC brought Xilinx into the heart of systems instead of handling peripheral functions and bridging connectivity.



Intel Acquisition of Altera

Press Kit

December 28, 2015

[Contact Intel PR](#)

Intel and Altera announced on June 1, 2015, that they had entered into a definitive agreement under which Intel would acquire Altera for \$54 per share in an all-cash transaction valued at approximately \$16.7 billion. The transaction closed December 28, 2015. The acquisition will couple Intel's leading-edge products and manufacturing process with Altera's leading field-programmable gate array (FPGA) technology. The combination is expected to enable new classes of products that meet customer needs in the data center and Internet of Things (IoT) market segments. Intel plans to offer Altera's FPGA products with Intel Xeon® processors as highly customized, integrated products. The companies also expect to enhance Altera's products through design and manufacturing improvements resulting from Intel's integrated device manufacturing model.

The Mercury Research CPU market share results

Overall x86 CPU Share - ALL CPUs

Overall x86 CPU Share	2021 Q4	2021 Q3	2020 Q4	Change	Change
Includes IoT and SoC	Share	Share	Share	Quarter	Year
Intel	74.4%	75.4%	78.3%	- 1.0	- 3.9
AMD	25.6%	24.6%	21.7%	+ 1.0	+ 3.9
VIA	0.0%	0.0%	0.0%	- 0.0	- 0.0
Total	100.0%	100.0%	100%		

By Tom's Hardware

Microsemi buys Actel for \$430 million

By Mark LaPedus 10.04.2010  0

SAN JOSE, Calif. – Microsemi Corp. has entered into the FPGA market.

The analog and mixed-signal chip maker has entered into a definitive agreement to acquire Actel Corp. for \$20.88 per share through a cash tender offer. The total transaction value is approximately \$430 million, net of Actel's projected cash balance at closing.

Based in Mountain View, Calif., Actel is a supplier of field programmable gate arrays (FPGAs). Actel competes against Altera, Lattice, SiliconBlue and Xilinx. Now, Microsemi will compete in the tough FPGA fray.

Observers were scratching their heads about the deal. It was a complete surprise. Analog and FPGAs are completely different markets.

“We believe the addition of Actel will deliver compelling synergies to Microsemi,” said James J. Peterson,

Microchip to Buy Microsemi for \$8.35 Billion

By Dylan McGrath 03.01.2018  1

SAN FRANCISCO — Microchip Technology plans to acquire Microsemi Corp. for about \$8.35 billion in cash under the terms of a definitive agreement announced Thursday (March 1).

The deal would significantly expand Microchip's presence in several end markets, including the communications and aerospace and defense markets, which make up about 60 percent of Microsemi's sales. Microchip said the acquisition would expand its serviceable market by about \$18 billion to more than \$50 billion.

The combined company would have annual sales of about \$5.8 billion, based on each firm's revenue in the fourth quarter of 2017.



“Microchip continues to view accretive acquisitions as a key strategy to deliver incremental growth and stockholder value,” said Steve Sanghi, Microchip's

FPGA & SoC

- ▶ FPGA & SoC Design Tools
- ▶ FPGAs
- ▶ Rad-Tolerant FPGAs
- ▶ Getting Started
- ▶ Antifuse FPGAs
- ▶ FPGA Applications
- ▶ Mi-V RISC-V Ecosystem
- ▶ Key Technologies & Solutions
- ▶ FPGA & SoC Partners
- ▶ Low Power Leadership
- ▶ SoC FPGAs
- ▶ Boards and Kits

Join the LegUp High-Level Synthesis Compiler (HLS) Early Access Program for PolarFire FPGAs

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FPGA & SoC

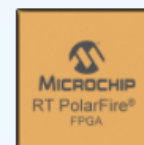
[Overview](#)[News & Videos](#)[TechBytes Archive](#)[Resources](#)[Applications](#)

We offer a broad portfolio of low and mid-range density FPGAs that lead the industry in their low power consumption, reliability, and security capabilities. Our devices are unique in their use of non-volatile, instant-on technology that enables our FPGAs to deliver 30 to 50 percent lower power than competing FPGAs, immunity to radiation, and unique security features. Use our FPGAs, SoC FPGAs, and Radiation Tolerant FPGAs to meet high-bandwidth connectivity and high-data throughput needs in applications such as Hybrid and Electric Vehicles, Communications IoT Infrastructure, Industrial Controls and Automation, Spacecraft, Commercial Aircraft, and Defense Equipment.

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TechBytes Newsletter](#)

PolarFire Mid-Range FPGAs

- Cost-optimized, lowest power in their class
- 250 Mbps to 12.7 Gbps transceivers
- 100K to 500K LE, up to 33 Mbits of RAM
- Best-in-class security and exceptional reliability

[New](#) [PolarFire FPGA Smart Embedded Vision](#)[>> All FPGAs](#)

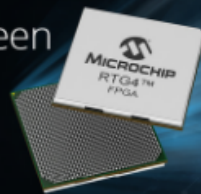
RT PolarFire Radiation Tolerant FPGAs

- Immune to configuration upsets
- 50% lower power than SRAM FPGAs
- 250 Mbps to 10 Gbps transceivers

[New](#) [RT Polar Fire FPGA Press Release](#)

Sub-QML FPGAs Bridge the Gap Between COTS and QML Components

Enabling Cost Reduction and Shorter Lead Times
for High Volume Satellite Constellations



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Rad-Tolerant FPGAs

[Overview](#) / [Family Comparison](#) / [Flight Heritage](#) / [Legacy](#) / [Discontinued](#) / [Applications](#) / [Parametric Search](#) / [Ordering](#) / [Support](#)



RT PolarFire FPGAs

- High-speed signal processing, SONOS-based FPGAs
 - 40 to 50% lower power, SEU immune, path to QML-V qualification
 - 481,000 LEs, 250 Gbps SERDES bandwidth, 33 Mbits memory, 1480 multipliers
- » [RT PolarFire FPGA](#)



RTG4 FPGAs

- Radiation Hardened by Design Flash-based FPGA with proven flight heritage
 - 300 MHz performance, up to 150,000 LEs, 5 Mbit SRAM
 - 462 multipliers, 75 Gbit / second SERDES bandwidth
- » [RTG4 FPGA](#)



RTAX FPGAs

- High-reliability, radiation-tolerant, antifuse-based FPGAs
 - Available with embedded multiply-accumulate blocks for DSP applications
 - Qualified to QML Class Q and QML Class V
- » [RTAX-S/SL](#) » [RTAX-DSP](#)



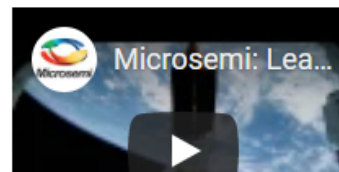
RT ProASIC3 FPGAs

- Reprogrammable, nonvolatile, radiation-tolerant, flash-based FPGAs
- For low-power space applications, requiring up to 350 MHz operation
- Qualified to QML Class Q

[Space System Managers are companion devices that can be used with any Radiation-Tolerant FPGAs](#)

Radiation-Tolerant Spacecraft
Telemetry Controller
LX7730

Radiation-Tolerant Spacecraft
Motor Controller with
Position Sensing
LX7720



Products



Silicon Devices

Xilinx offers the broadest lineup of FPGA, CPLD and EPP solutions that set the industry standard for the lowest cost, highest performance, and lowest power.

[+ Show Devices](#)

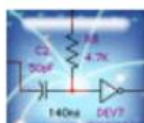
Design Tools

The ISE Design Suite unlocks the full potential of Xilinx Targeted Design Platforms with configurations for logic, embedded, and DSP designs – all available with tightly integrated design flows.

[+ Show Design Tools](#)

Boards & Kits

Xilinx development kits provide out-of-the box design solutions that help you cut your development time by up to 50%.



Intellectual Property

Xilinx Intellectual Property (IP) are key building blocks of Xilinx Targeted Design Platforms. An extensive catalog of base-level cores is available to address the general needs of FPGA designers, as well as robust domain- and market-specific cores to address requirements found in DSP, Embedded, and Connectivity



Technology

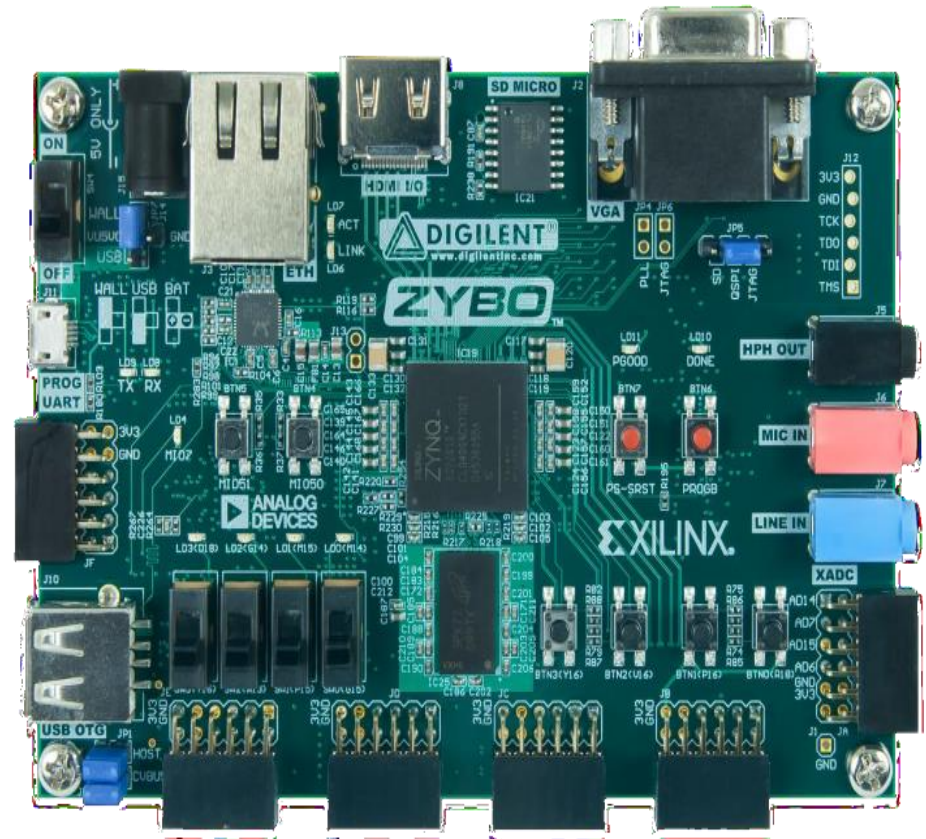
Xilinx offers the most complete suite of technology solutions available in the industry including Embedded Processing, DSP, Connectivity and more.

[+ Show Technologies](#)

Zybo Development Board

The ZYBO (ZYnq BOard) is a feature-rich, ready-to-use, entry-level embedded software and digital circuit development platform built around the smallest member of the Xilinx Zynq-7000 family, the Z-7010.

The Z-7010 is based on the Xilinx All Programmable System-on-Chip (AP SoC) architecture, which tightly integrates a dual-core ARM Cortex-A9 processor with Xilinx 7-series Field Programmable Gate Array (FPGA) logic.

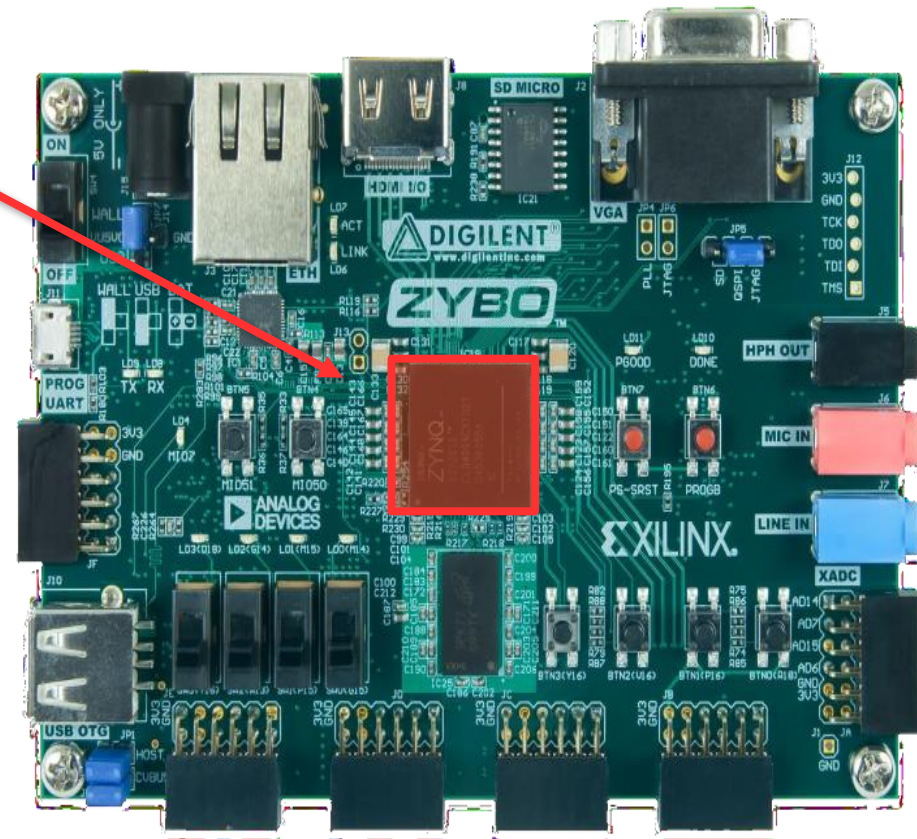


Zybo Development Board



The Board

- **ZYNQ XC7Z010-1CLG400C**
- 512MB x32 DDR3 w/ 1050Mbps bandwidth
- Dual-role (Source/Sink) HDMI port
- 16-bits per pixel VGA source port
- Trimode (1Gbit/100Mbit/10Mbit) Ethernet PHY
- MicroSD slot (supports Linux file system)
- OTG USB 2.0 PHY (supports host and device)
- External EEPROM (programmed with 48-bit globally unique EUI-48/64™ compatible identifier)
- Audio codec with headphone out, microphone and line in jacks
- 128Mb Serial Flash w/ QSPI interface
- On-board JTAG programming and UART to USB converter
- GPIO: 6 pushbuttons, 4 slide switches, 5 LEDs
- Six Pmod connectors (1 processor-dedicated, 1 dual analog/digital, 3 high-speed differential, 1 logic-dedicated)

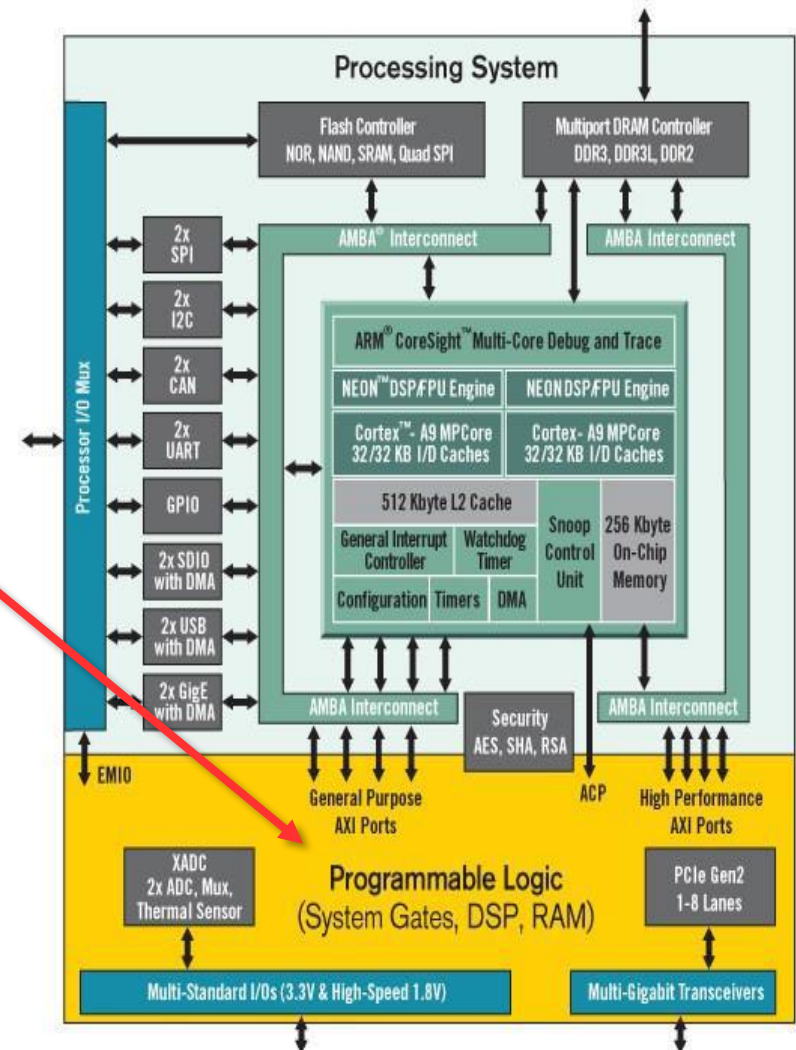


Zybo Development Board

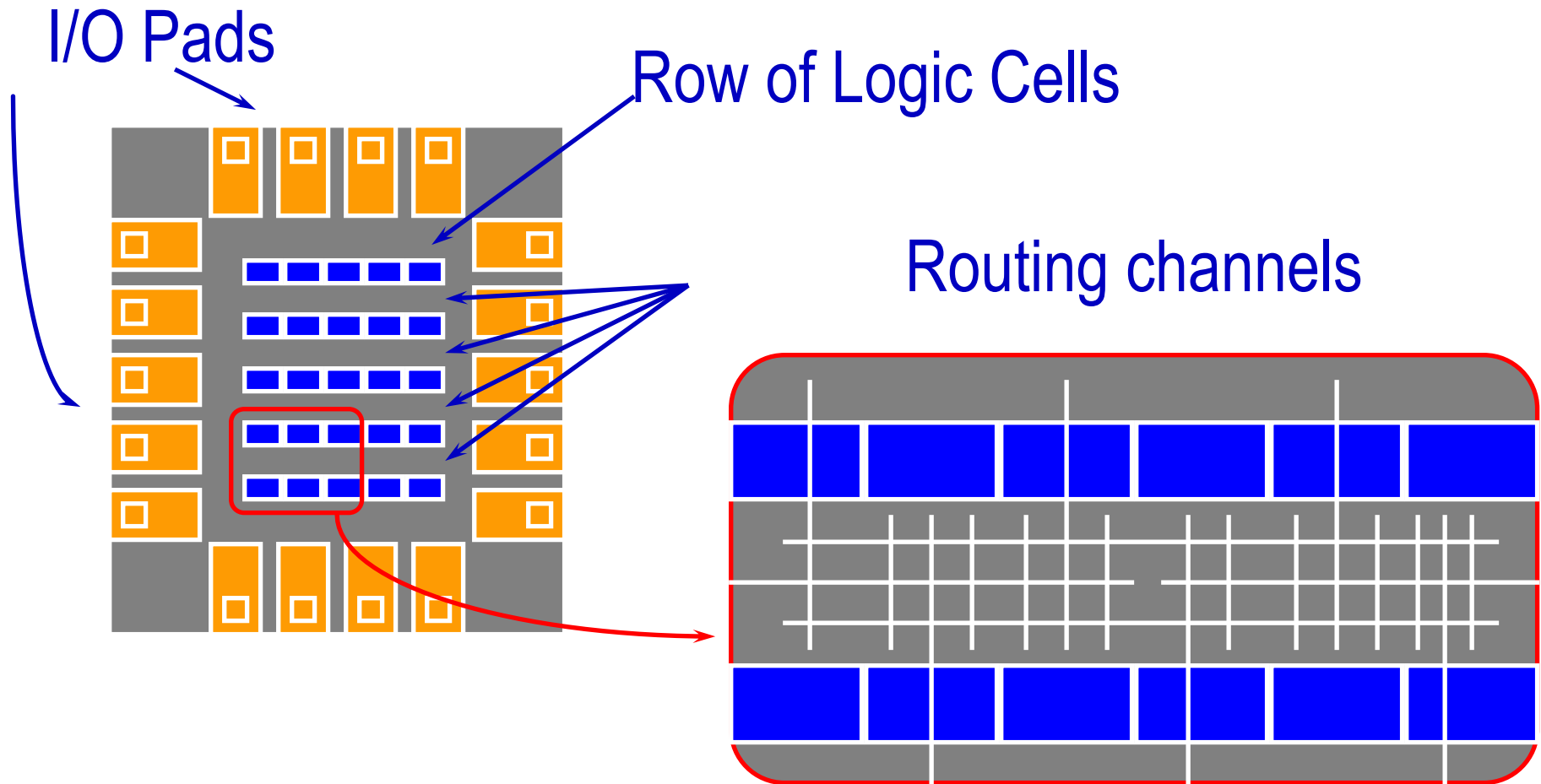


The ZYNQ Z-7010 AP SoC

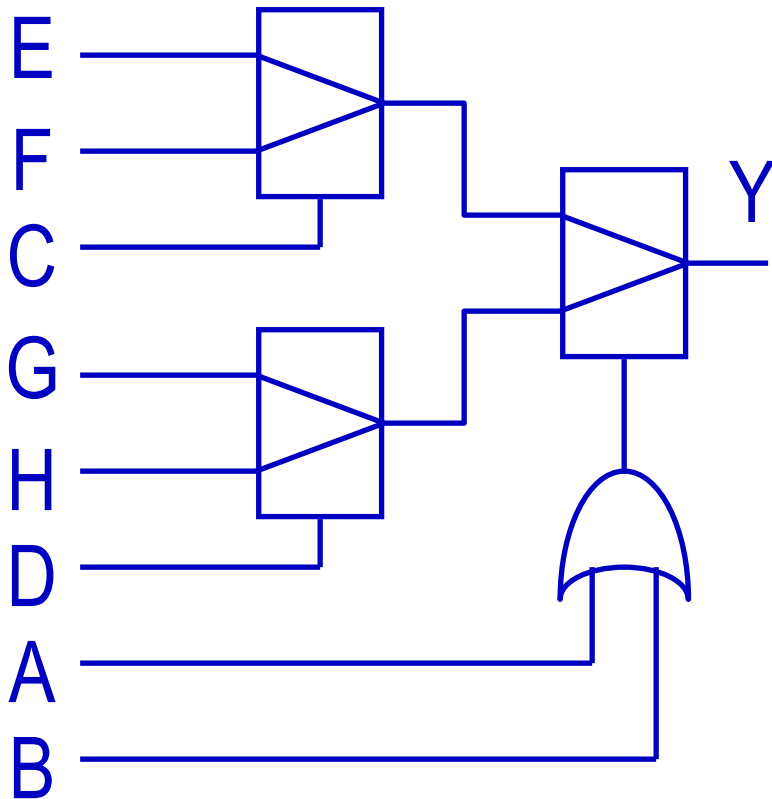
- 650Mhz dual-core Cortex-A9 processor
- DDR3 memory controller with 8 DMA channels
- High-bandwidth peripheral controllers: 1G Ethernet, USB 2.0, SDIO
- Low-bandwidth peripheral controller: SPI, UART, CAN, I2C
- Reprogrammable logic equivalent to Artix-7 FPGA
 - 4,400 logic slices, each with four 6-input LUTs and 8 flip-flops
 - 240 KB of fast block RAM
 - Two clock management tiles, each with a phase-locked loop (PLL) and mixed-mode clock manager (MMCM)
 - 80 DSP slices
 - Internal clock speeds exceeding 450MHz
 - On-chip analog-to-digital converter (XADC)



FPGA Actel

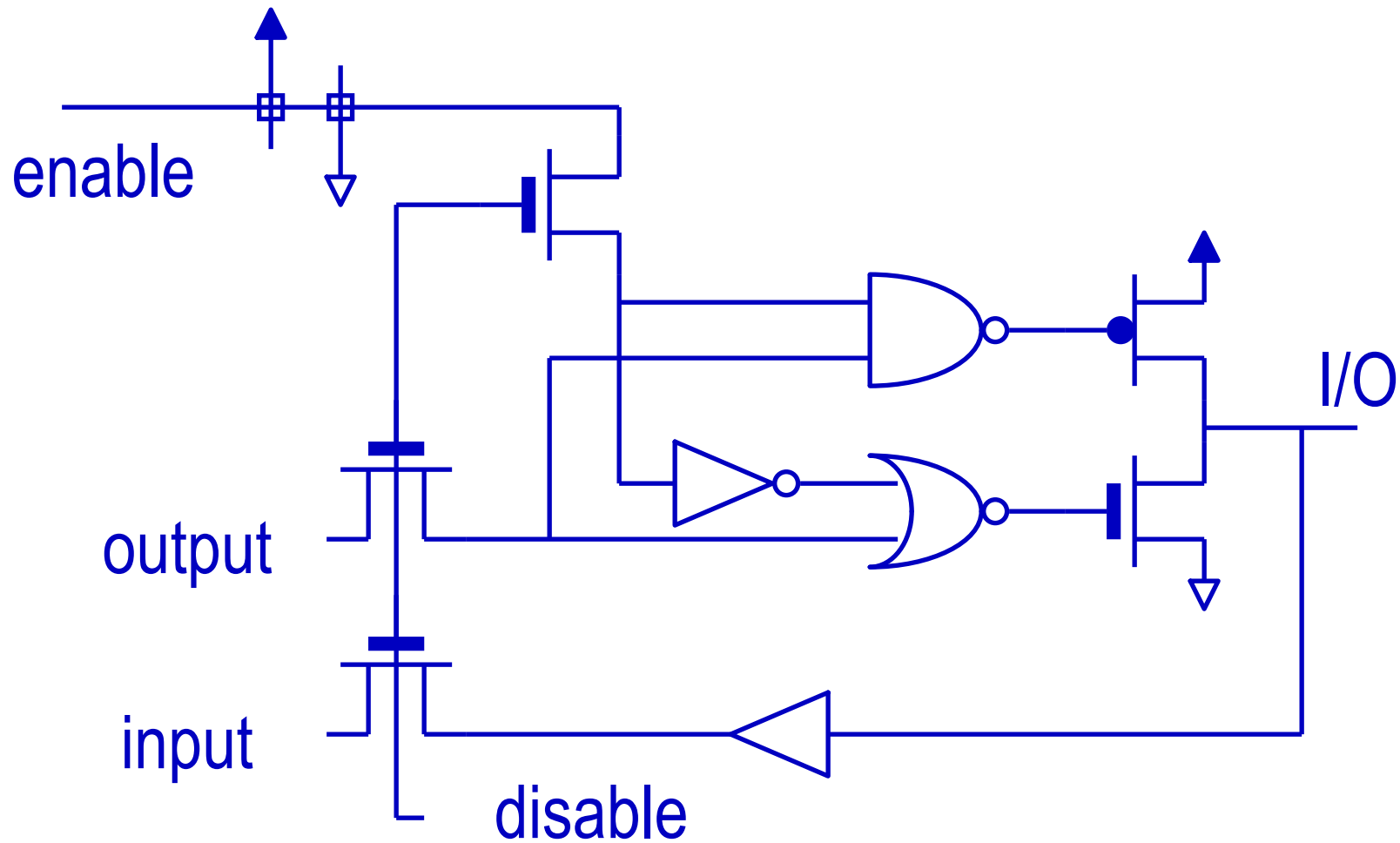


FPGA Actel: Basic Logic Cell

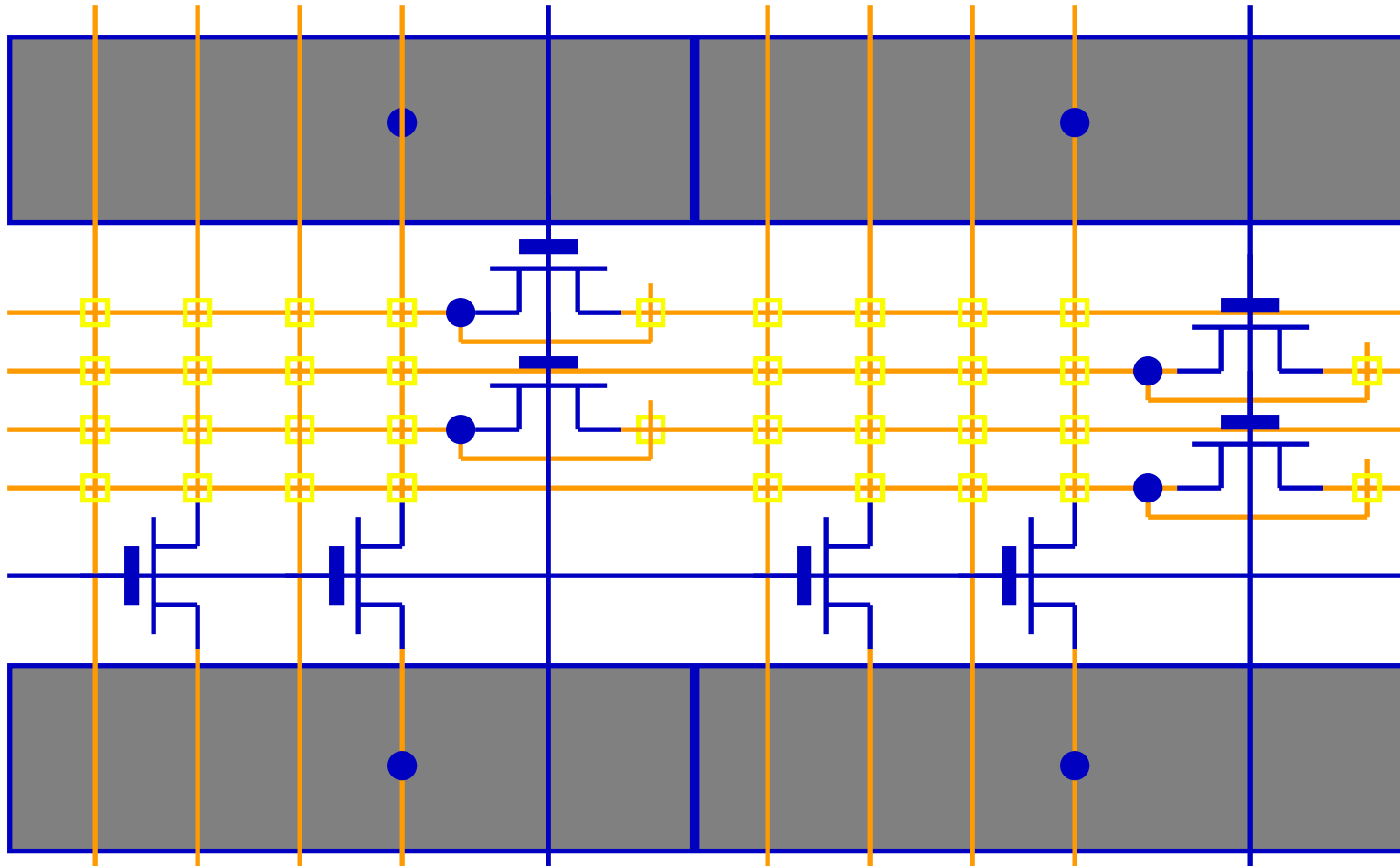


- 8 inputs
- Implements all functions with 2 or 3 inputs and many with 4 inputs
- Sequential Functions are obtained with two logic cells in feedback

FPGA Actel: I/O Pads



FPGA Actel: Interconnection

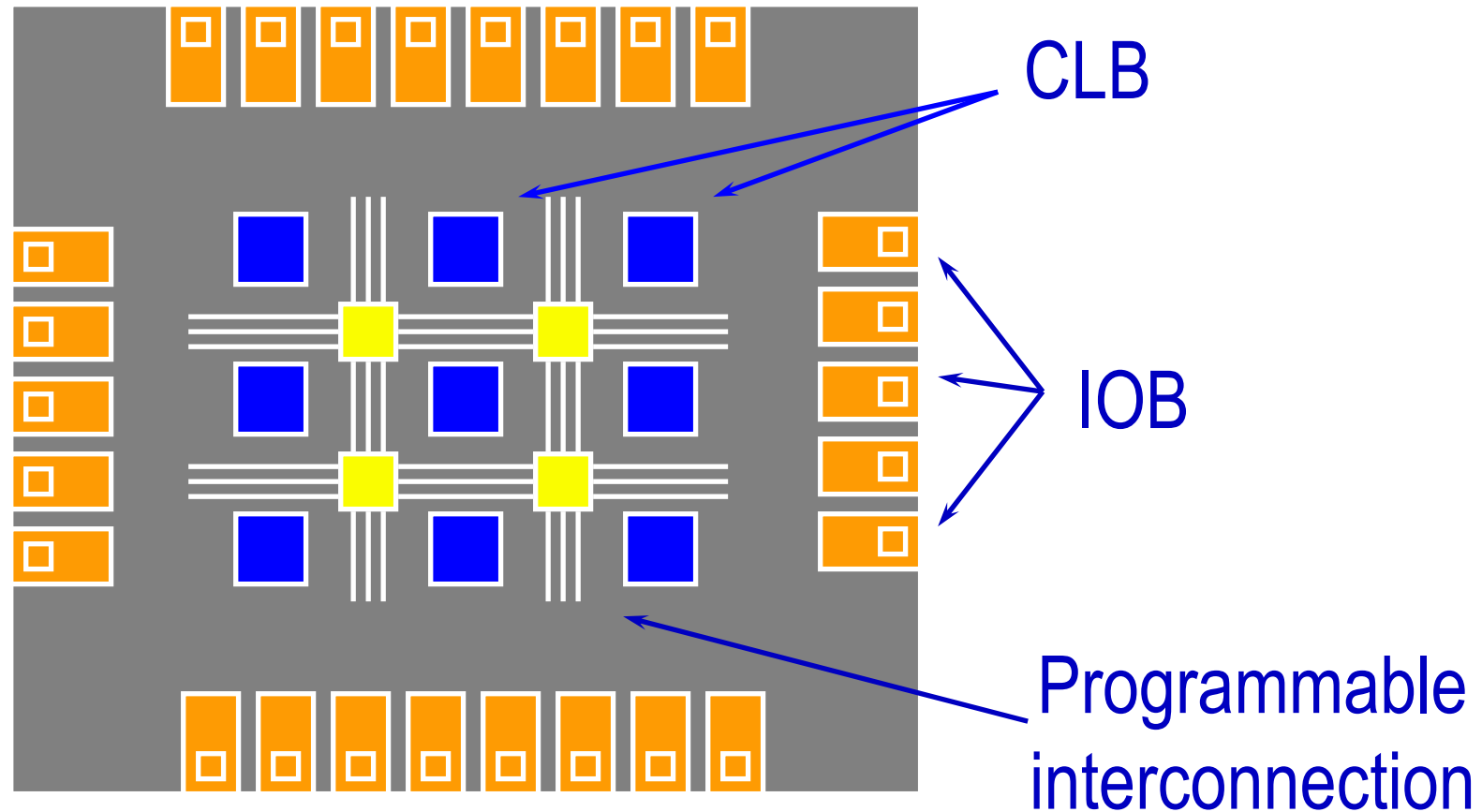


FPGA SRAM-based
(Xilinx)

Properties of FPGA SRAM-based

- Volatile Static Memory
- ☹ Configuration Load at power-on
- ☹ Required an External Memory
- ☹ Sensibility on external noise (soft errors)
- ☺ Re-programmable
- ☺ Technology CMOS standard
 - ☺ Anti-fuse technology is not required

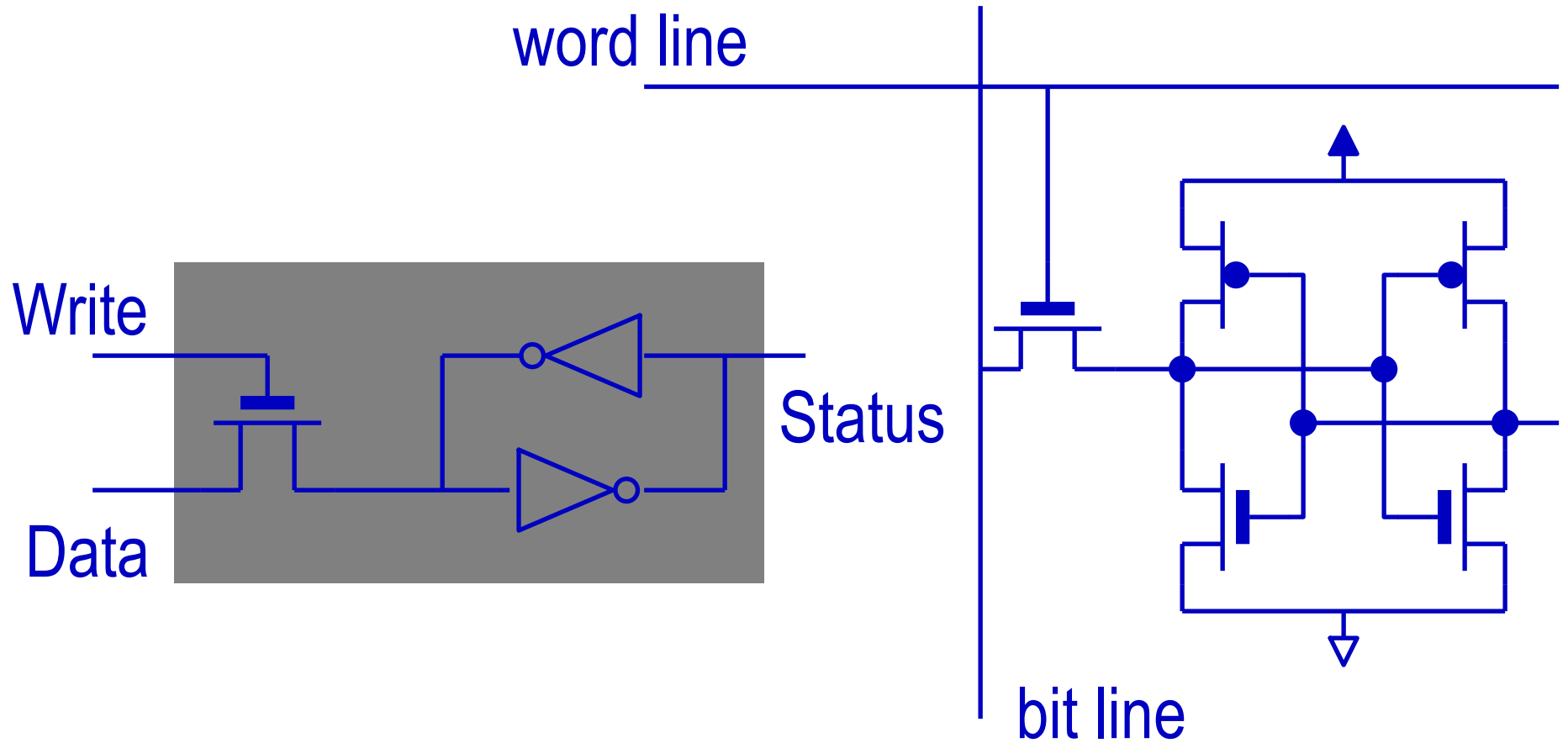
FPGA Xilinx Architecture



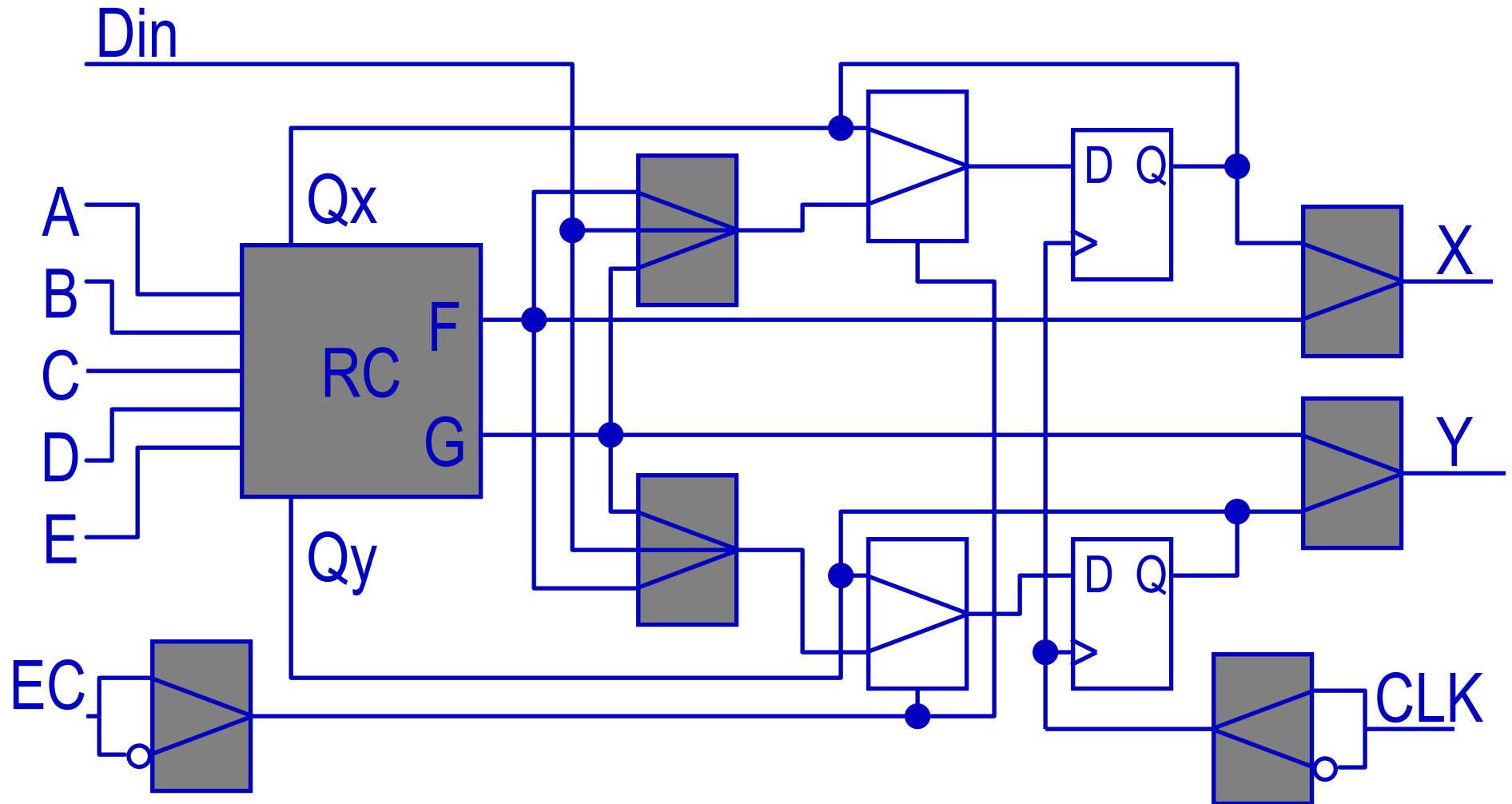
Building blocks

- Configurable Logic Blocks (CLB)
 - Where logic functions are implemented
- Input-output circuits (IOB)
 - Programmability of functionality and direction
- Routing Channels horizontal and vertical
- Memory Cell and Routing Cell are spread along the entire matrix

Memory Cell



CLB Structure



CLB properties (1/2)

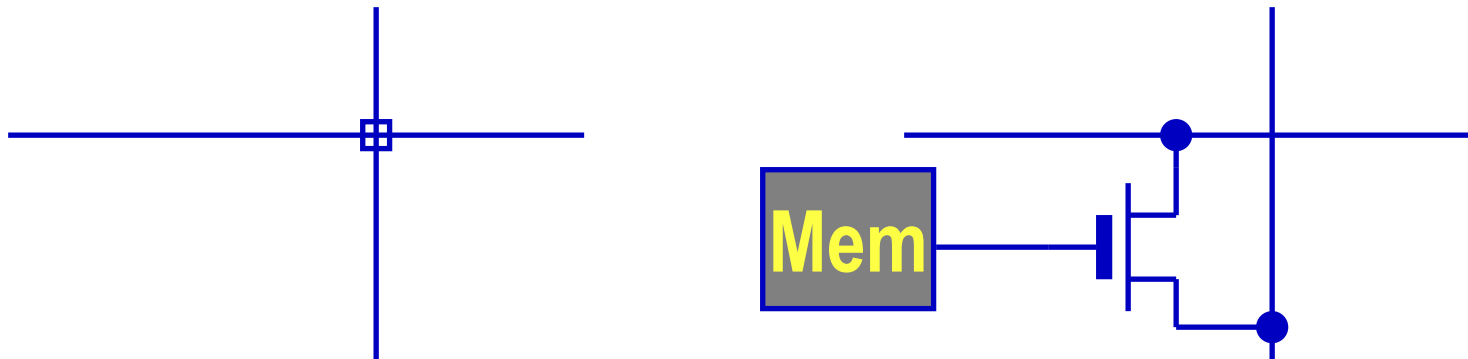
- 5 inputs, data in, clock and enable
- 2 symmetric outputs (X e Y)
- Logic Function Generator (PROM)
 - 32 cells (all functions with 5 inputs)
 - 2 splitted outputs (F e G) or a common one ($F = G$)
- 2 flip-flop D per logic block
 - positive or negative edge-triggered with enable clock

CLB properties (2/2)

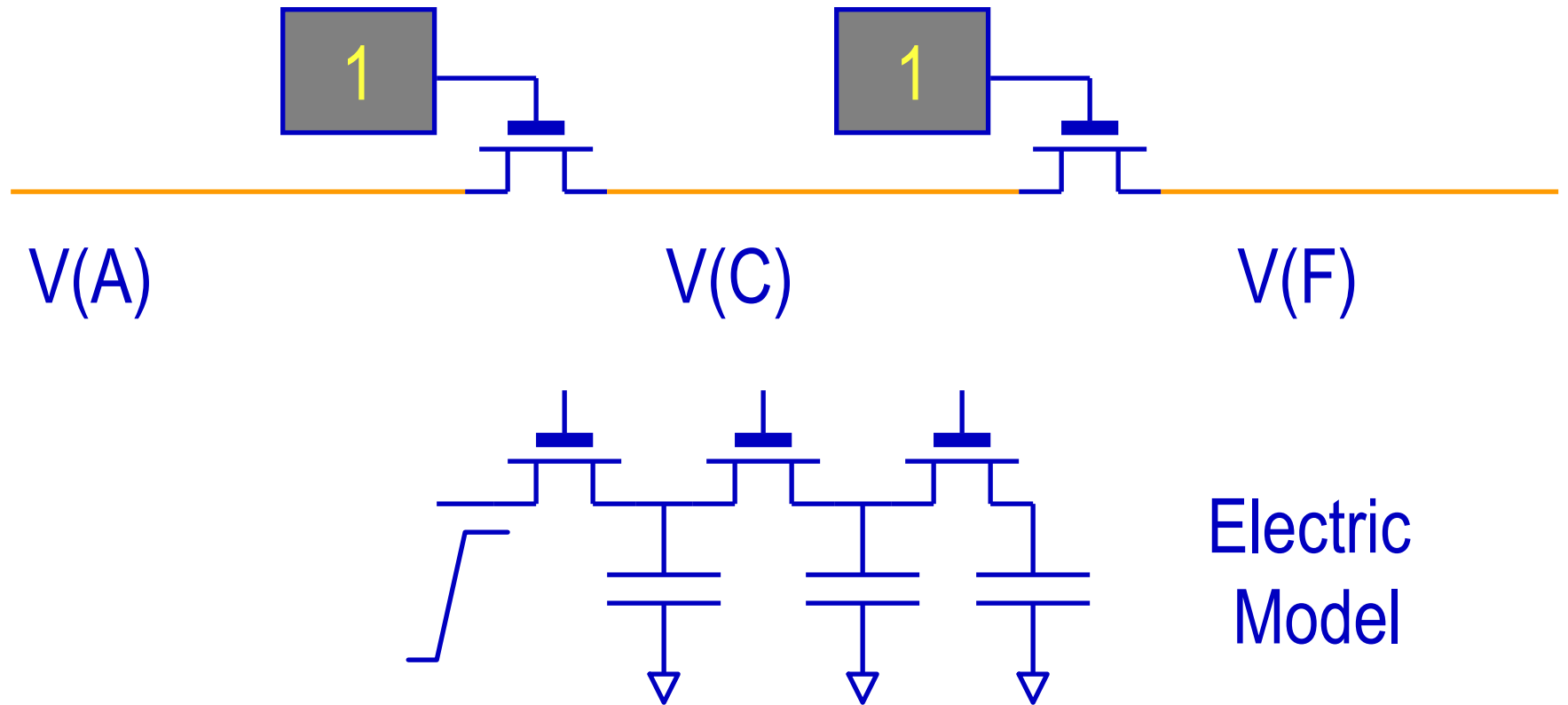
- Sequential functions with internal feedback (Q_x e Q_y)
 - To implement Finite state machine
 - Synchronous Design Style
- Din input avoiding the combinatory network
 - CLB as a building element of a CLB
 - Reduction of the propagation delay

Programmable interconnection

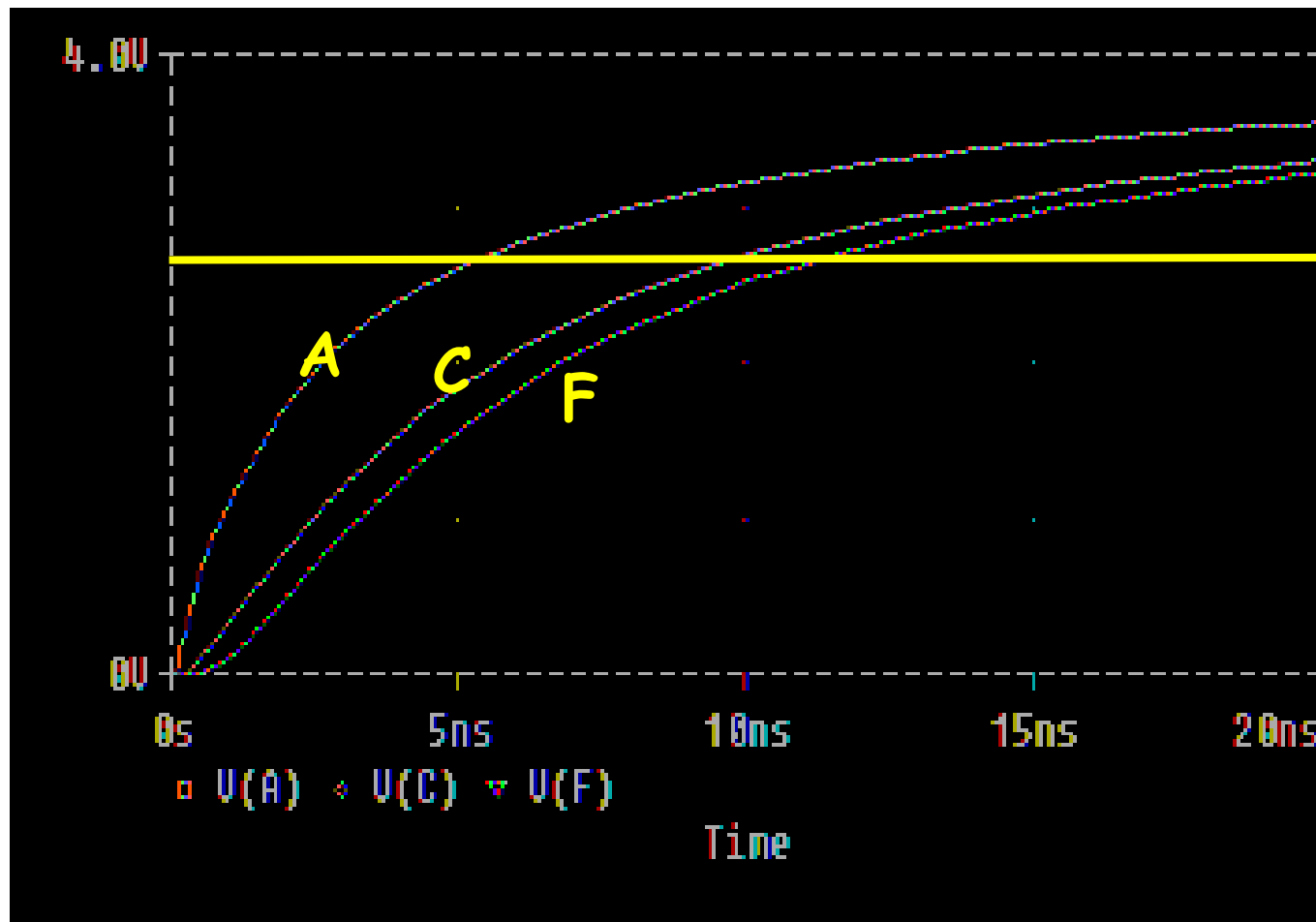
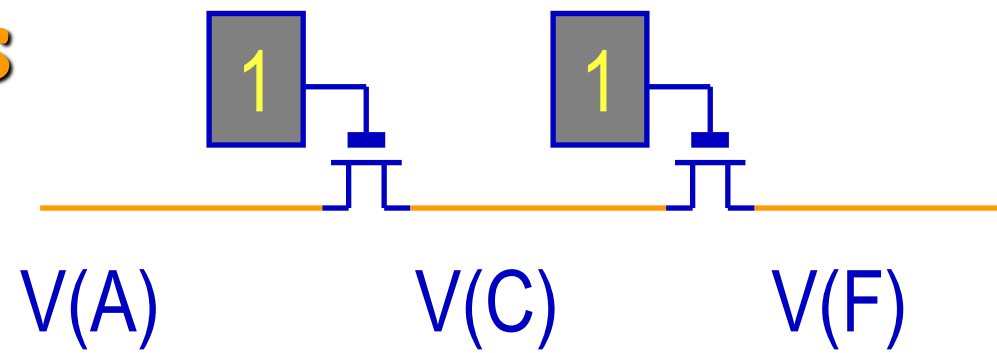
- Programmable interconnections are obtained through a pass transistor driven by a memory cell
- Programmable Interconnect Point (PIP)



Interconnection



Simulation Waveforms



Routing Problem

- Many short segments (high flexibility) determine long delay (many transistors in series)
- Few long segment (smaller propagation delay) makes connectivity more difficult

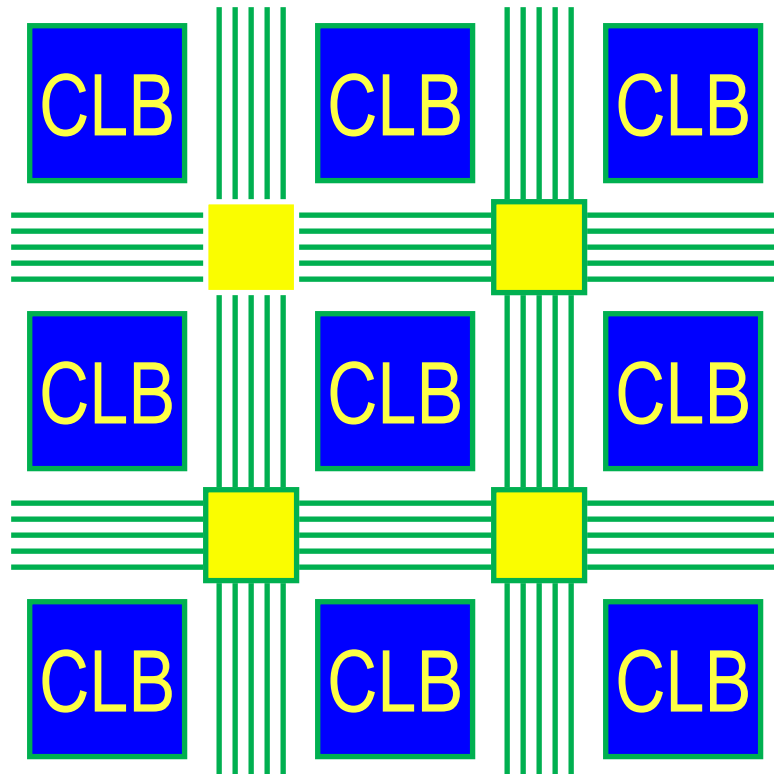
Required routing resources with different characteristics:

Short and long segments

Possible Interconnection

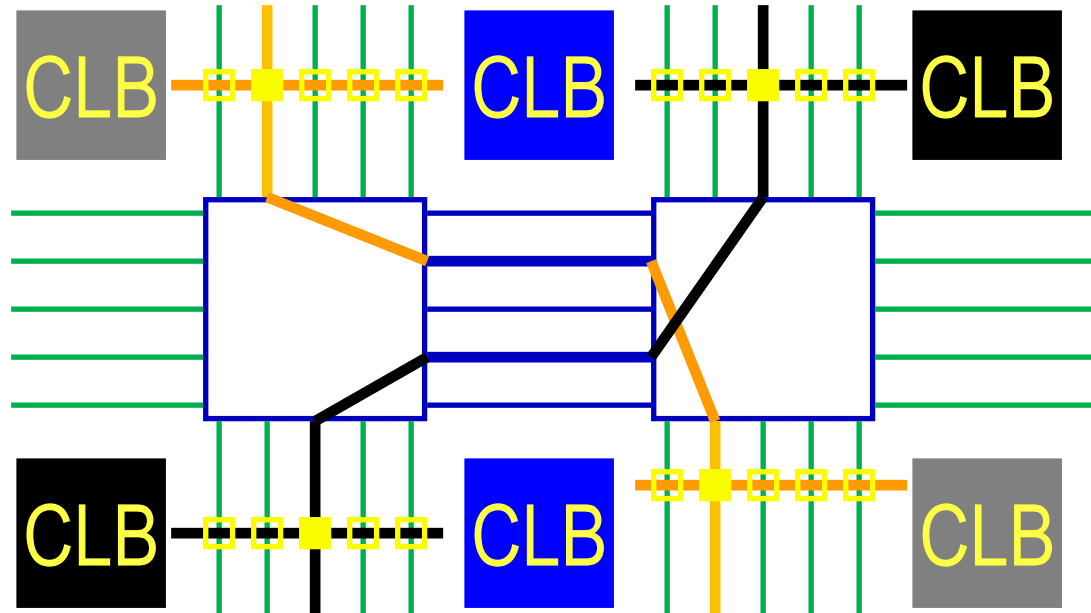
- **General**
 - Segment routed through small exchange matrix (magic box)
- **Direct or Local**
 - To link the output of a block to the input of a neighborhood one
- **Global**
 - Long interconnection lines for global use

General Interconnection

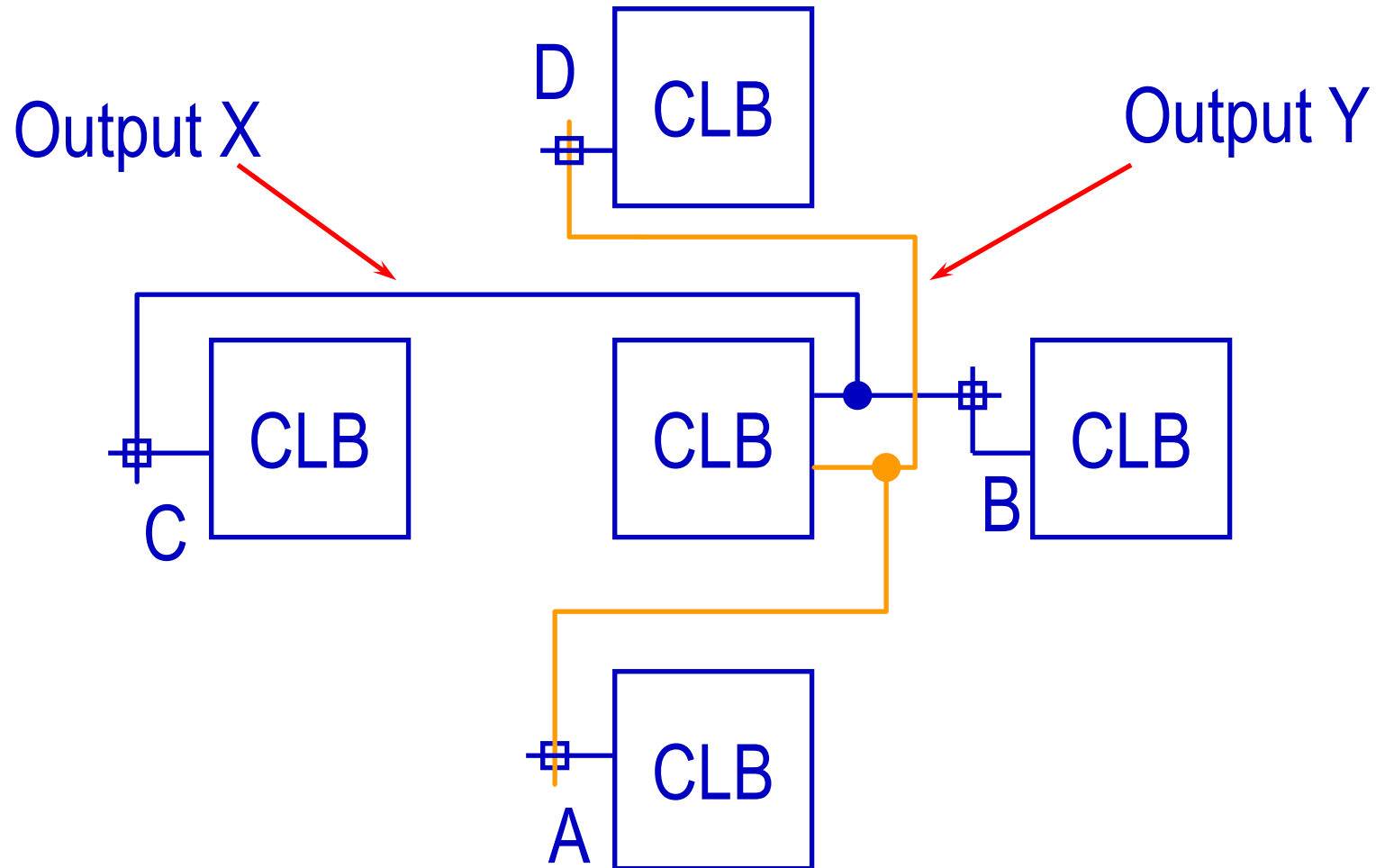


- 5 horizontal and Vertical Segments
- Exchange Matrix (5 x 5)
- CLB inputs and outputs are available on the channels through PIP

Example of a General Interconnection



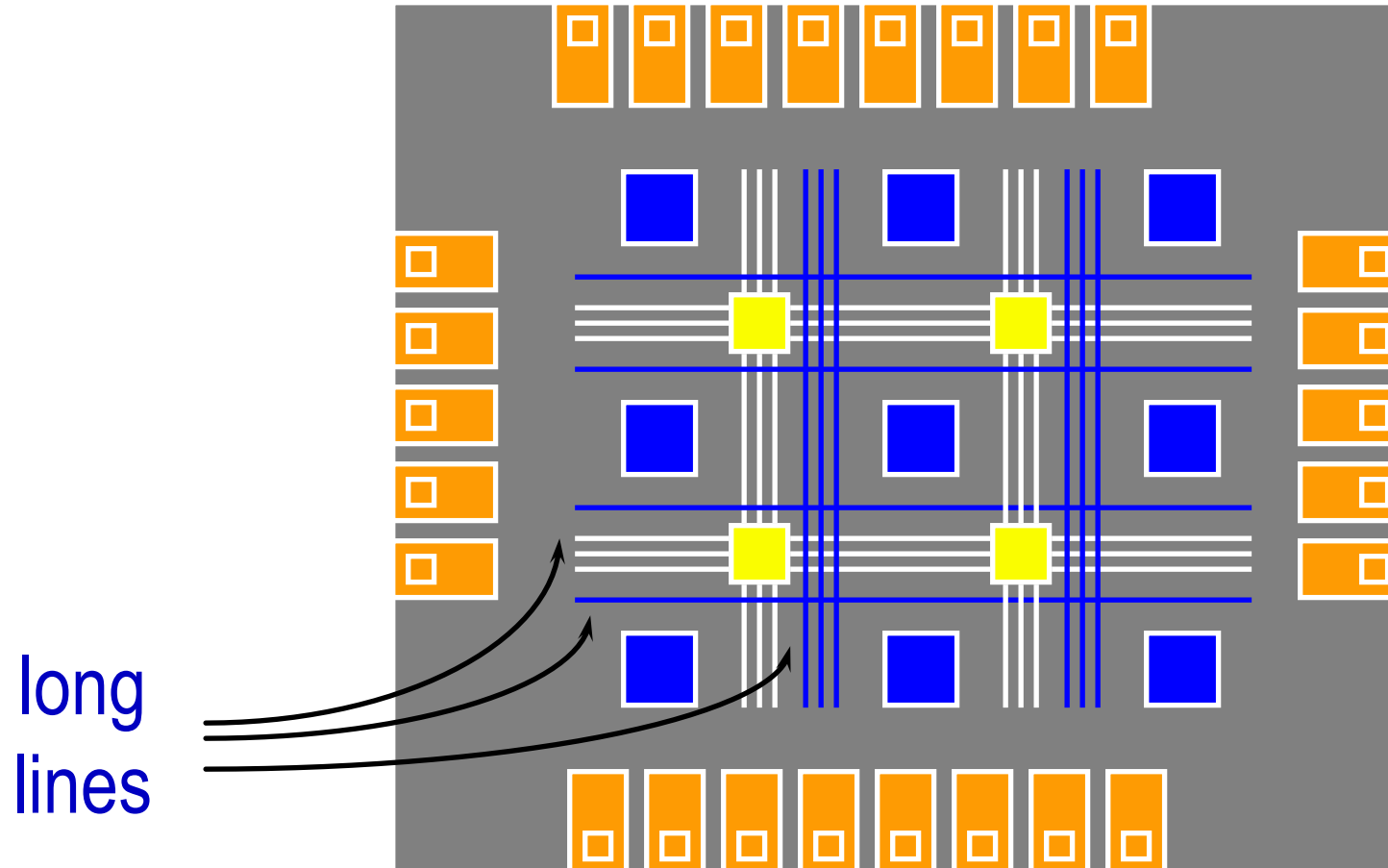
Direct Interconnections



Direct Interconnections

- Dedicated Local Connection among adjacent blocks
 - Very likelihood to happen
- Minimum and Defined Propagation Delay
- No use of general resources
- Output X is propagated to CLB on both right and left sides
- Output Y is propagated to CLB on upper and lower sides

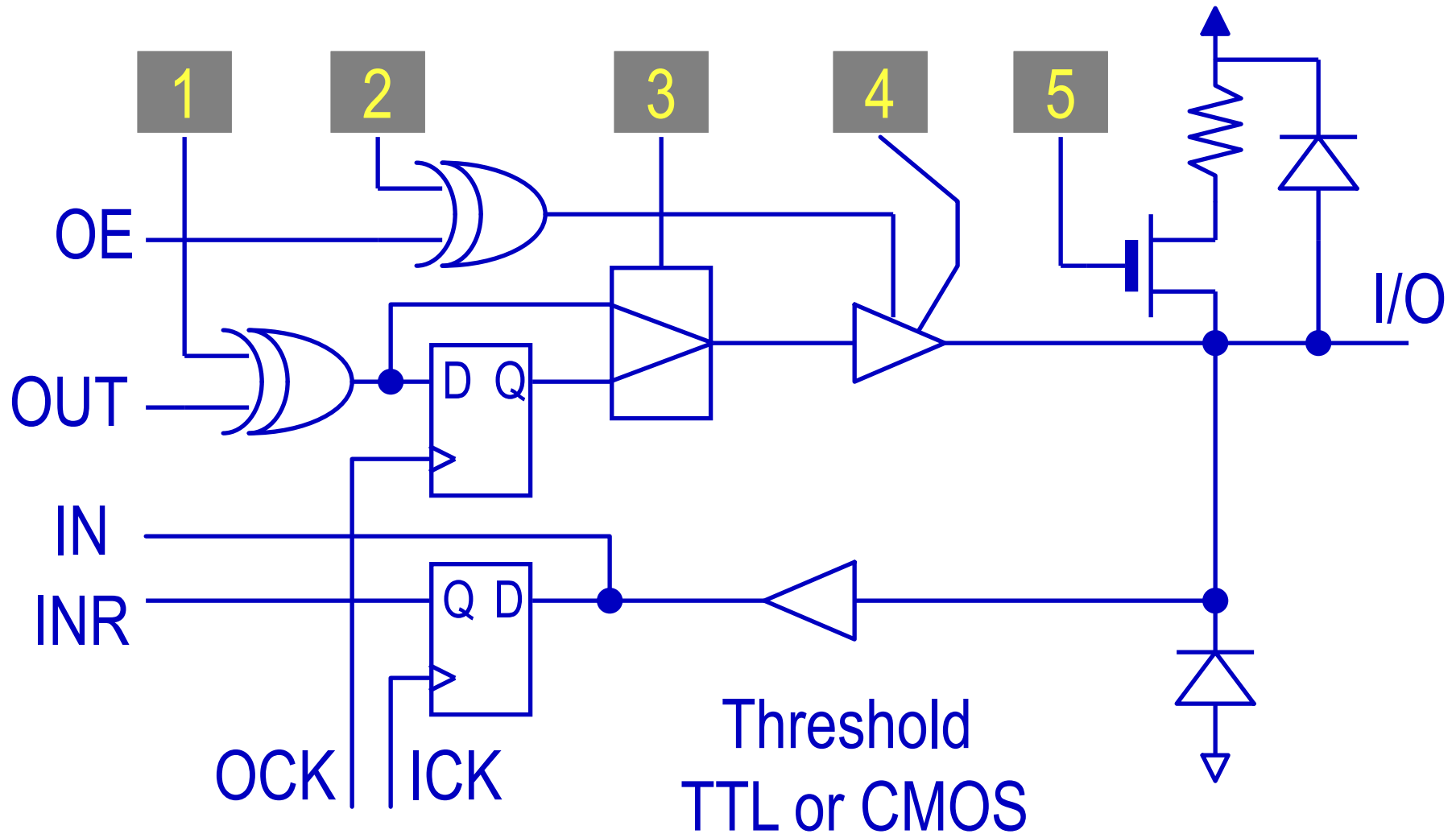
Global Interconnections (long lines)



Global Interconnections

- Long lines across the entire circuit vertically (3) and horizontally (2) without using exchange matrix
- Used for global signal (clock) or with long distance
 - Minimum clock skew
 - Possibly to be used as FPGA internal buses

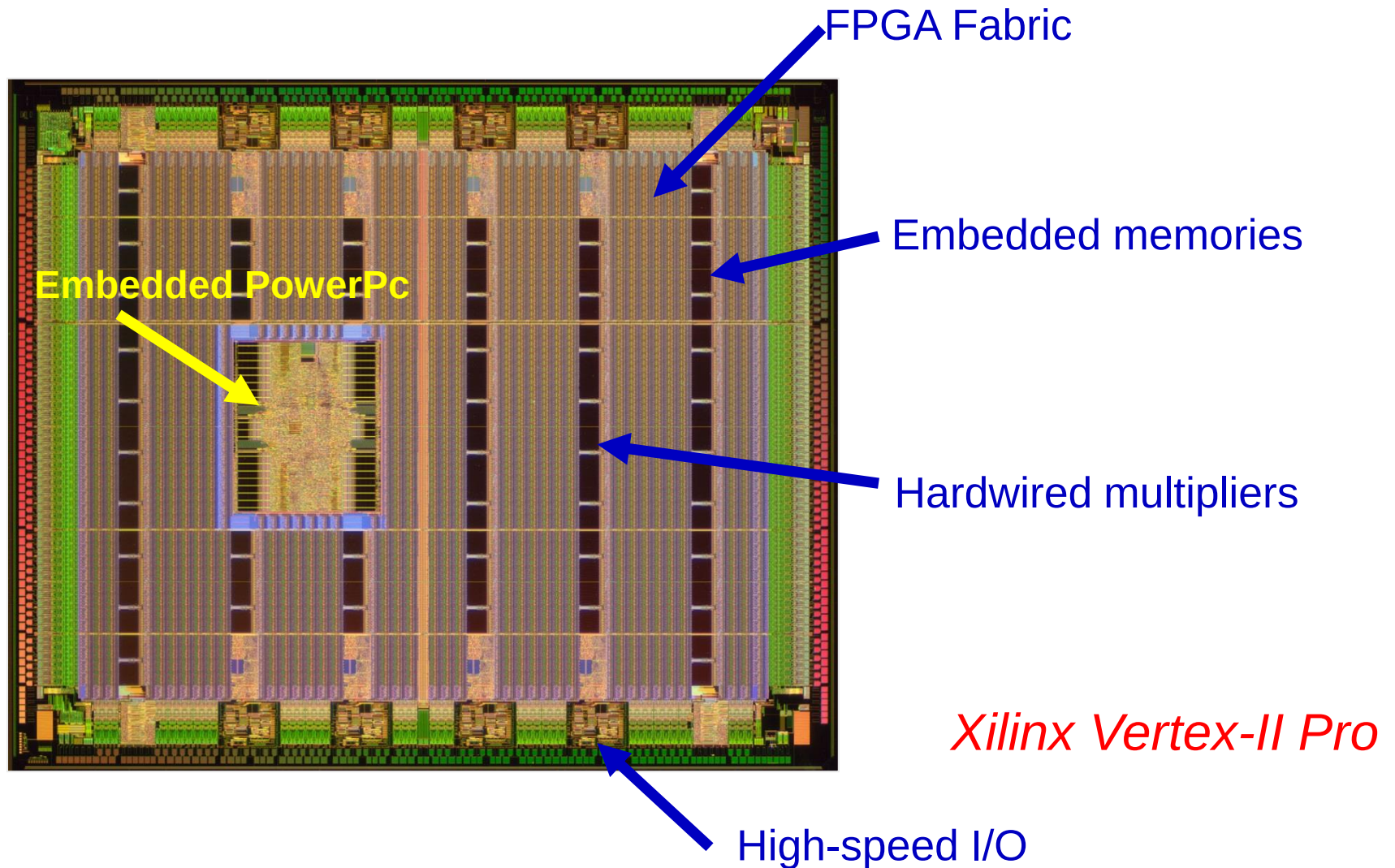
IOB Structure



IOB Functionalities

- Input-Output Terminal
- Possible to select:
 - 1 Output signal polarity
 - 2 Output enable signal polarity
 - 3 Output registered or Direct
 - 4 Output Buffer slew-rate
 - 5 Resistive pull-up
 - Registered input data (synchronization)

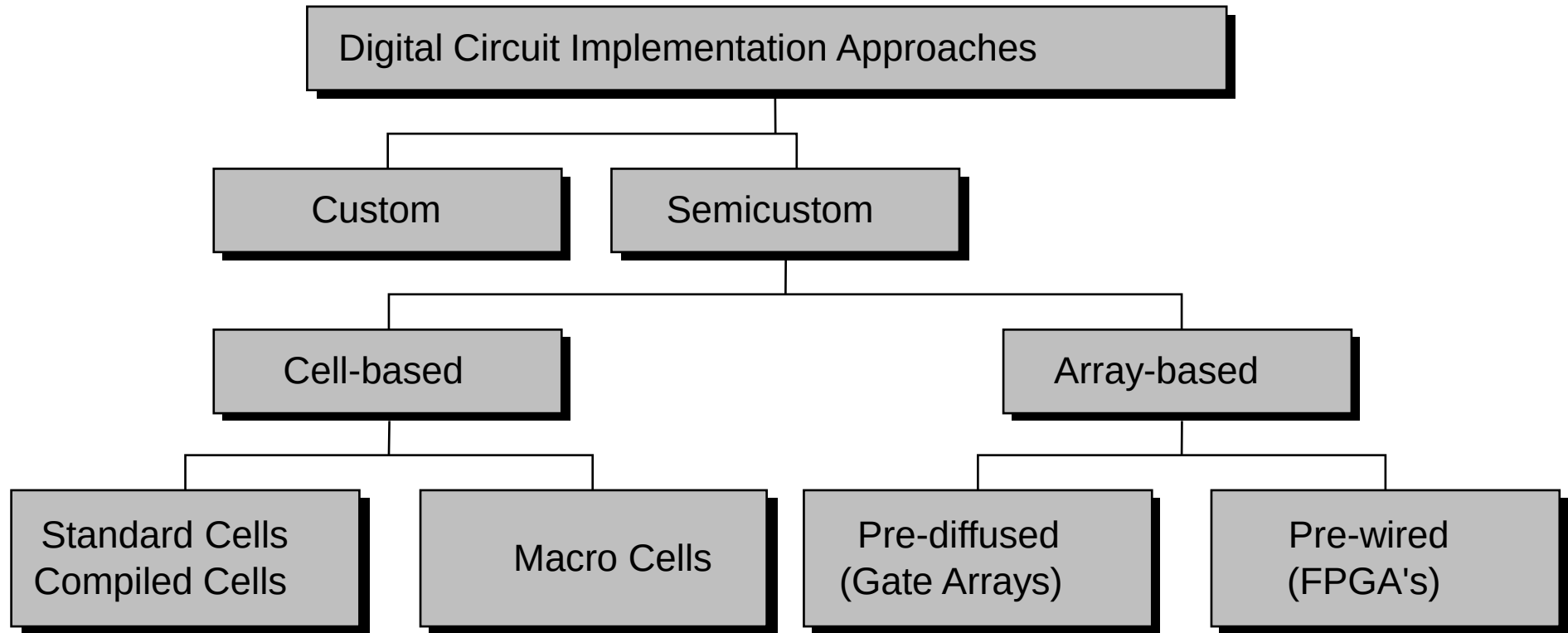
Heterogeneous Programmable Platforms



Outline

- Electronic Abstraction Level Design
- Integrated Circuit Design Styles
- Design Methodologies and Flows
- Cost-Performance Trade-off
- Field Programmable Gate Array
- Design Productivity
- Evolution of EDA Industry
- How to map a system/algorithm to a System on Chip

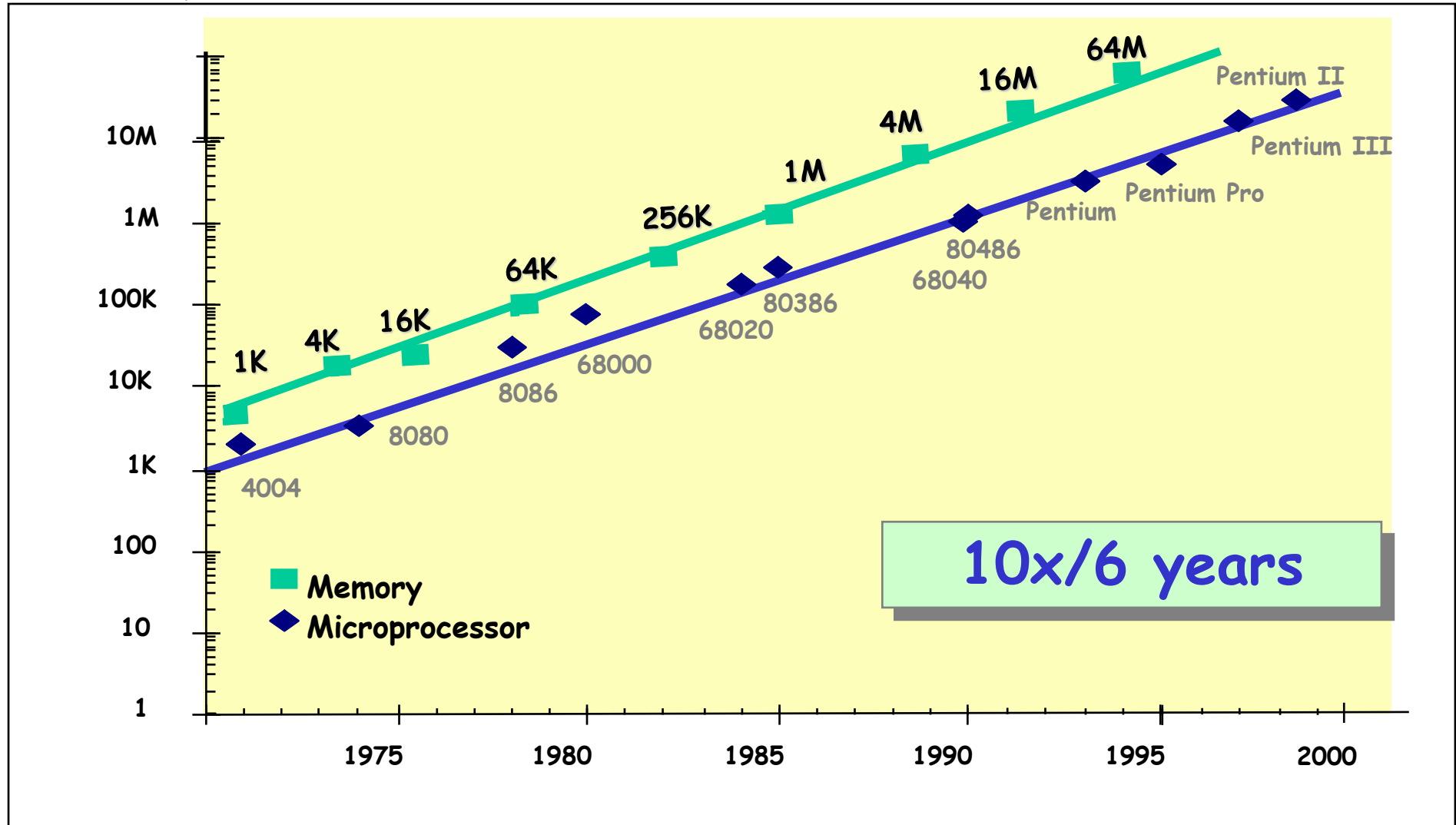
Integrated Circuit Design Styles



Let's compare the Design Productivity !!

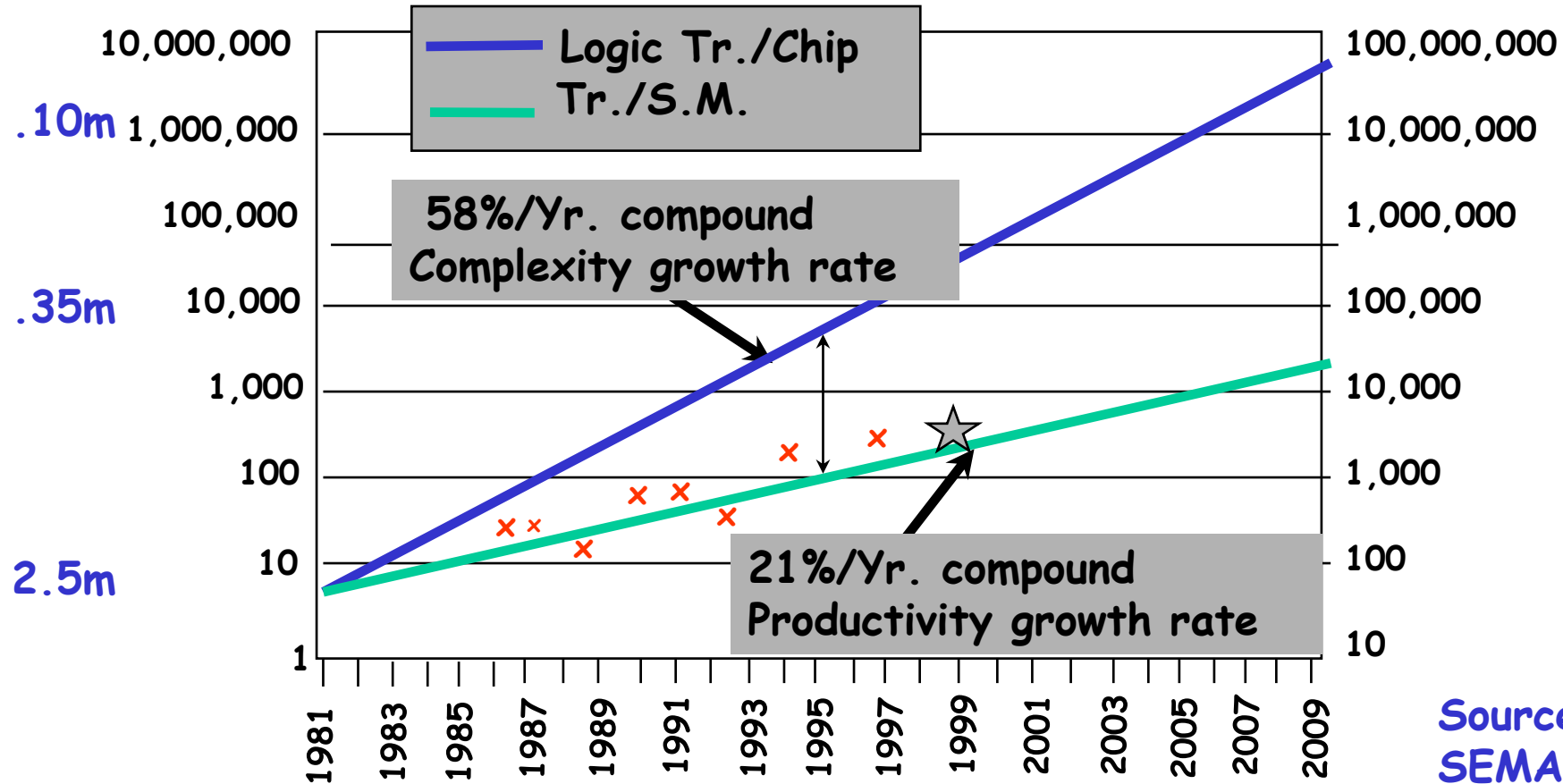
VLSI Technology: Moore's Law

Transistors per die



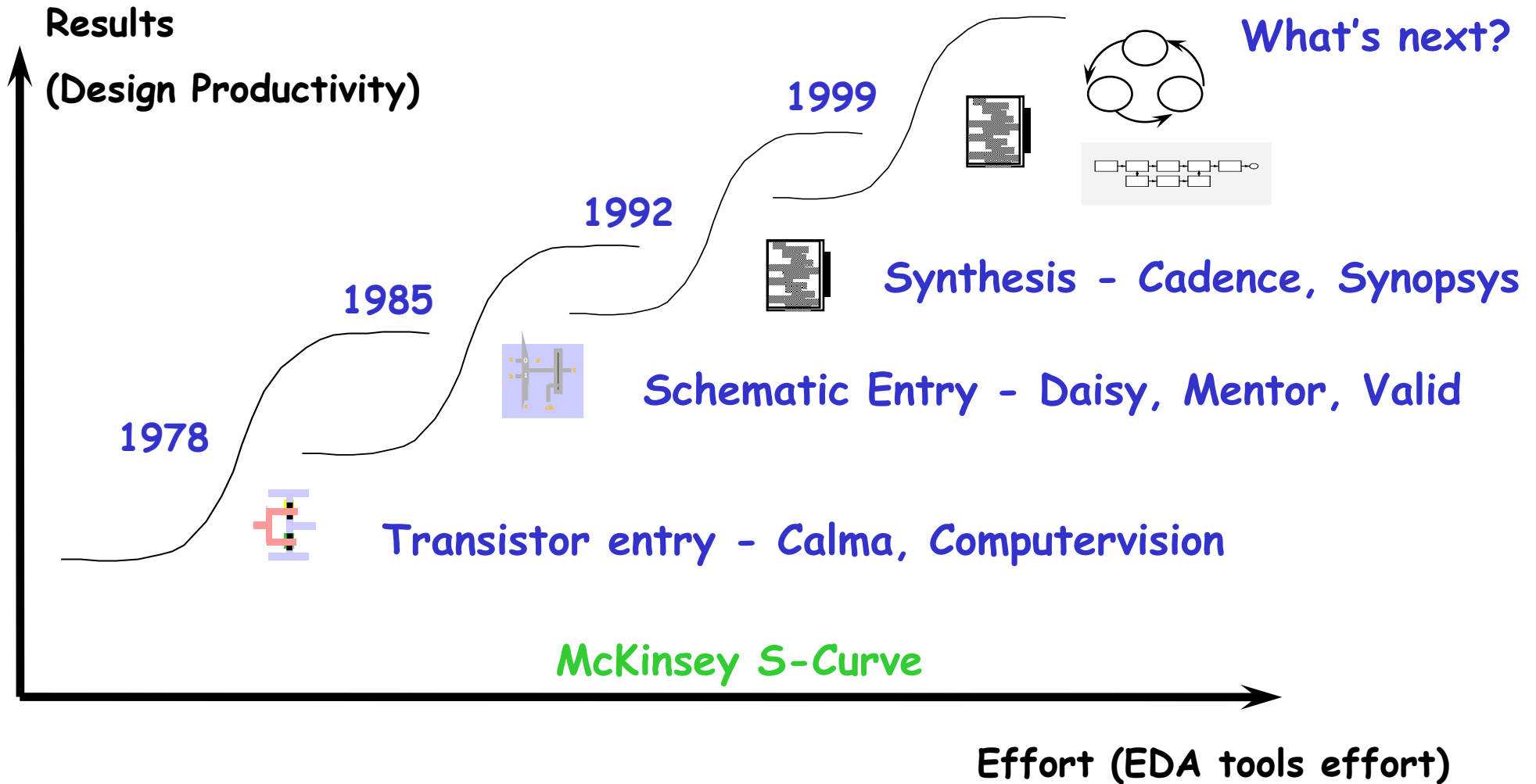
Design Productivity Gap

Logic Transistors per Chip (K)

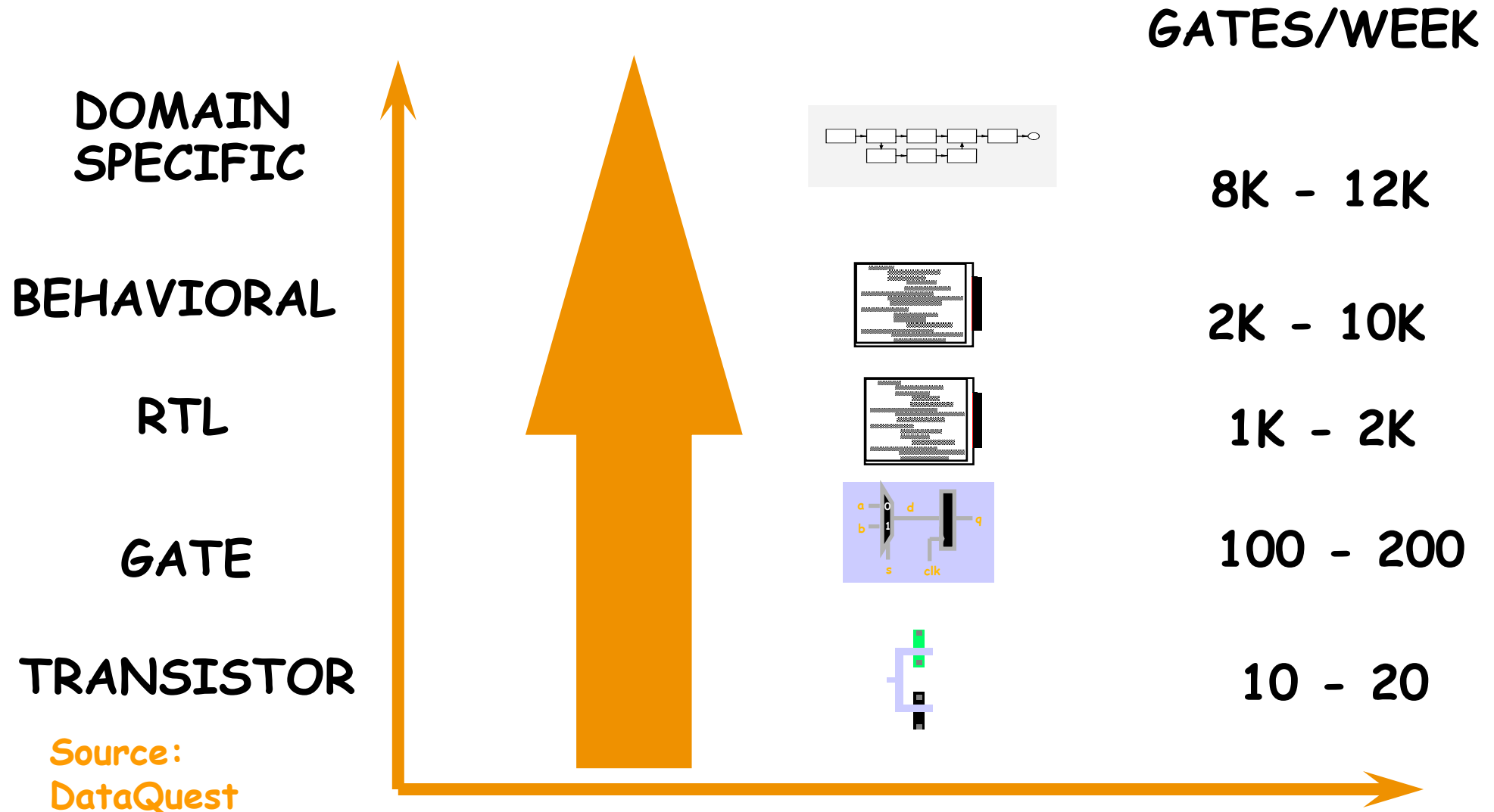


Source:
SEMATECH

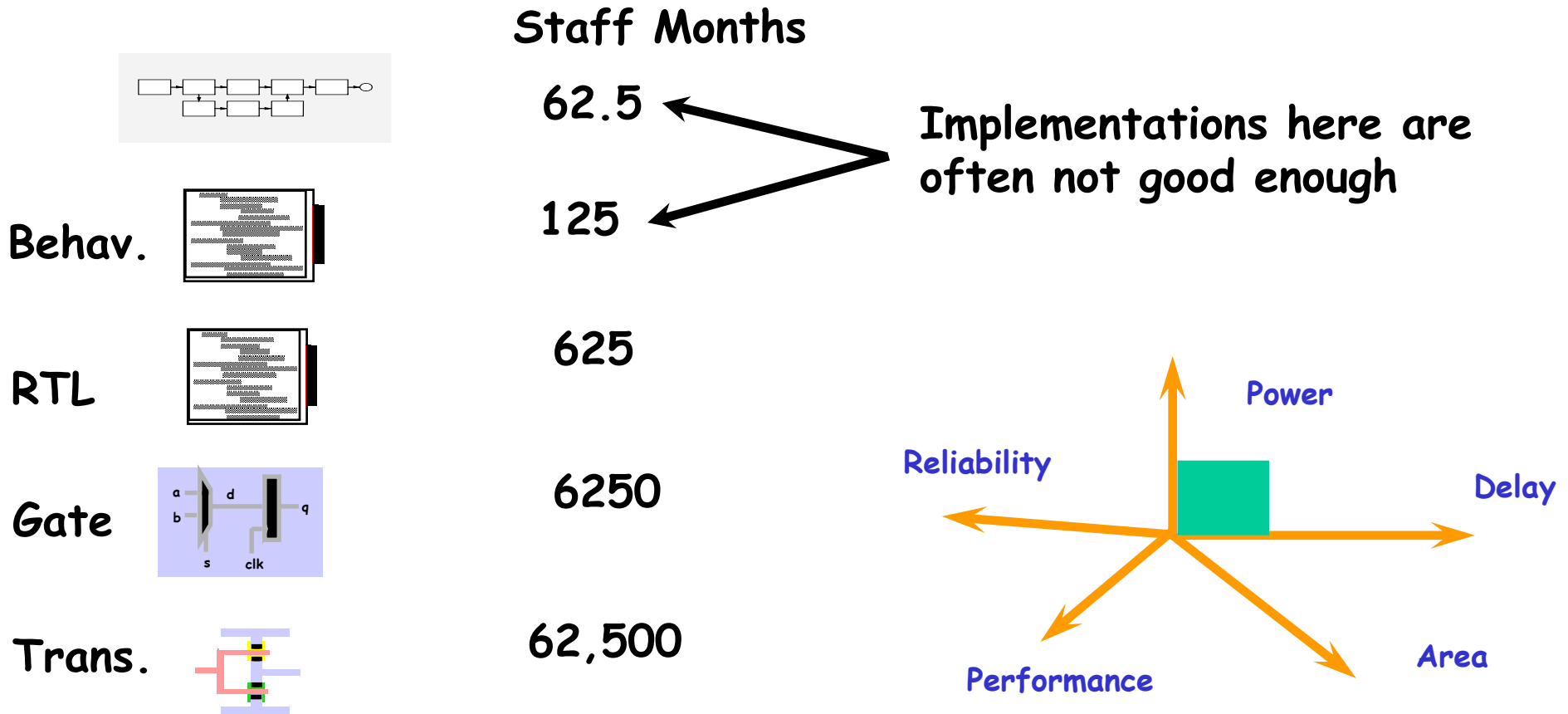
Evolution of the EDA Industry



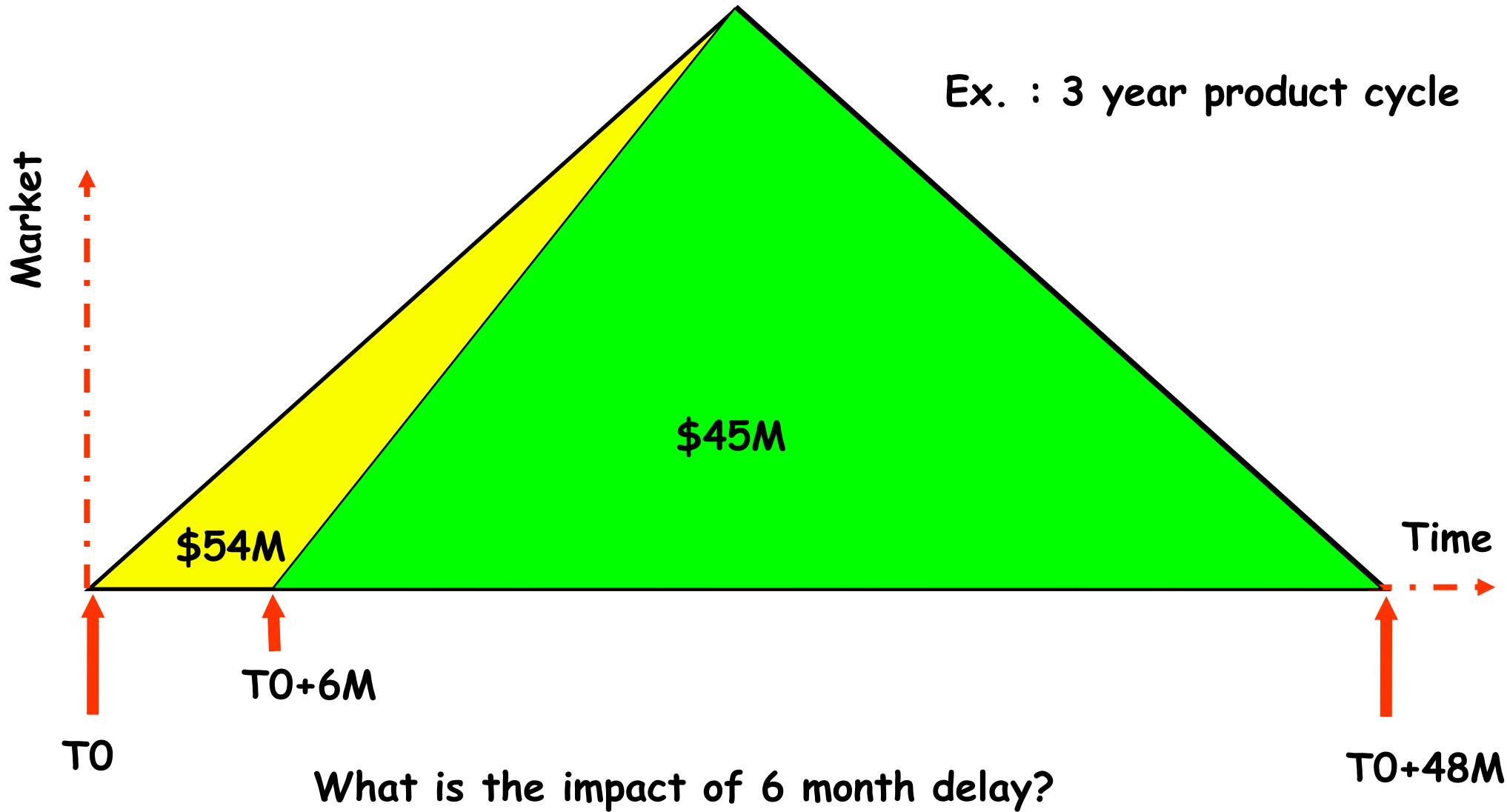
Design Productivity by Approach



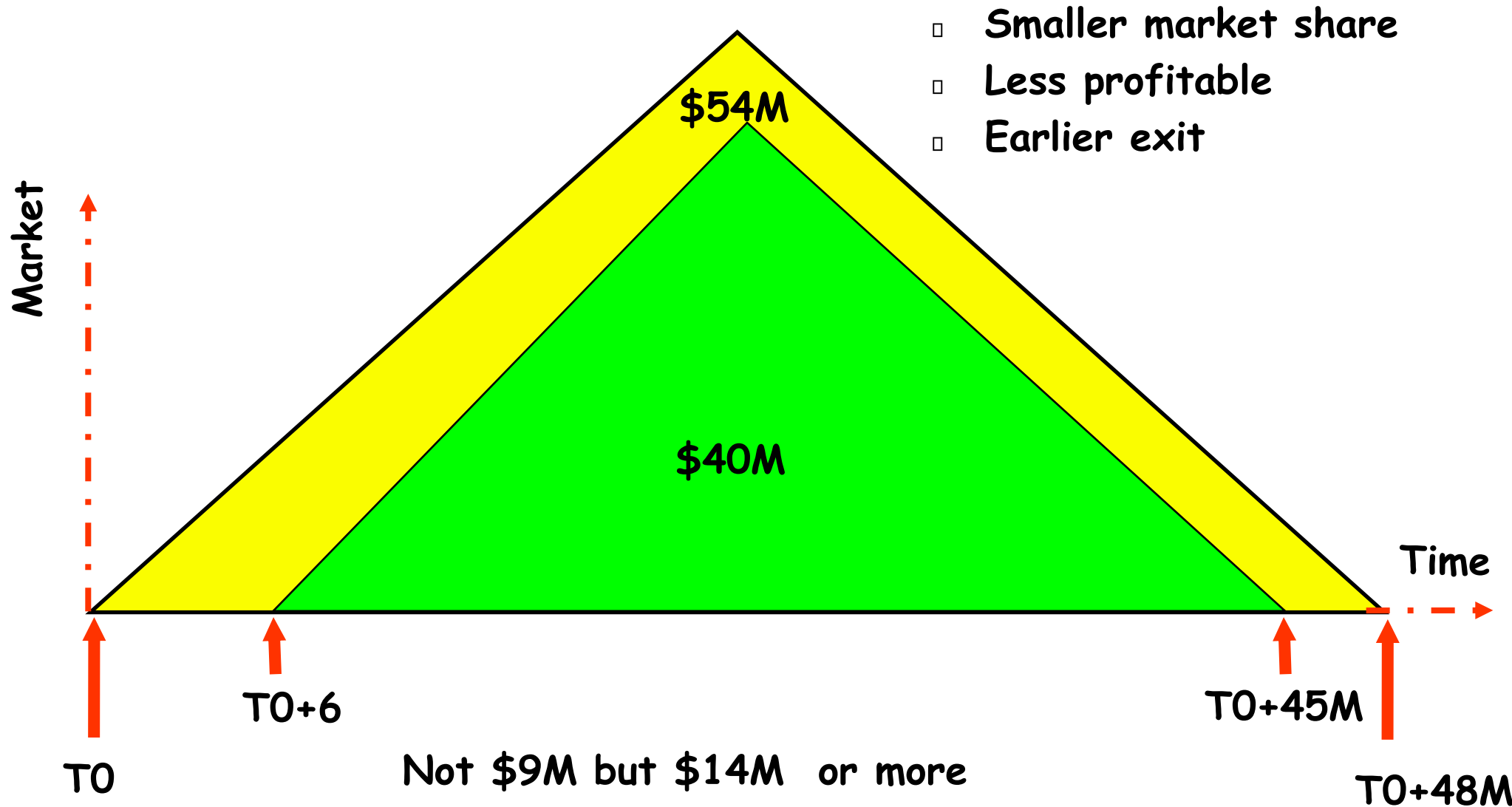
To Design, Implement, Verify 10M transistors



Time to Market (1/2)



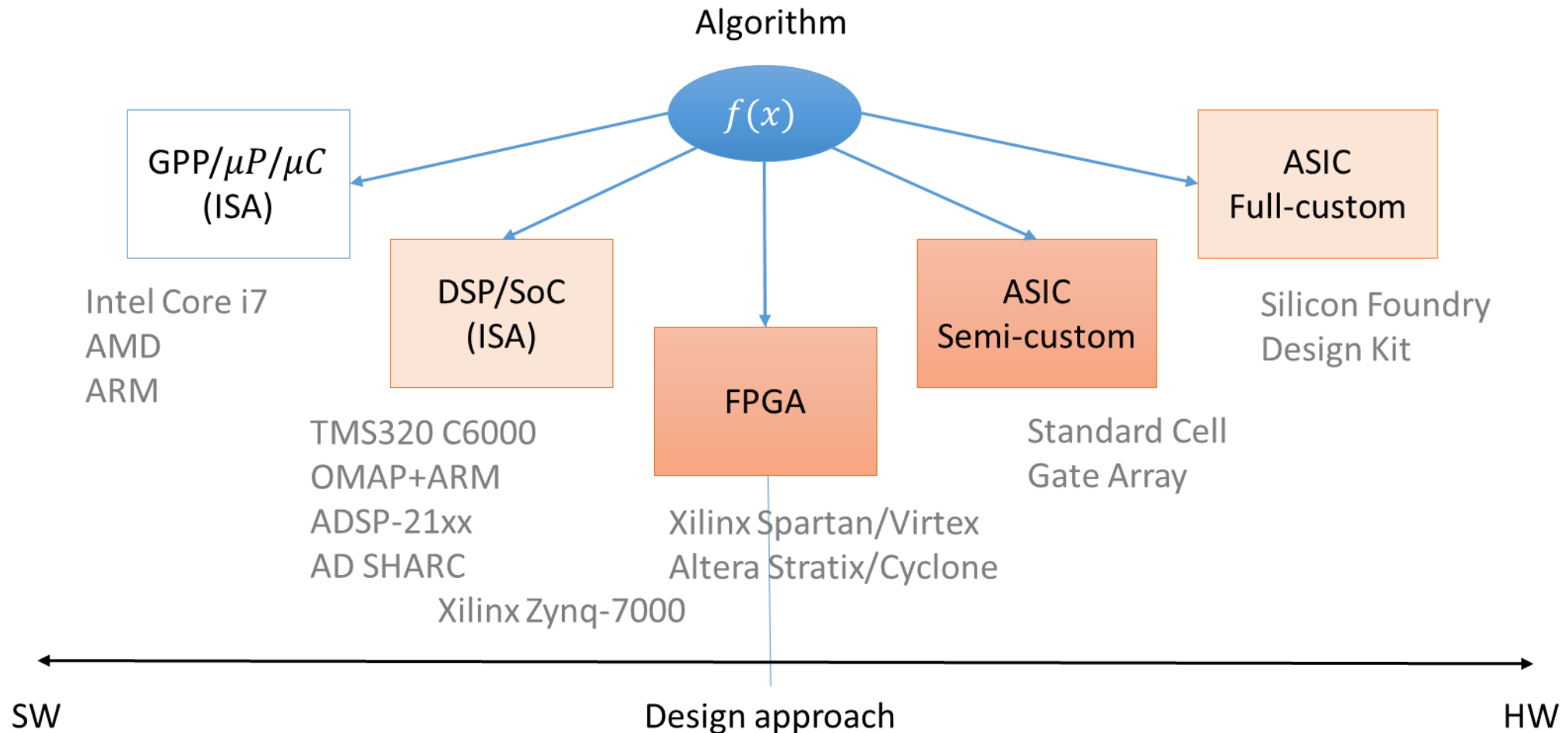
Time to Market (2/2)



Outline

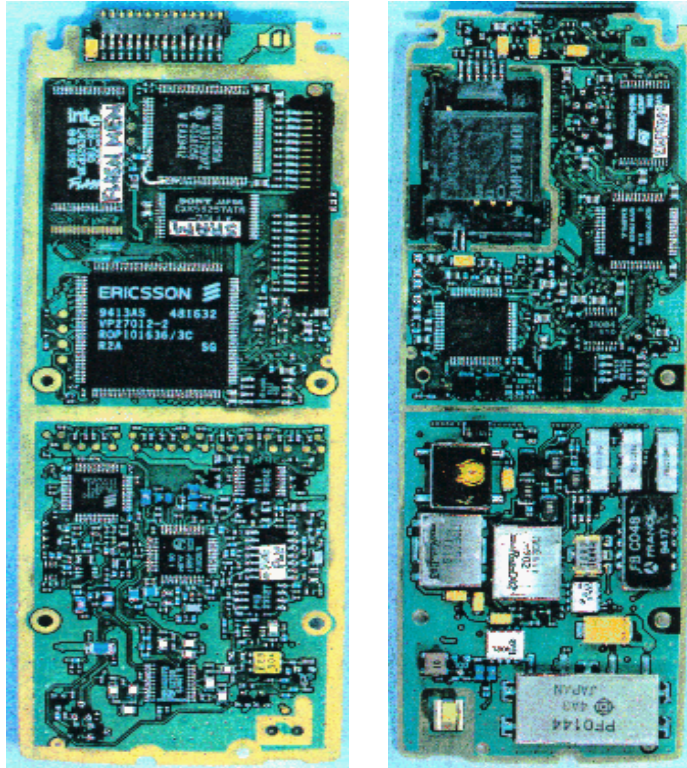
- Electronic Abstraction Level Design
- Integrated Circuit Design Styles
- Design Methodologies and Flows
- Cost-Performance Trade-off
- Field Programmable Gate Array
- Design Productivity
- Evolution of EDA Industry
- How to map a system/algorithm to a System on Chip

How to map a system/algorithm to hw/sw design

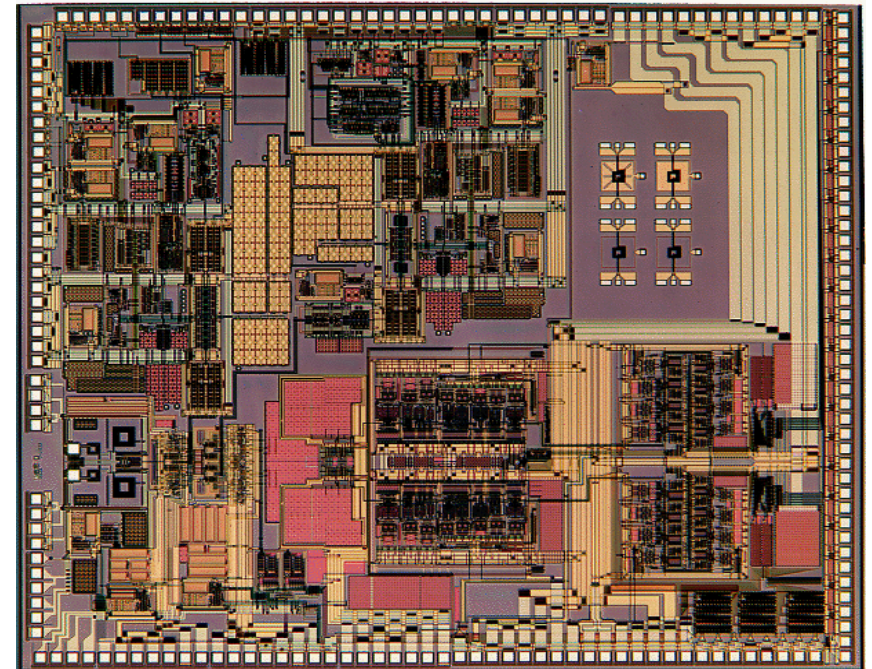


System on a Chip: GSM System

Source:
BWRC

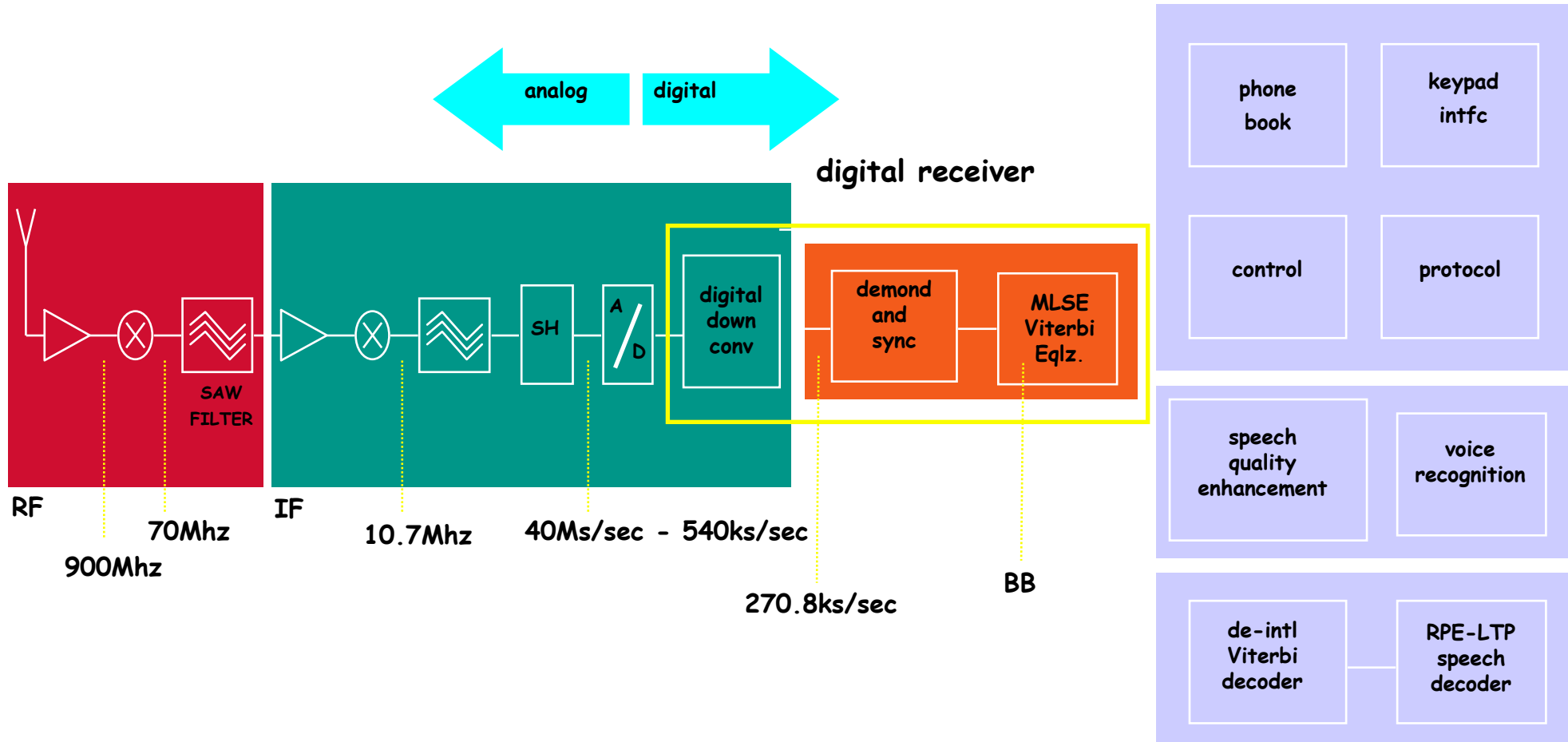


Conventional cellular
phone solution

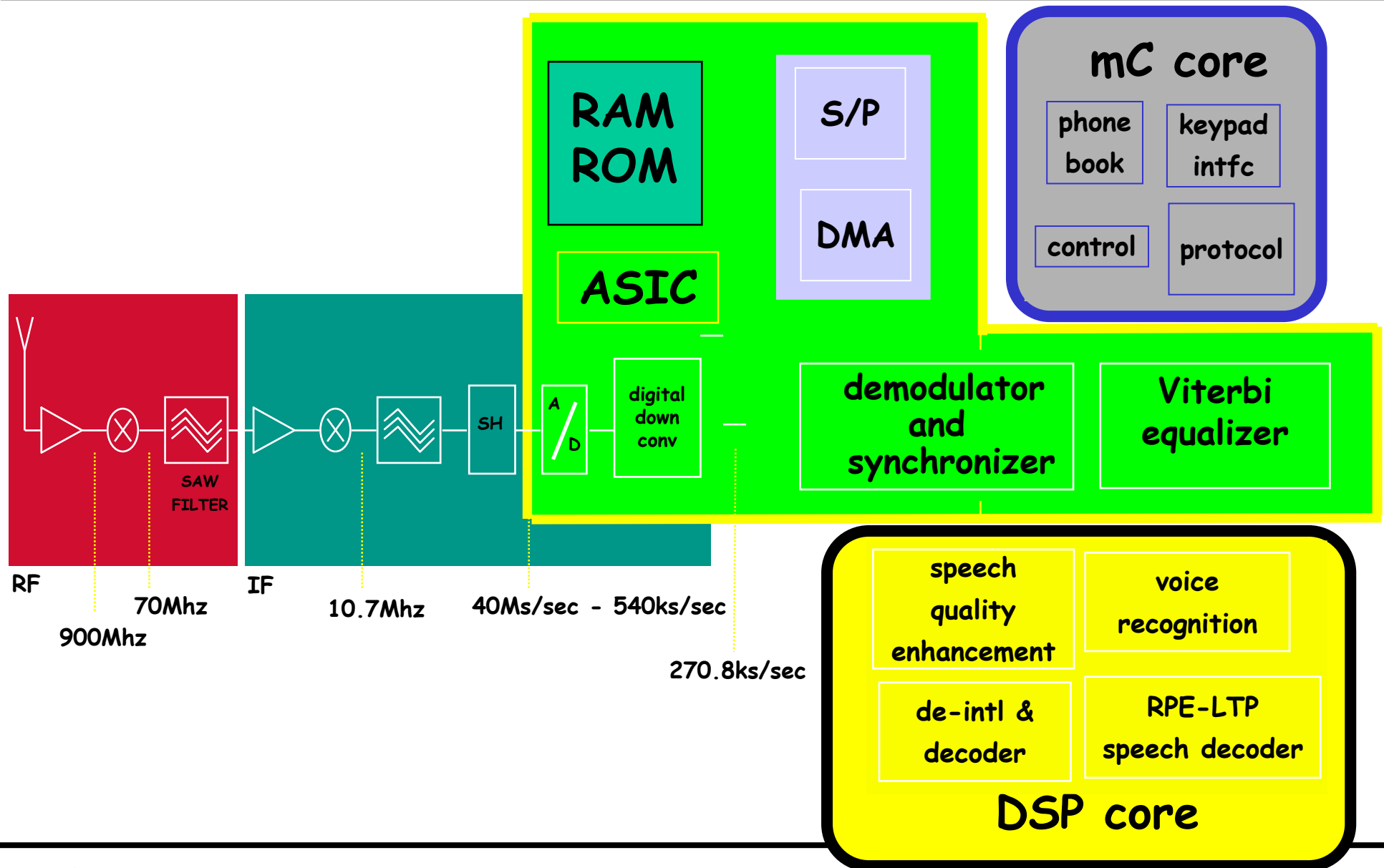


Research into technology and
design methodologies for
CMOS single chip radios

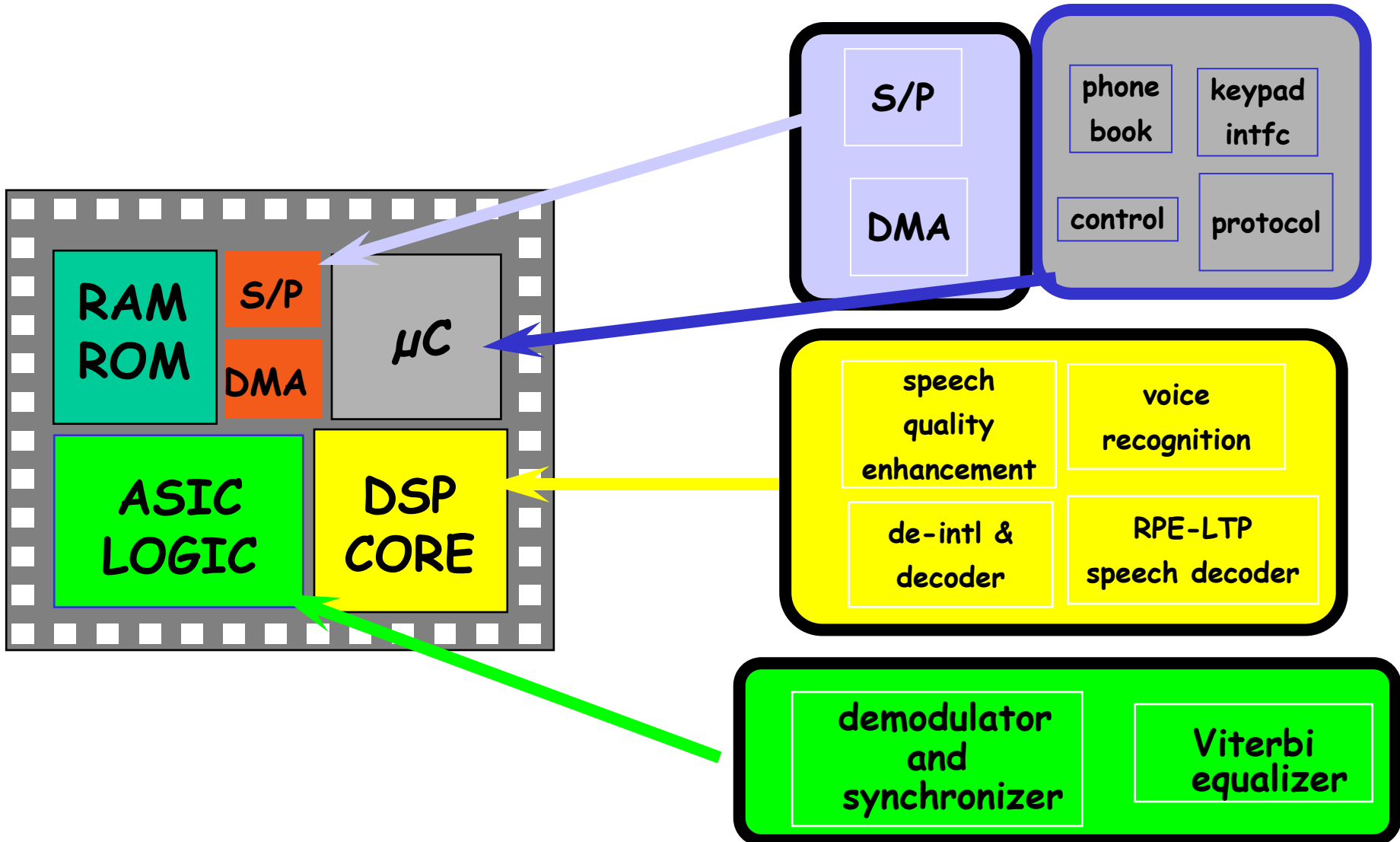
System level Modeling



Targeting an IC Implementation

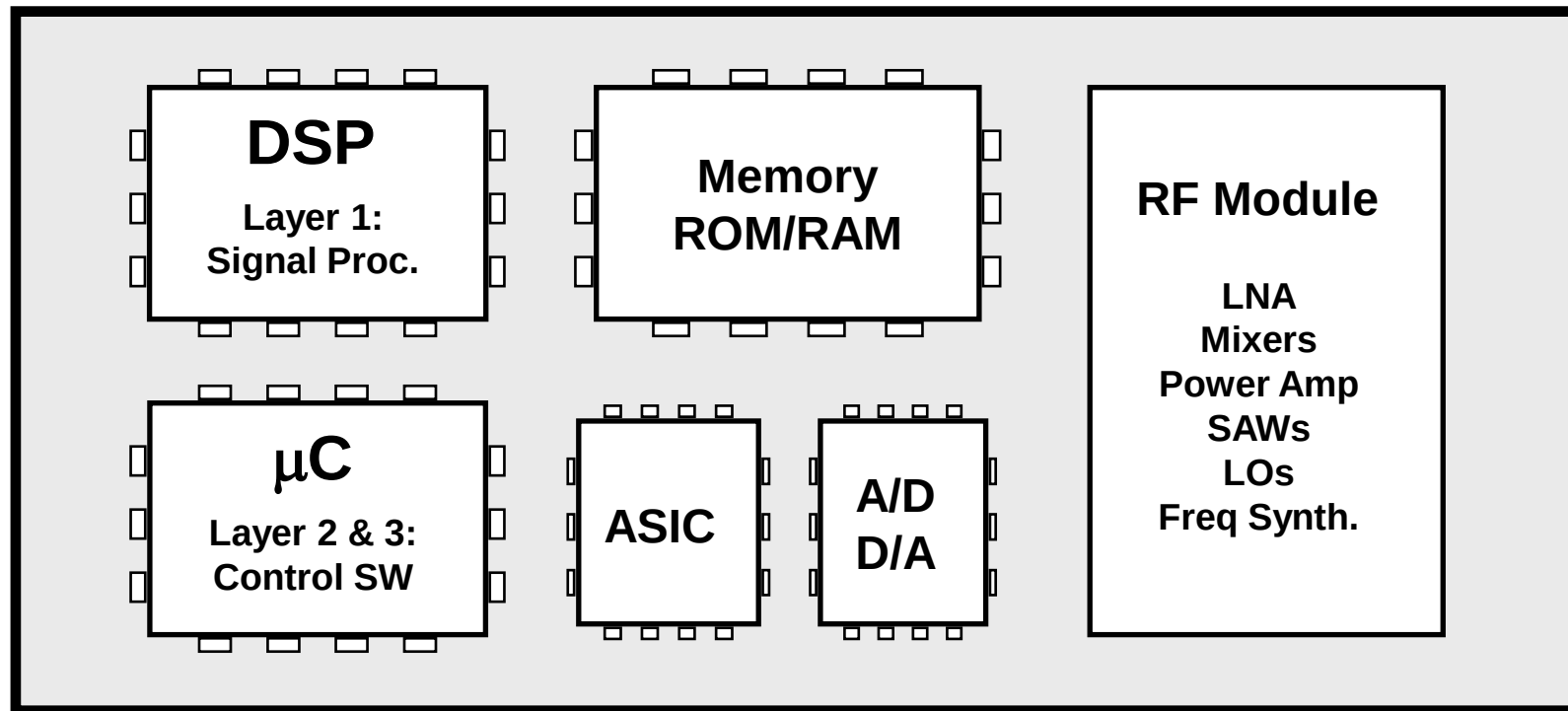


Mapping onto a system on a chip

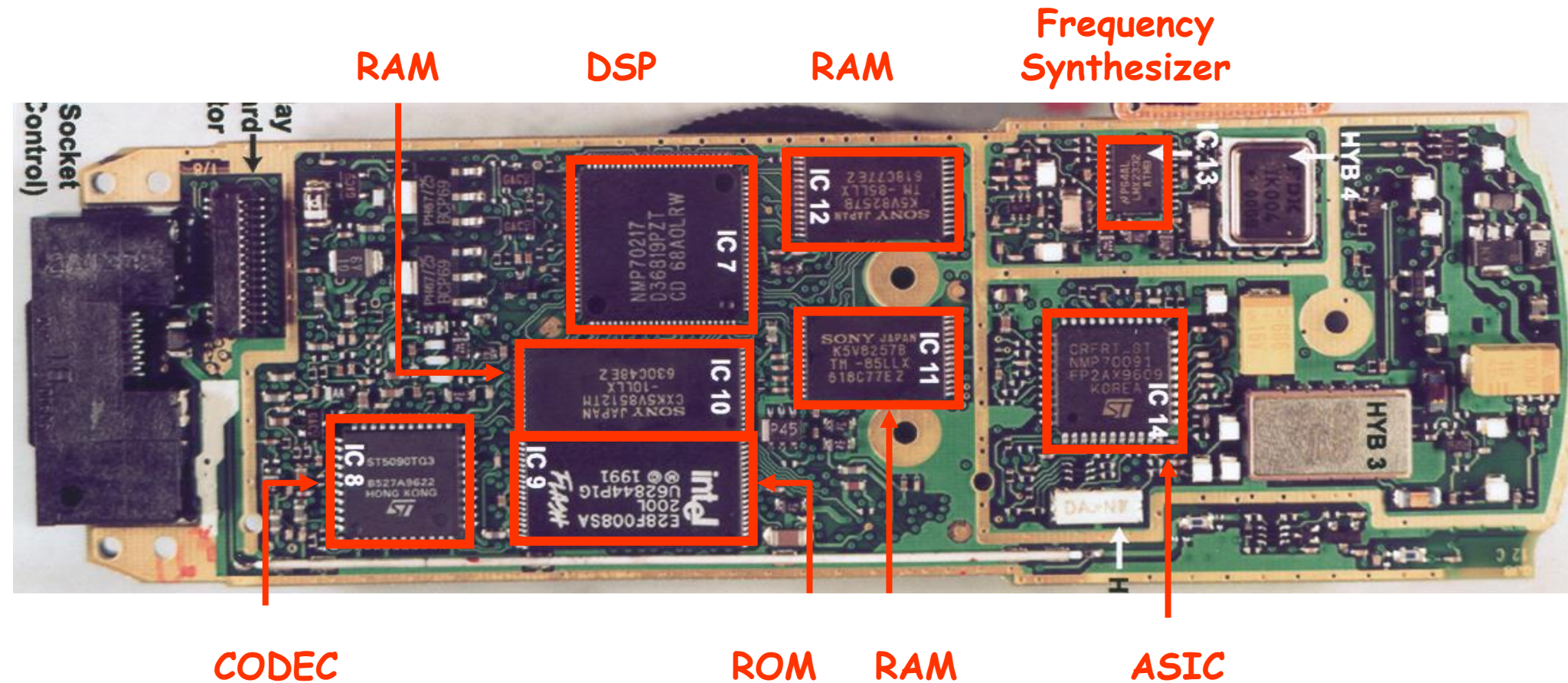


GSM Receiver Implementation (2/3)

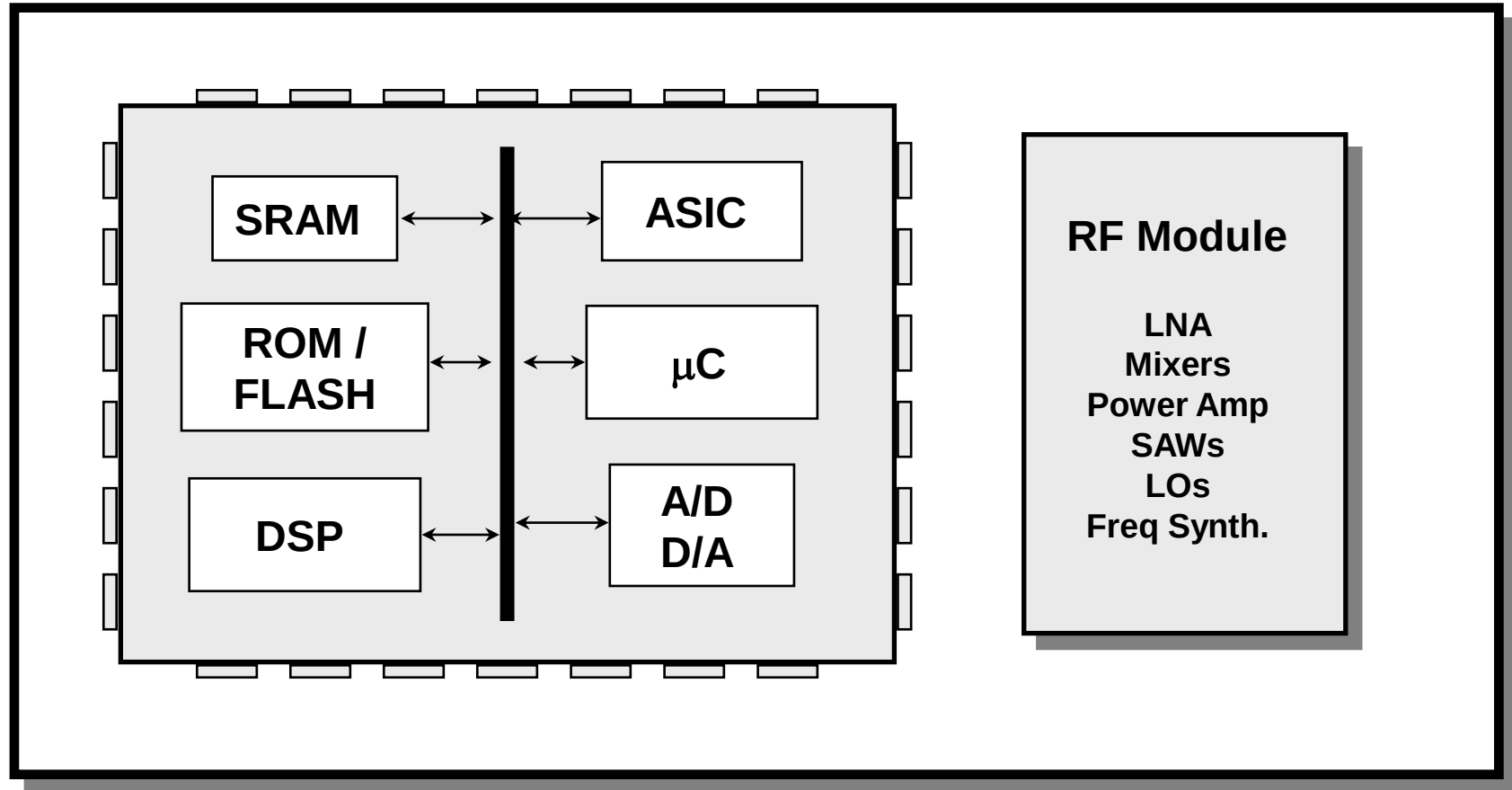
DSP - centric Solution



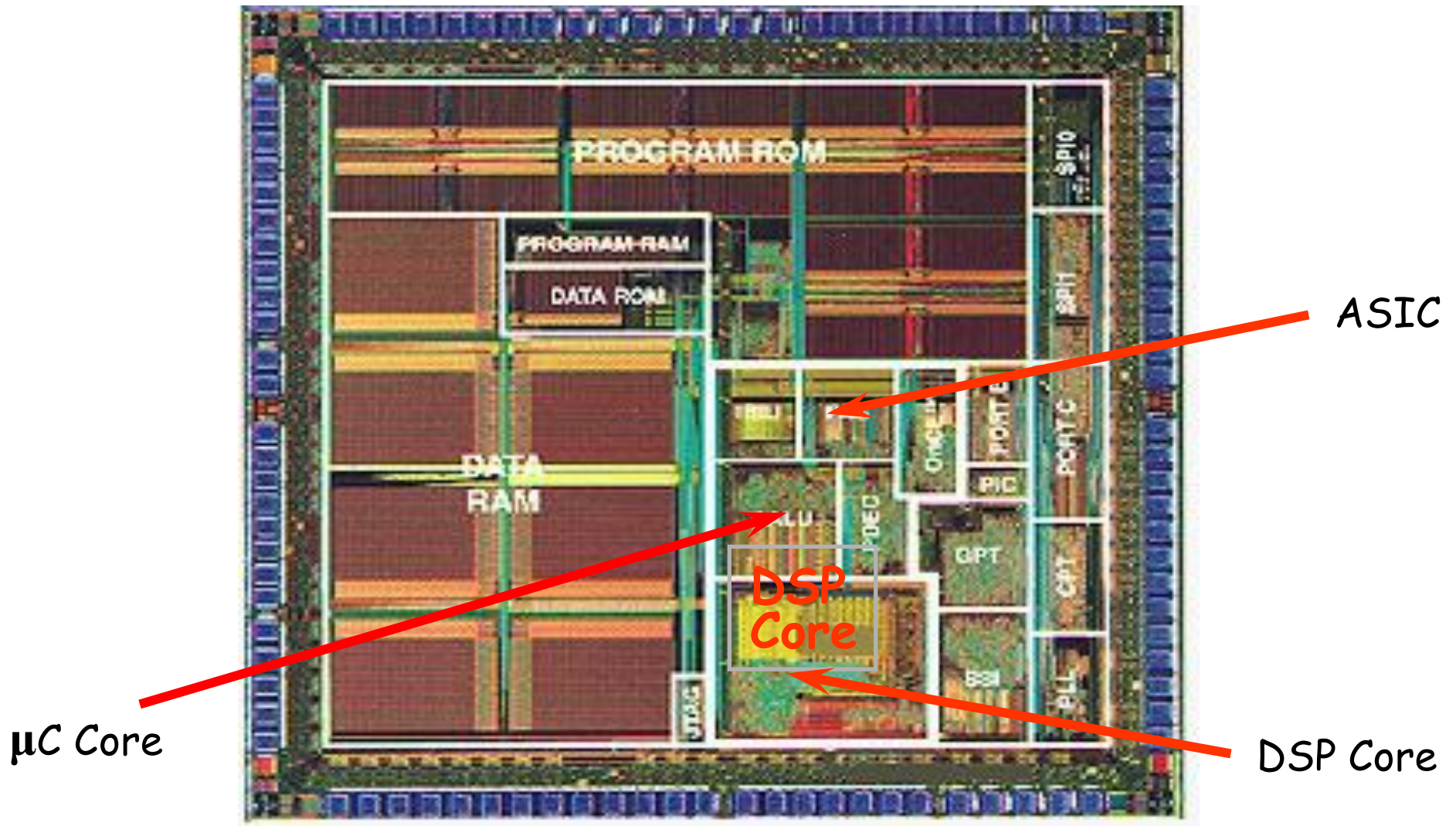
GSM Receiver Implementation (3/3)



GSM SoC Receiver (1/2)

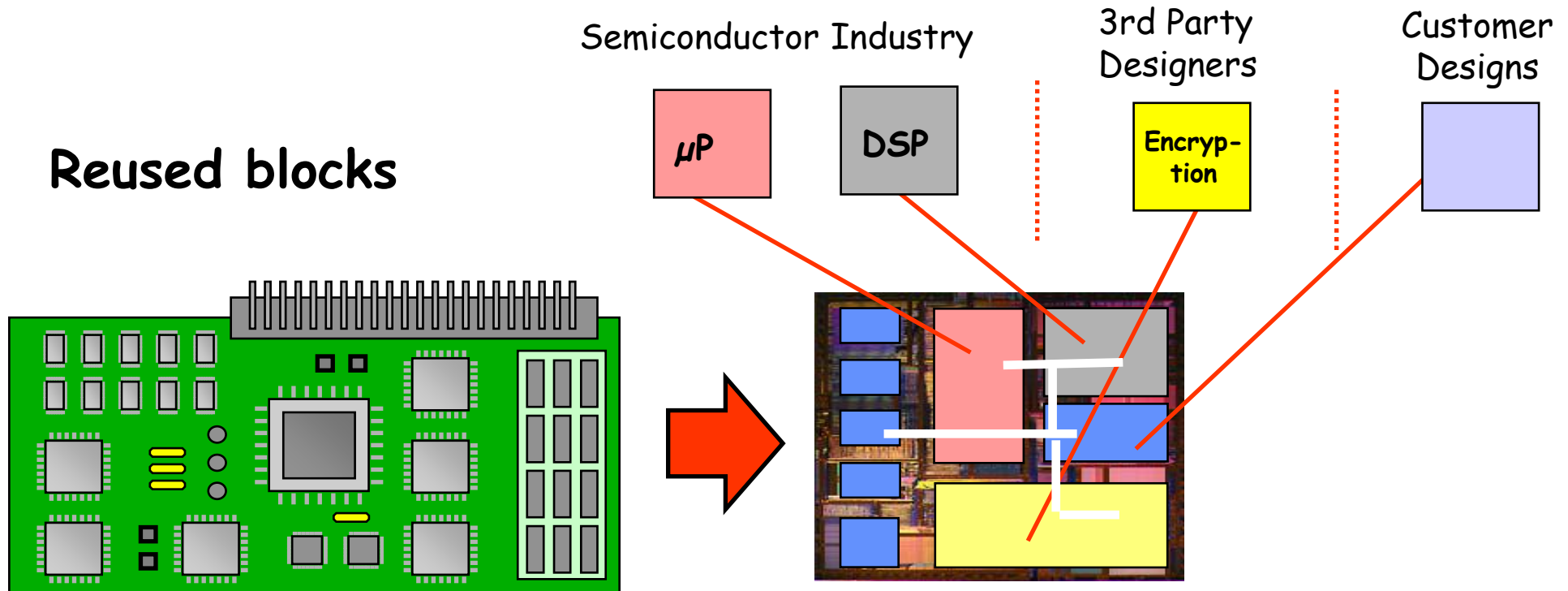


GSM SoC Receiver (2/2)



Re-useable IP

Reused blocks



Achieve required design productivity by assembling re-useable blocks of *intellectual property (IP)*

Source: BWRC

Growing list of IP Blocks

- Video: MPEG, DVD, HDTV
- Audio: MP3, voice recognition
- Processors: CPUs, DSPs, Java
- Networking: ATM, Ethernet, ISDN, FibreChannel, SONET
- Bus: PCI, USB, IEEE 1394
- Memory: SRAM, ROM, CAM
- Wireless: CDMA, TDMA
- Communication: modems, transceivers
- Coding: speech, Viterbi, Reed-Solomon
- Display drivers/controllers: TFT
- Other: sensors, encryption/decryption, GPS
- Power PC core: 3.1mm² in 0.35μ
- ARM Core: 3.8 mm² in 0.35μ
- MPEG2 Decoder: ~65k gates
- PCI Bus: ~8k gates
- Ethernet MAC: ~7k gates (soft)
- RSA Encryption: ~7k gates

Re-useable IP: Limiting factors

- Company Culture: from design for *one* use to design for *multiple* reuse.
- Legal Protection
- Contract Delays
- IP/VC Availability
- IP/VC Quality
- Mix & Match



Virtual Component Exchange

to organize a marketplace for the buying and selling of semiconductor IP/VC.



Virtual Socket Interface Alliance (VSIA) addresses the mix and match of IP/VC facilitating reuse for SoC designs. From 1996 to 2008 (<http://www.vsi.org/>).

Now refers to the Spirit Consortium (<http://www.spiritconsortium.org>)

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About Us

Who We Are

Accellera Systems Initiative is an independent, not-for profit organization dedicated to create, support, promote, and advance system-level design, modeling, and verification standards for use by the worldwide electronics industry. We are composed of a [broad range of members](#) that fully support the work of our [technical committee](#) to develop technology standards that are balanced, open, and benefit the worldwide electronics industry. Leading companies and semiconductor manufacturers around the world are using our electronic design automation (EDA) and intellectual property (IP) standards in a wide range of projects in numerous application areas to develop consumer, mobile, wireless, automotive, and other "smart" electronic devices. Through an ongoing partnership with the IEEE, standards and technical implementations developed by Accellera Systems Initiative are contributed to the IEEE for formal standardization and ongoing governance.

Our Mission



Accellera Systems Initiative Chair gives an update on Accellera activities at DAC 2014.

Upgrade
your

Memory VIP

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- On-the-fly reconfiguration
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USB 3.0 femtoPHY - TSMC
16FF+LL x1 OTG,
North/South Poly Orientation



DVB-T2 Modulator



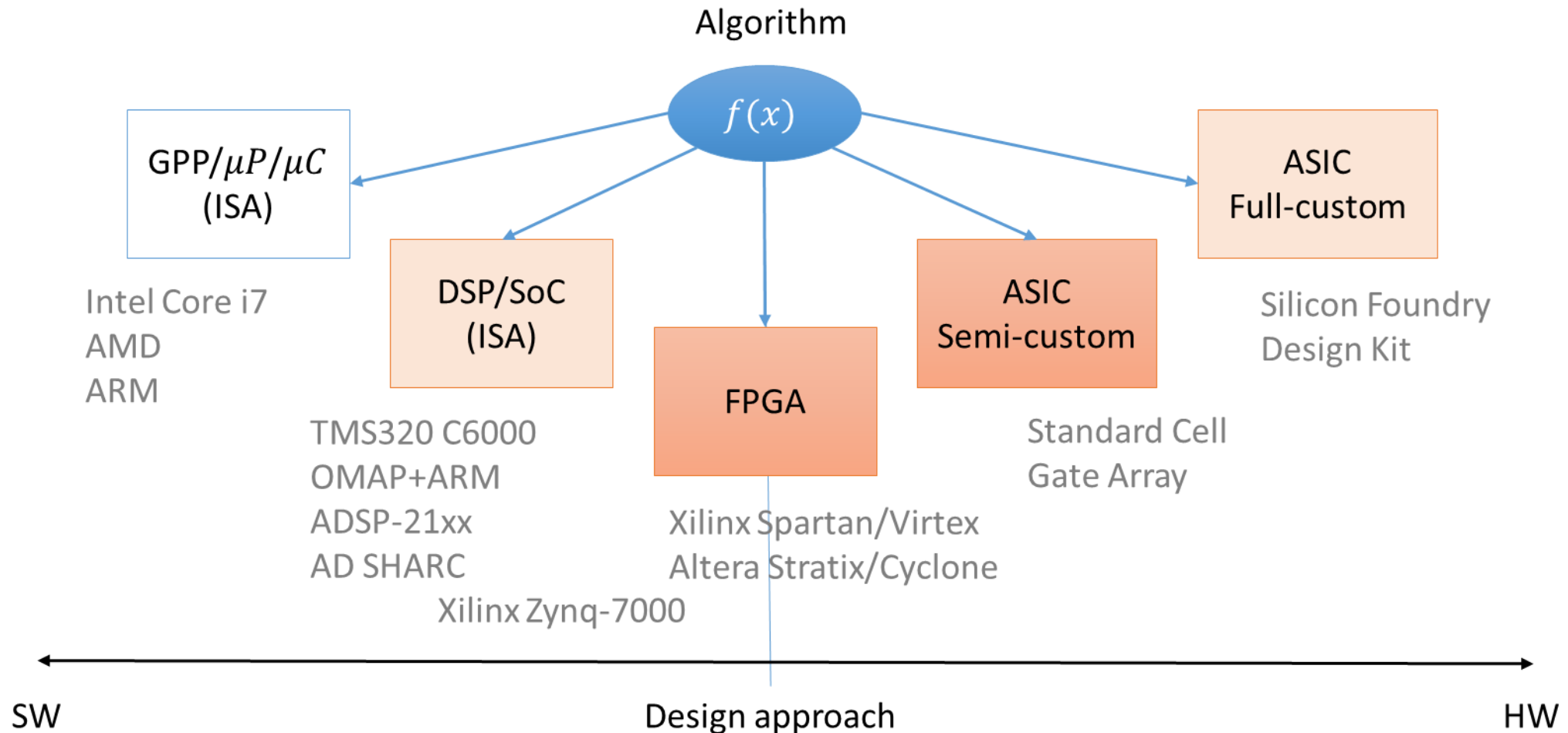
FPGA core in TSMC 28HPM

Featured

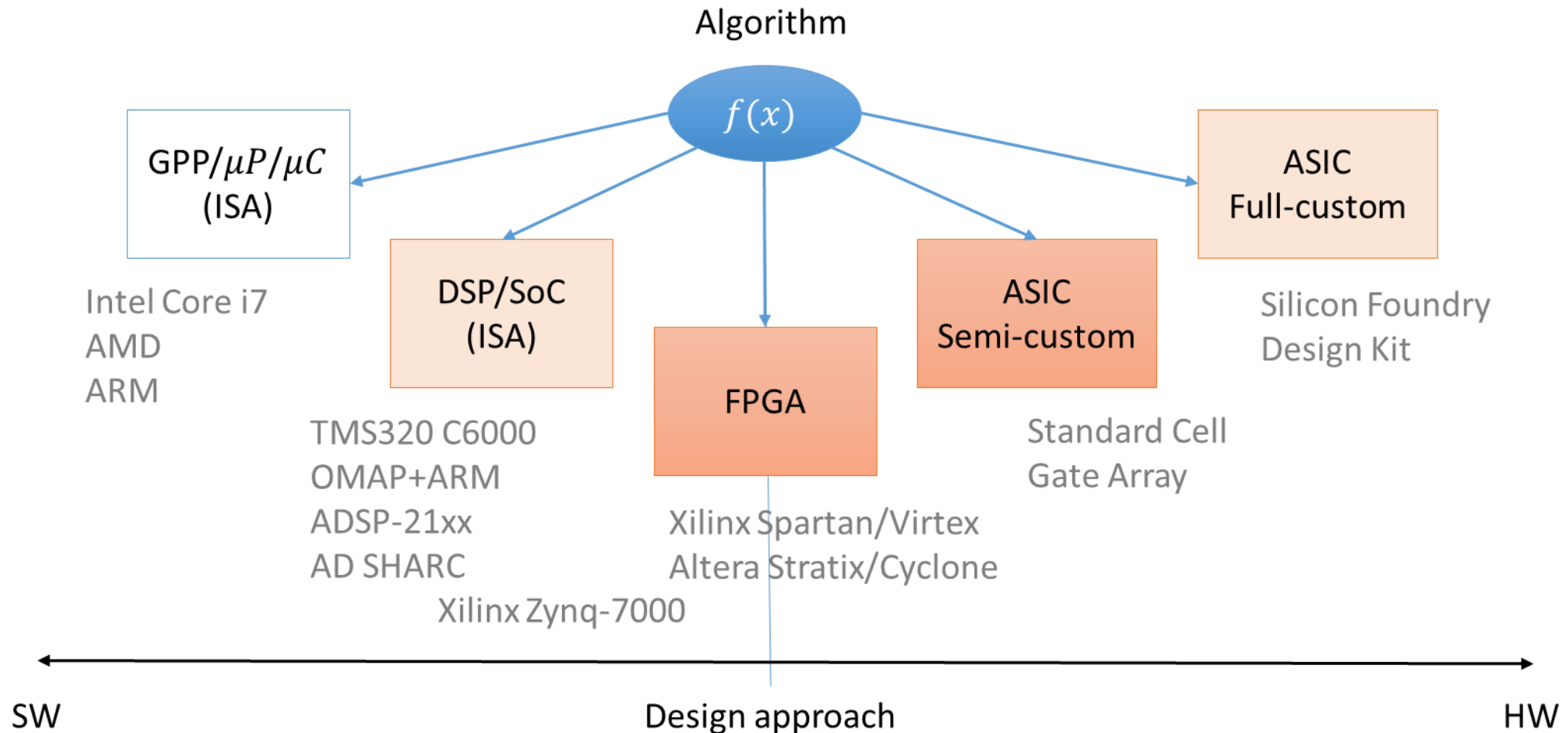
Requirements for re-use

- In order to be re-usable, I.P.'s must be properly validated:
 - Documented
 - Compliant with design tools
 - Fully debugged
 - Testable
 - Manufacturable
 - Possibly verified on silicon
- Re-use in place of re-development can reduce costs, but:
 - There is an additional cost in making the IP re-usable
 - Who pays the cost is not always the same who profits from re-use
- An I.P. risks being obsolete before is completely validated
- Two main trends:
 - Design blocks for re-use (e.g. ARM)
 - High level I.P.'s (algorithms, software blocks, synthesizable blocks have a better chance to pay their cost

How to map a system/algorithm to hw/sw design

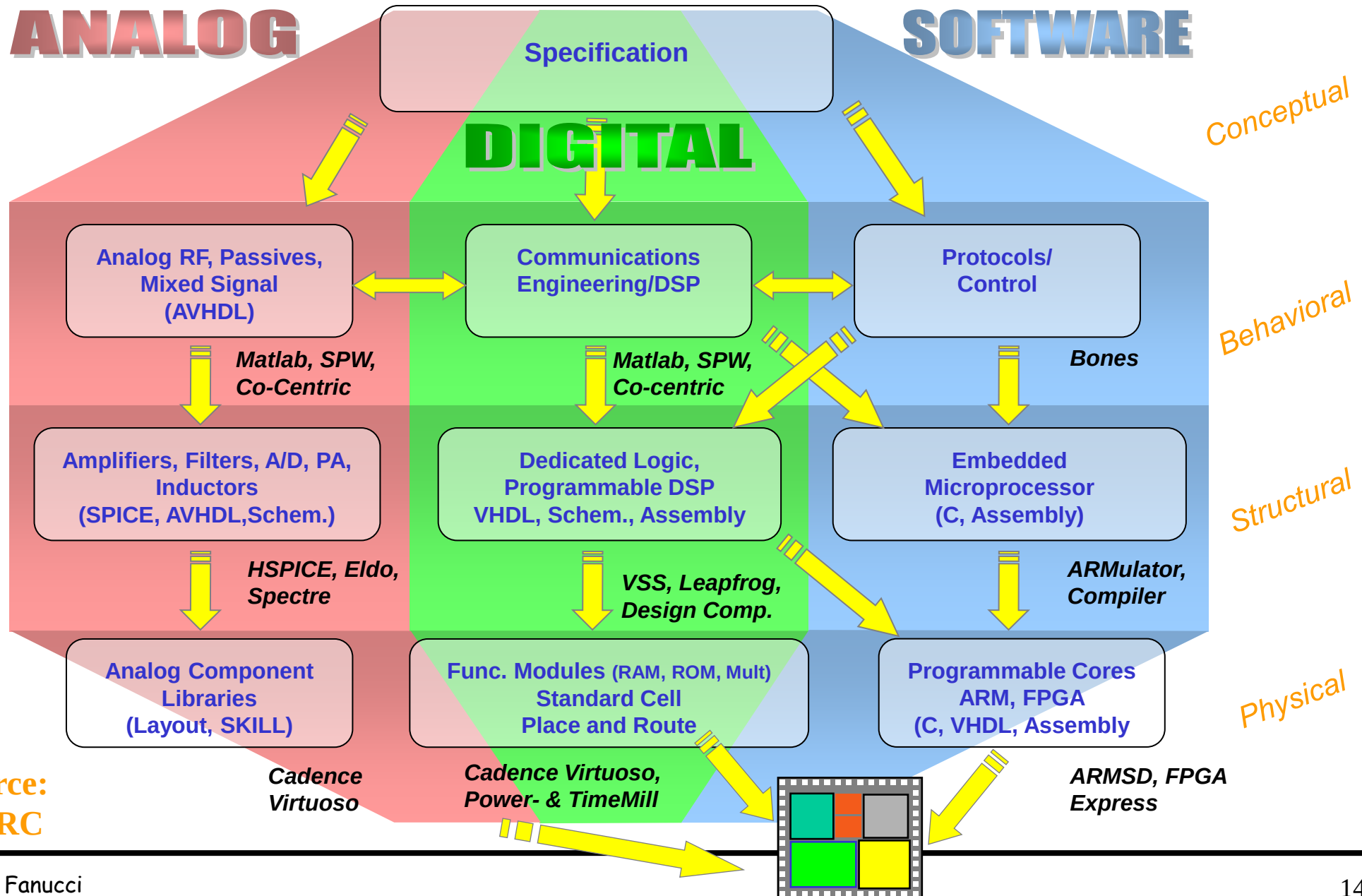


How to map a system/algorithm to hw/sw design



Design is the art of managing engineering trade-offs !

System on a Chip: Design Flow Example



Summary

- ▣ The capabilities of CMOS technology have now reached a threshold which allows System Level Integration
- ▣ The development of proper design methodology and CAD tools are fundamental for future System On a Chip :
 - ▣ Efficient management of design complexity
 - ▣ Reduced design cycle (time to market)
 - ▣ Efficient exploration of the solution space
 - ▣ Re-usability of whole designs or part of them
 - ▣ Technology independence

End, Questions ?

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- Integrated Circuit Design Styles
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